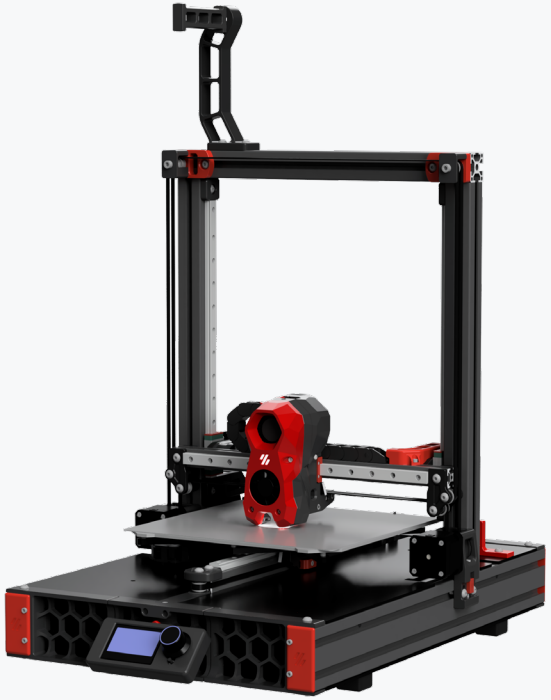
VORON



SWITCHWIRE ASSEMBLY MANUAL

We build space shuttles with gardening tools
so anyone can have a space shuttle of their own.

VERSION 2025-9-24

Before you begin on your journey, a word of caution.

In the comfort of your own home you are about to assemble a robot. This machine can maim, burn, and electrocute you if you are not careful. Please do not become the first VORON fatality. There is no special Reddit flair for that.

Please, read the entire manual before you start assembly. As you begin wrenching, please check our Discord channels for any tips and questions that may halt your progress.

Most of all, good luck!

THE VORON TEAM

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PART PRINTING GUIDELINES

The Voron Team has provided the following print guidelines for you to follow in order to have the best chance at success with your parts. There are often questions about substituting materials or changing printing standards, but we recommend you follow these.

3D PRINTING PROCESS

Fused Deposition Modeling (FDM)

MATERIAL

ABS

LAYER HEIGHT

Recommended: 0.2mm

EXTRUSION WIDTH

Recommended: Forced 0.4mm

INFILL TYPE

Grid, Gyroid, Honeycomb, Triangle or Cubic

INFILL PERCENTAGE

Recommended: 40%

WALL COUNT

Recommended: 4

SOLID TOP/BOTTOM LAYERS

Recommended: 5

PRINT IT FORWARD (PIF)

Often times our community members have issues printing ABS will bootstrap themselves into a VORON using our Print It Forward program. This is a service where approved members with VORON printers can make you a functional set of parts to get your own machine up and running. Check Discord if you have any interest in having someone help you out.

HOW TO GET HELP

If you need assistance with your build, we're here to help. Head on over to our Discord group and post your questions. This is our primary medium to help VORON Users and we have a great community that can help you out if you get stuck.

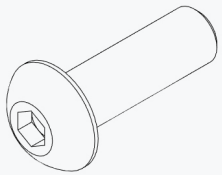
https://discord.gg/voron

THIS IS JUST A REFERENCE

This manual is designed to be a simple reference manual. Building a Voron can be a complex endeavour and for that reason we recommend downloading the CAD files off our Github repository if there are sections you need clarification on. It can be sometimes be easier to follow along when you have the whole assembly in front of you.

https://github.com/vorondesign

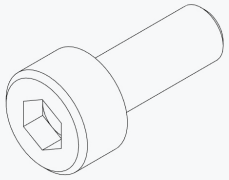
https://docs.vorondesign.com



BUTTON HEAD CAP SCREW (BHCS)

Metric fastener with a domed shape head and hex drive. Most commonly found in locations where M5 fasteners are used.

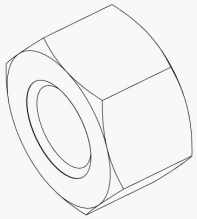
ISO 7380-1



SOCKET HEAD CAP SCREW (SHCS)

Metric fastener with a cylindrical head and hex drive. The most common fastener used on the Voron.

ISO 4762



HEX NUT

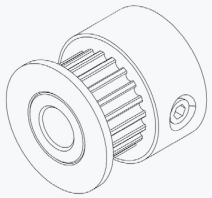
Hex nuts couple with bolts to create a tight, secure joint. You'll see these used in both M3 and M5 variants throughout this guide.

ISO 4032

FLAT HEAD COUNTERSUNK SCREW (FHCS)

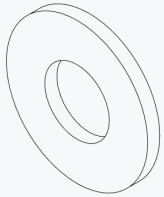
Metric fastener with a cone shaped head and a flat top.

ISO 10642



PULLEY

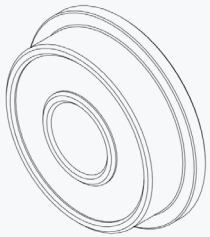
GT2 pulley used on the motion system of the Voron.



SHIM

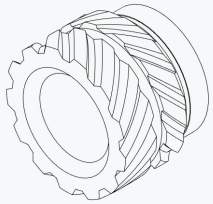
Not to be confused with stamped washers. These are used in all M5 call-out locations in this manual.

DIN 988



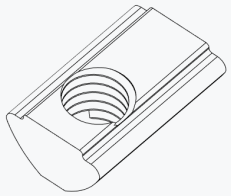
F695 BEARING

A ball bearing with a flange used in various gantry locations.



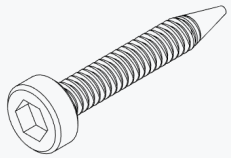
HEAT SET INSERT

Heat inserts with a soldering tip so that they melt the plastic when installed. As the plastic cools, it solidifies around the knurls and ridges on the insert for excellent resistance to both torque and pull-out.



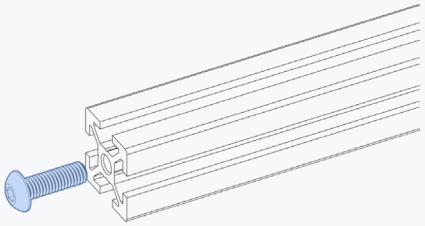
POST INSTALL T-SLOT NUT

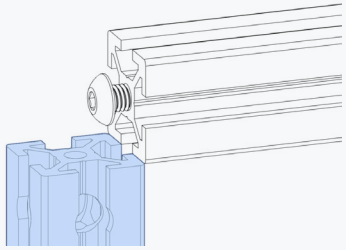
Nut that can be inserted into the slot of an aluminium profile. Used in both M3 and M5 variants throughout this guide.



SELF TAPPING SCREW

Fastener with a pronounced thread profile that is screwed directly into plastic.



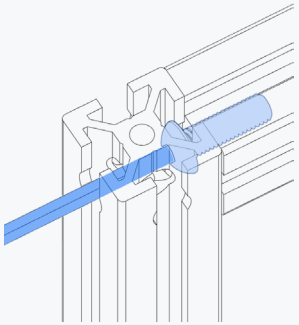


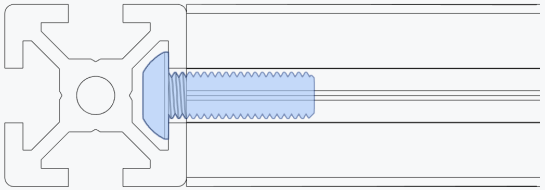
BLIND JOINT BASICS

Blind Joints provide a cost effective and rigid assembly method.

The head of the BHCS is slid into the channel of another extrusion and securely fastened through a small access hole in the extrusion.

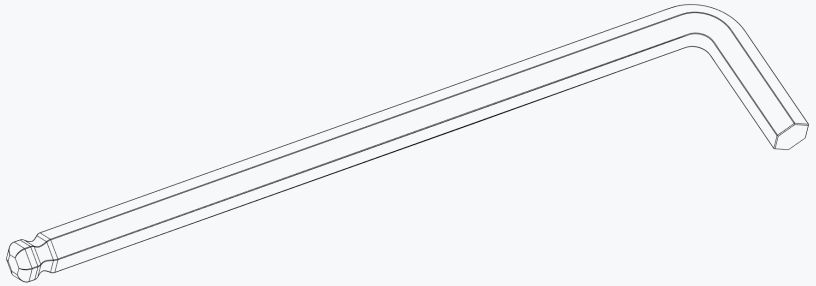
https://voron.link/onjwmcd





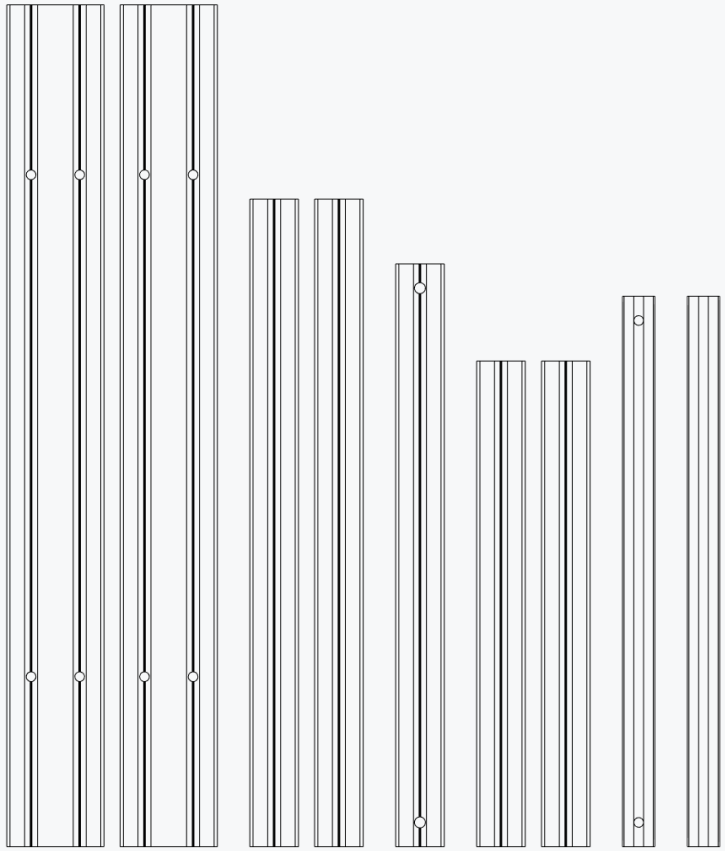
BALL-END DRIVER

Some parts of this design require the use of a ball-end hex driver for assembly. We recommend you get a 2.5mm and 3mm one.



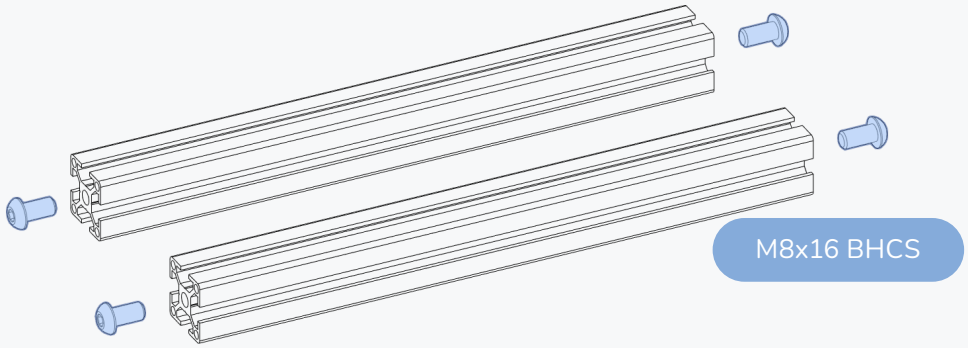
2.5MM HEX DRIVER

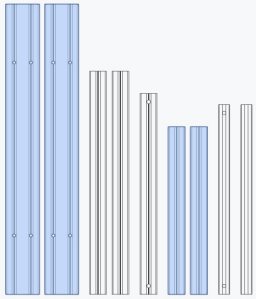
The 2.5mm hex driver will see a lot of use in this build. A quality driver is strongly recommended. Refer to the sourcing guide for suggestions.



GETTING EXTRUSIONS TOGETHER

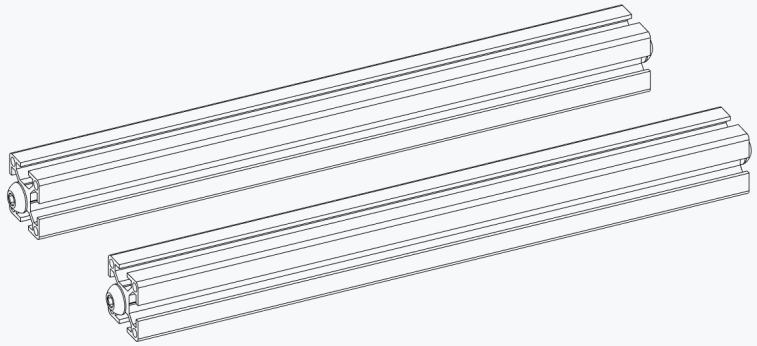
Separate the extrusions you're going to need for this section of the build. We've laid out all the parts you should have and highlighted the ones that will be used in the following sections.

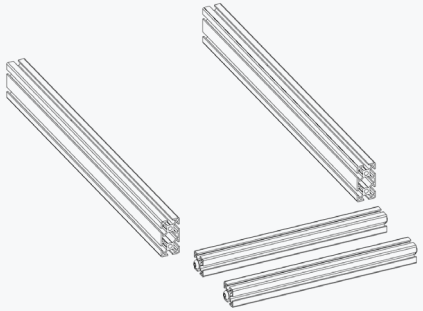




BLIND JOINT BASICS

Blind Joints provide a cost effective and rigid assembly method. The head of the BHCS is slid into the channel of another extrusion and securely fastened through a small access hole in the other extrusion.





FIRST BLIND JOINT

This design relies on blind joints to assemble the frame. We outlined the basics of blind joints on page 8.

If you've never assembled one before we recommend you watch the linked guide.

https://voron.link/onjwmcd

UPSIDE DOWN ASSEMBLY

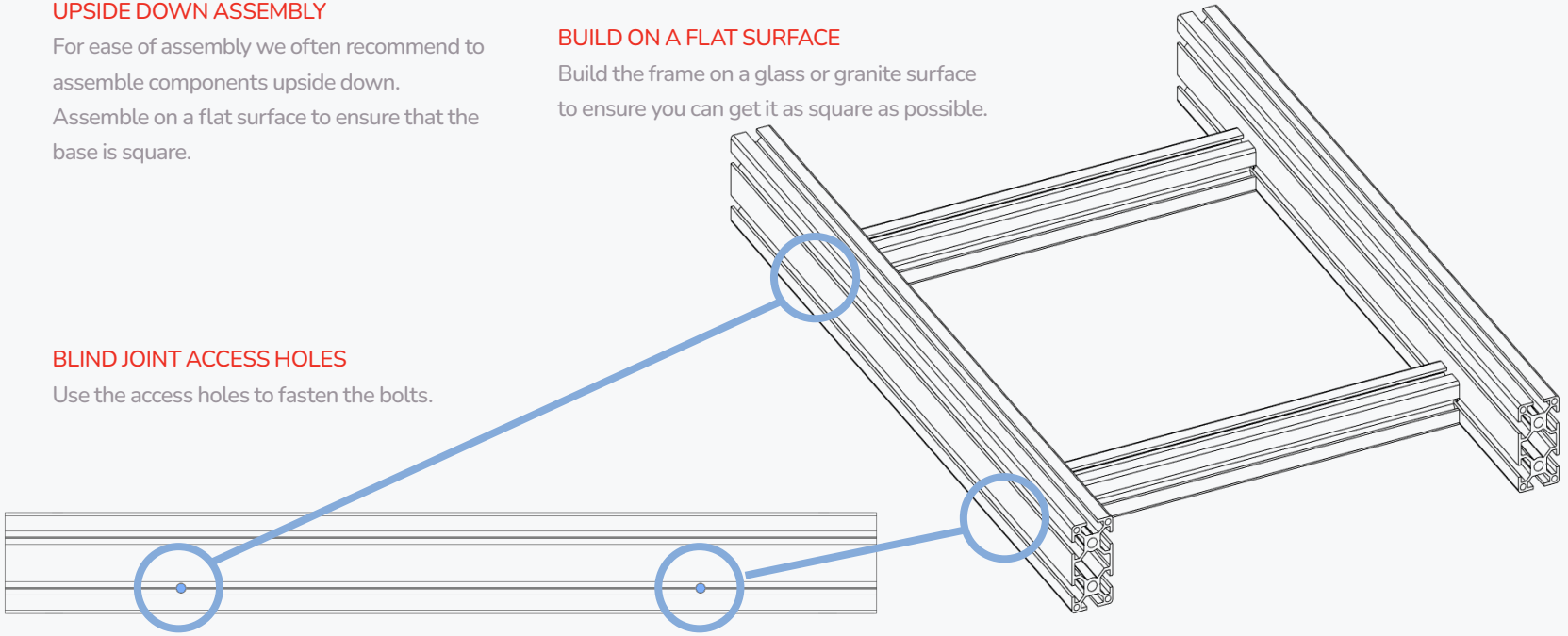
For ease of assembly we often recommend to assemble components upside down. Assemble on a flat surface to ensure that the base is square.

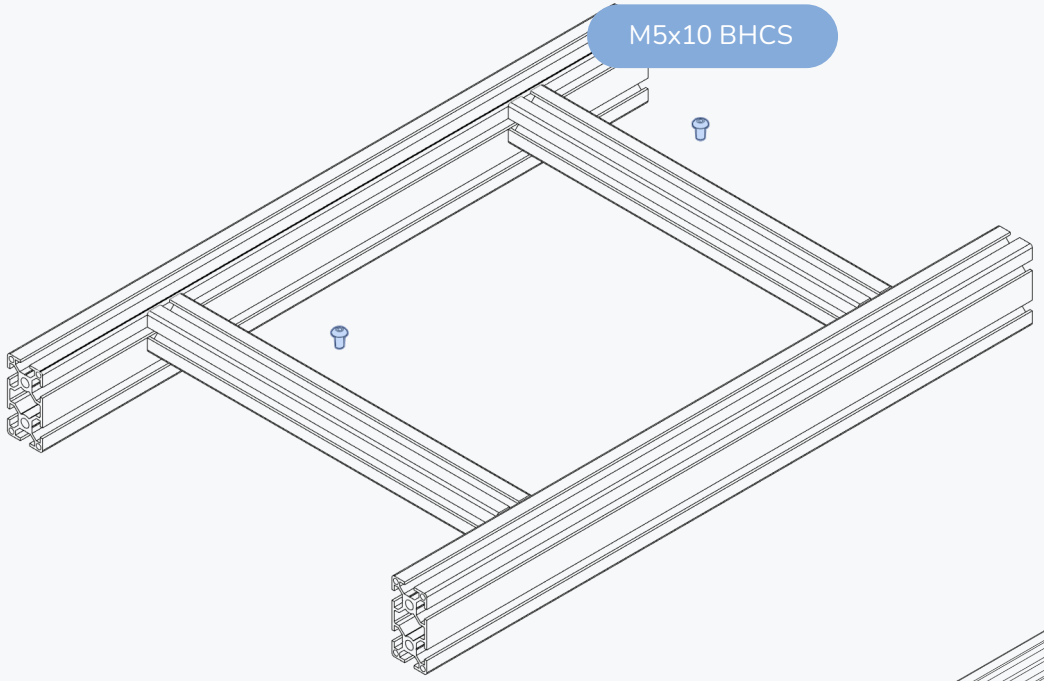
BUILD ON A FLAT SURFACE

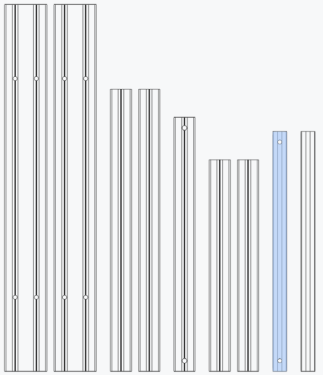
Build the frame on a glass or granite surface to ensure you can get it as square as possible.

BLIND JOINT ACCESS HOLES

Use the access holes to fasten the bolts.



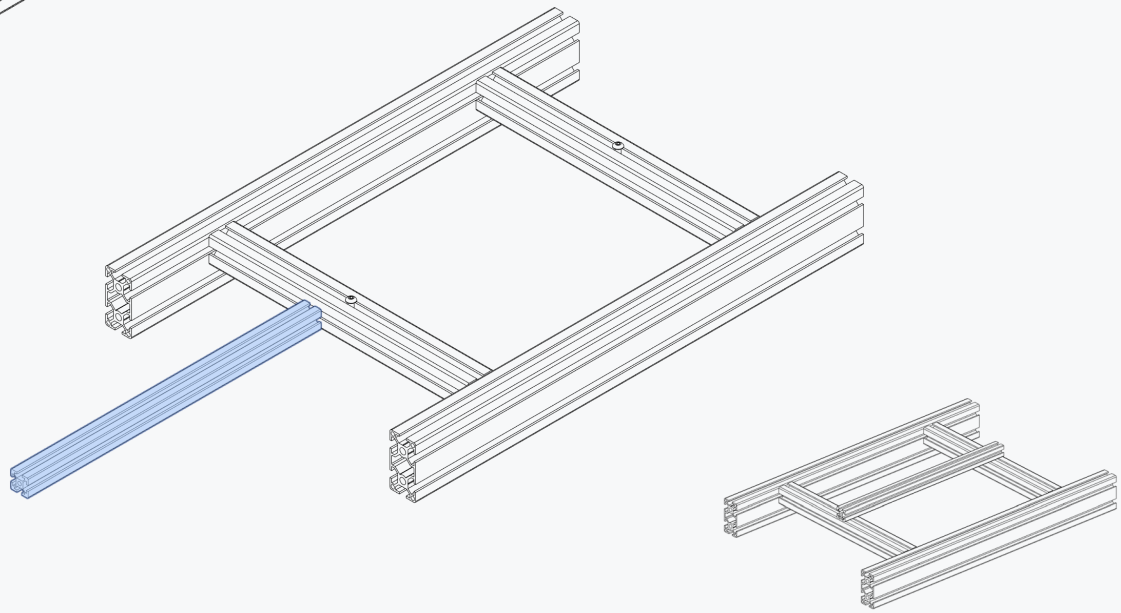


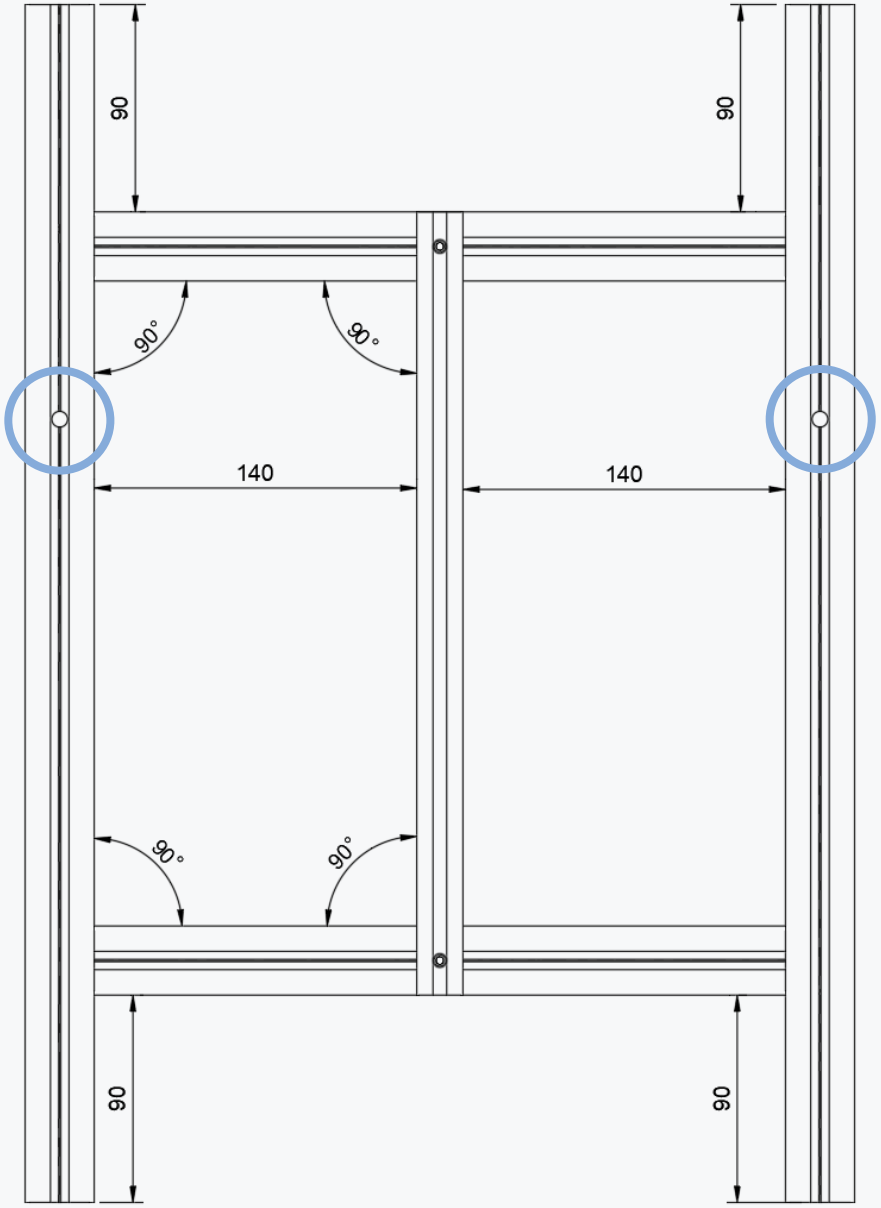


T-NUTS?

We don't show the insertion of T-Nuts in our assembly manuals.

The T-Nuts we recommend can be inserted at any point during the installation.



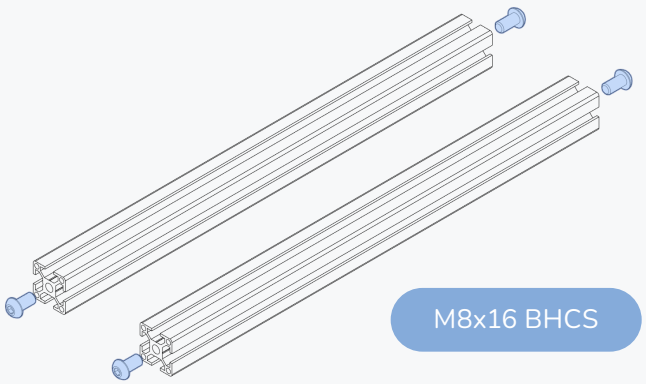


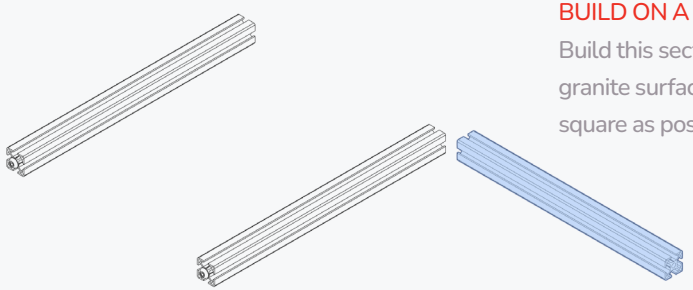
VERIFY DIMENSIONS

Make sure the frame is square and that all positions are set correctly before moving on. Check the 90° angles with a machinist square.

MIND ACCESS HOLE POSITION

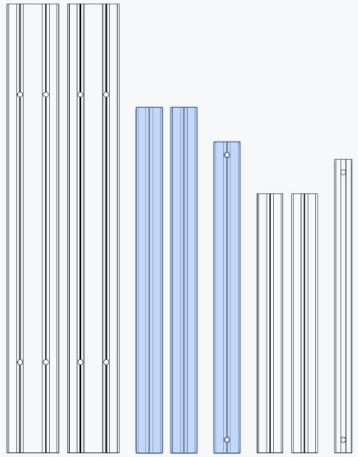
We do our best to call out things that may bite you later in the assembly process but may skip things that seem obvious to us. If in doubt please refer to the CAD model, it might save you some considerable time down the road.

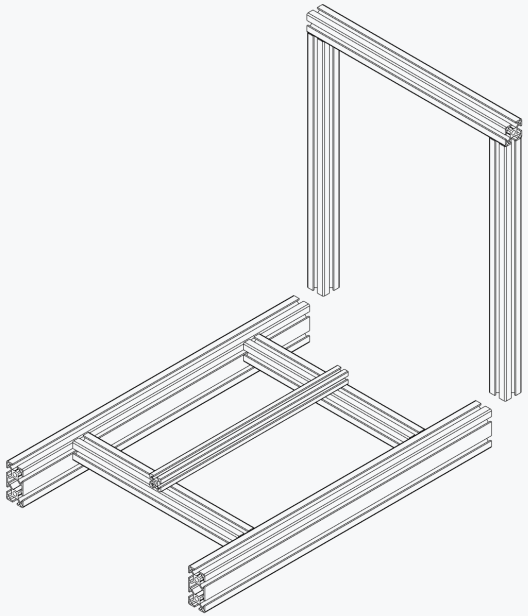




BUILD ON A FLAT SURFACE

Build this section of the frame on a glass or granite surface to ensure you can get it as square as possible.





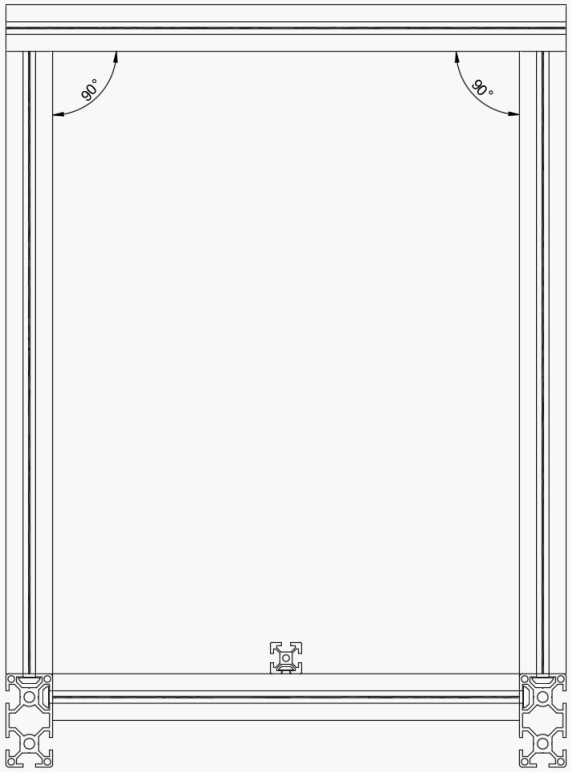
TILT FRAME 90°

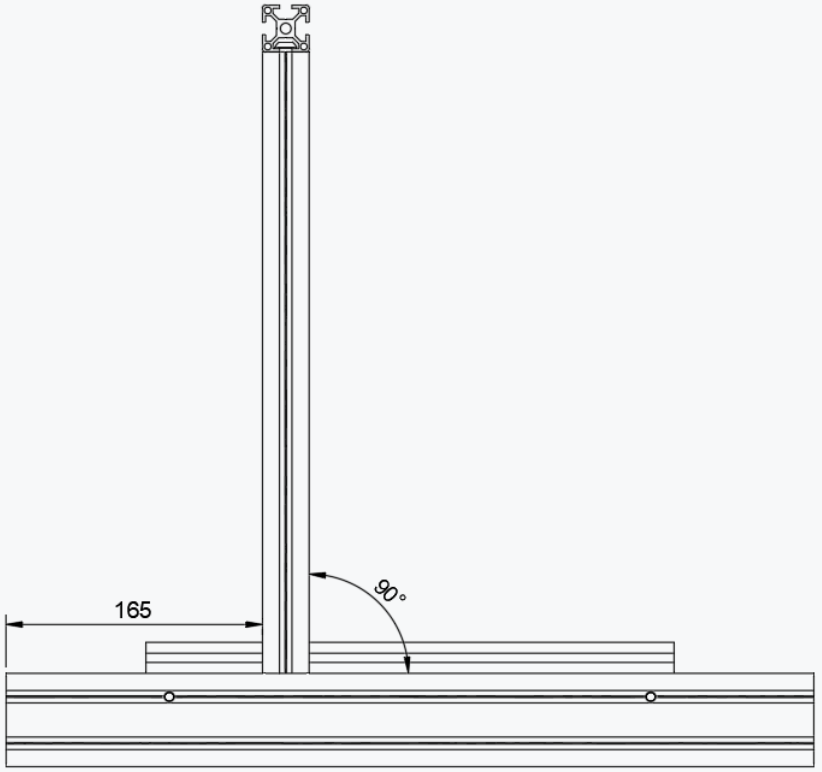
The access holes for the Blind Joints are at the bottom. Lay the frame flat on it's side to access them. This also helps in mounting the uprights flush to the base.

VERIFY DIMENSIONS

Check the corners and make sure that they are at 90° angles. Use a small machinist square.
If you have the measuring equipment verify that both uprights are dead on parallel.

Correct the angle is required. Failure in doing so may result in a sticky Z Axis.

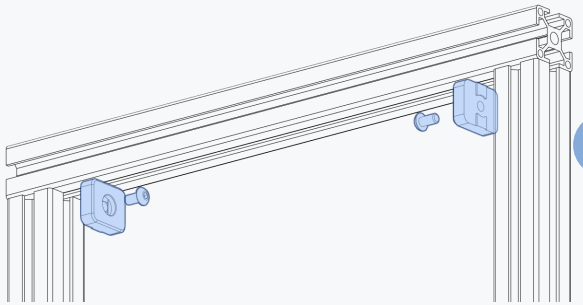




MIND THE CARRIAGE

The carriages are designed to slide along the rail easily. This unfortunately also includes sliding off the rails.

Dropping the carriage likely irreparably damages it.



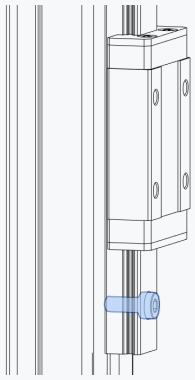
M5x10 BHCS

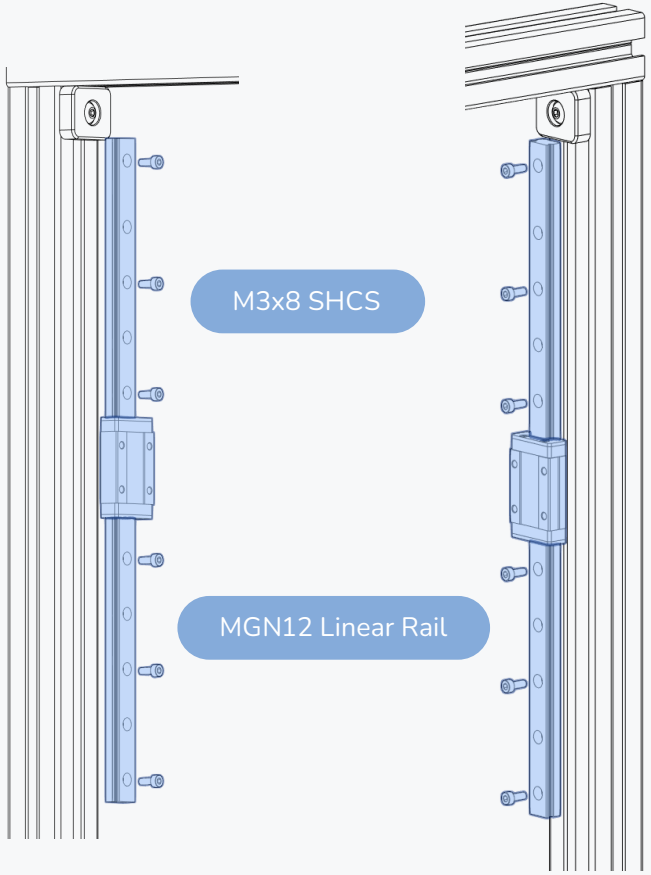
NO STOPPER ON LEFT RAIL

The left rail does not have a stopper in place to prevent the carriage from sliding off the rail. Some rails come with little plastic stop pins, you can leave the bottom one in place.

If your rail does not have these stop pins leave the last bolt slightly loose to act as a stopper.

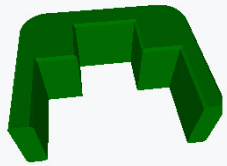
Tip: Mark the location with some tape to draw your attention to it later.





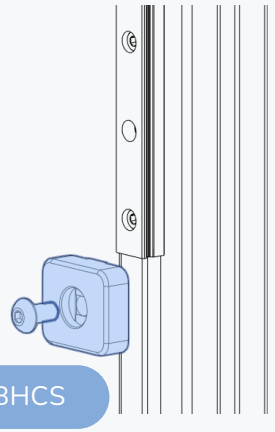
M3x8 SHCS

MGN12 Linear Rail



CENTERED RAIL INSTALLATION GUIDE

Use the guides to position the rail in the center of the extrusion prior to fastening the bolts.



M5x10 BHCS

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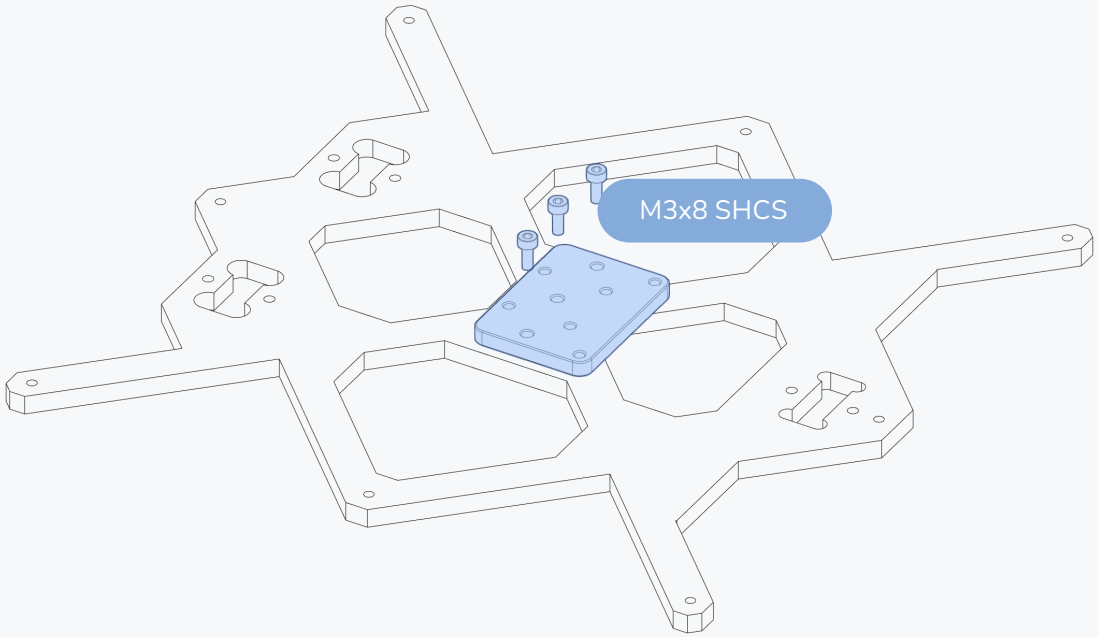


CARRIAGE MODIFICATION

If you sourced an unmodified carriage (e.g. a Prusa replacement part) you'll need to modify the hole pattern.

Temporarily mount the drill jig to the carriage.
Drill the 6 holes marked on the right with a 3mm or 3.2mm drill bit.

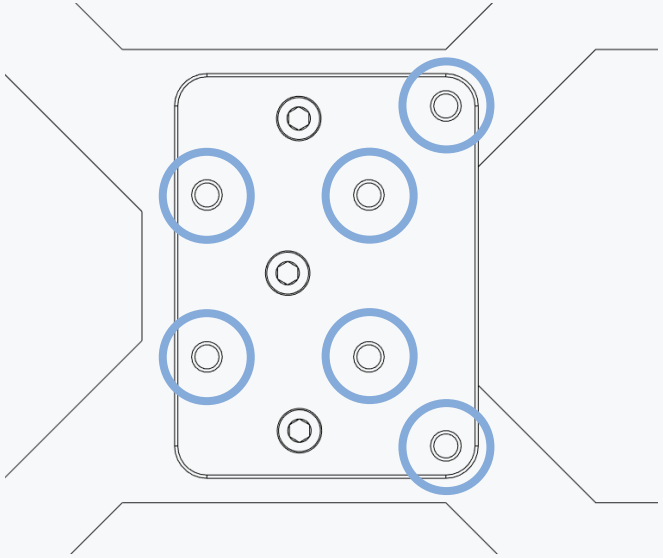
Remove the jig after you have drilled the holes.

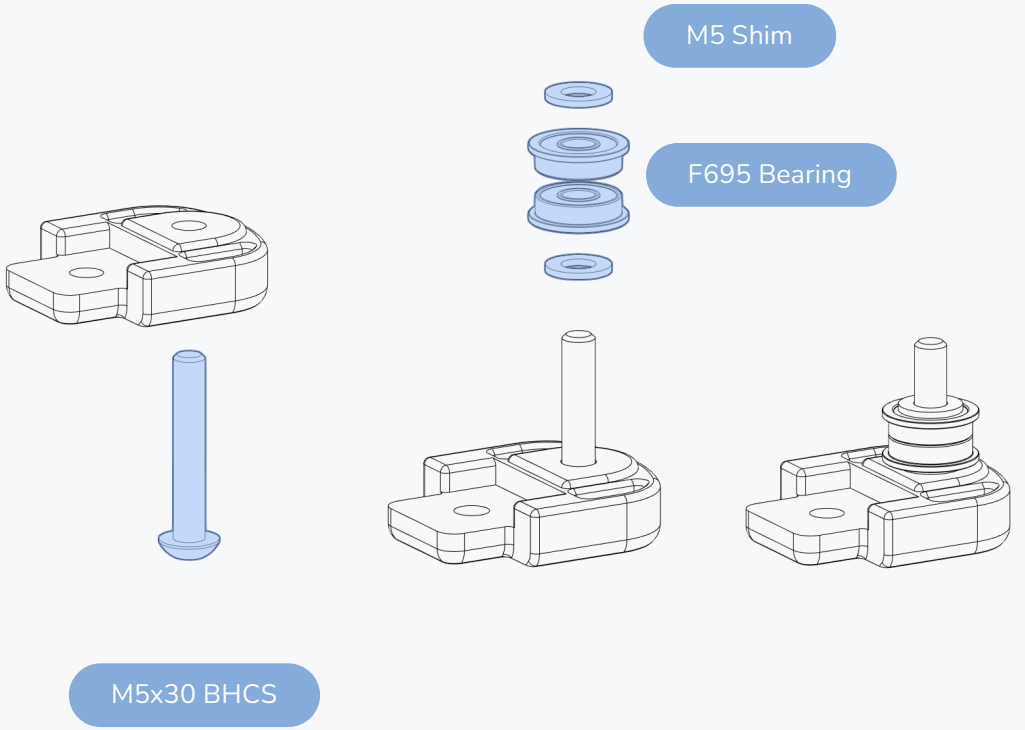
FINISHED CARRIAGE

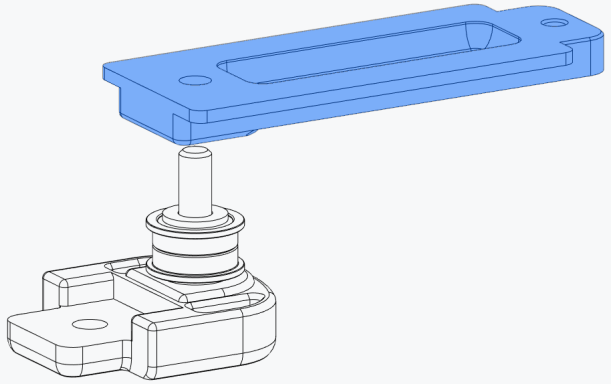
LDO MOTORS

Some vendors offer carriages with the bolt pattern required for the Voron Switchwire.

If you sourced a premachined carriage you can skip this step.





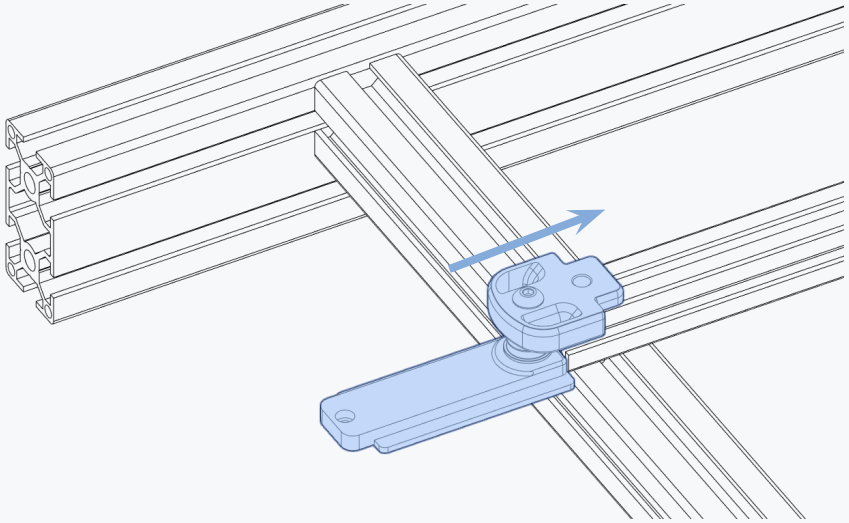


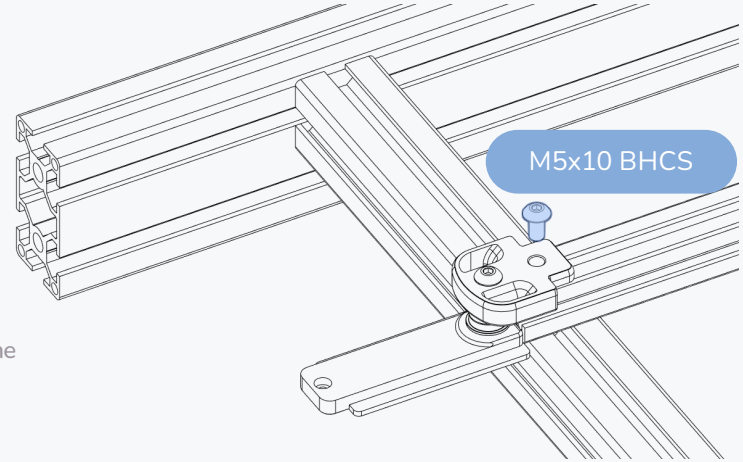
DON'T OVER TIGHTEN

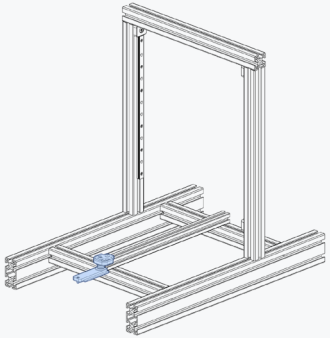
The M5 screws are threaded directly into plastic.
The plastic part has threads modeled into it.

UPSIDE DOWN ASSEMBLY

Why fight gravity if it can work for us.



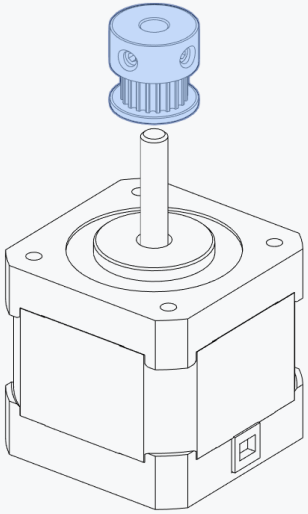


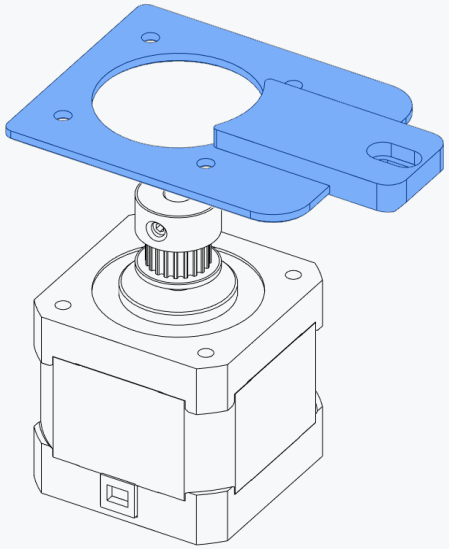


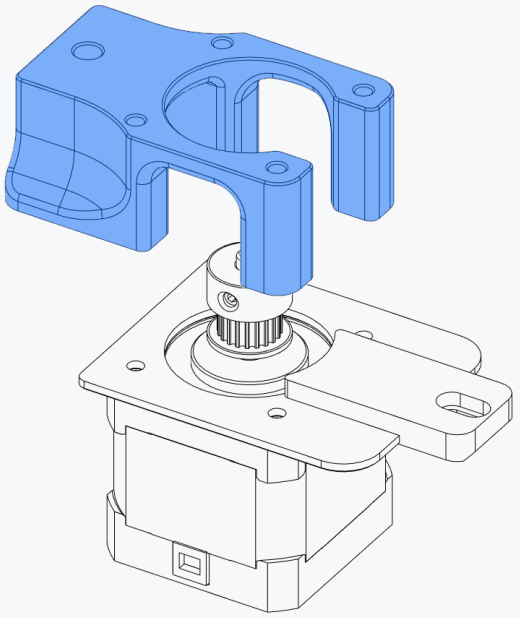
FULLY SEAT

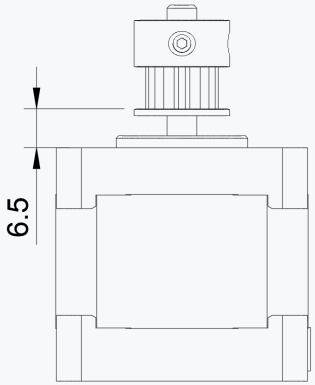
Make sure the part is installed flush against the extrusion.

GT2 20T Pulley









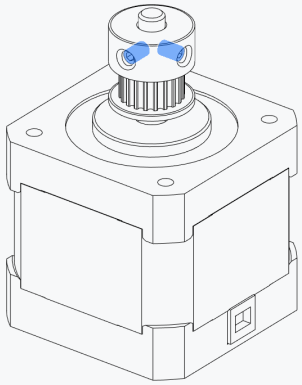
GRUB SCREWS

aka the root of all issues

Use thread locker on all grub screws after you set their position.

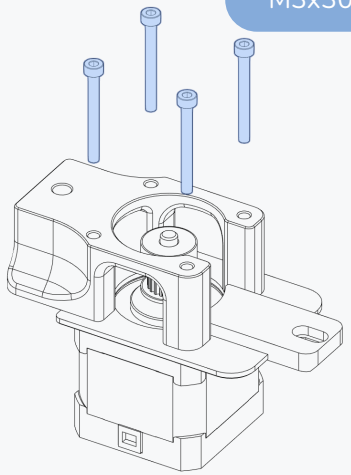
Loose grub screws account for the majority of issues that our users report. Save yourself hours of troubleshooting and apply thread locker to all grub screws during the build.

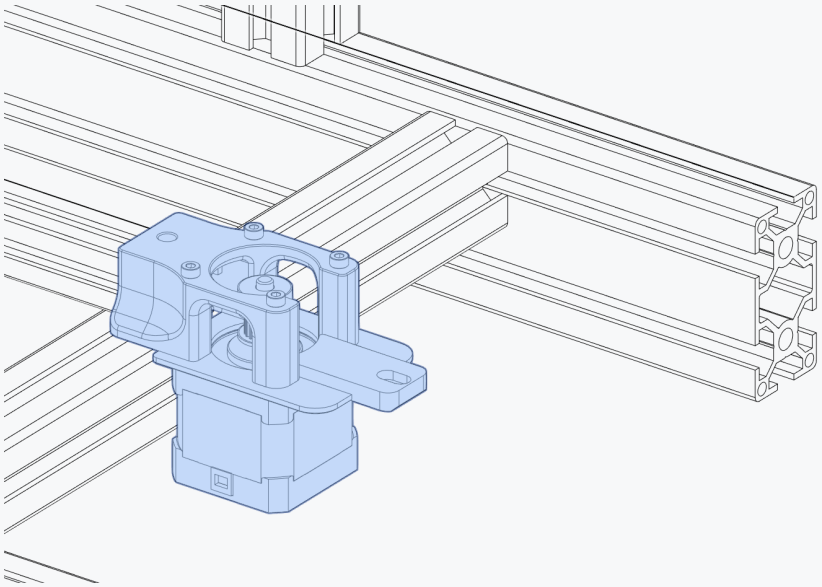
See the products application notes for instructions.

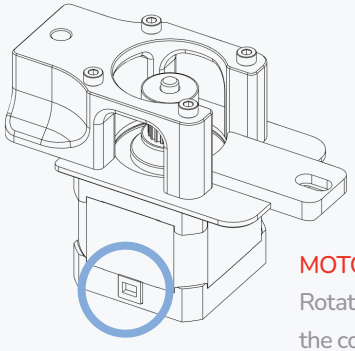


Y MOTOR

M3x30 SHCS



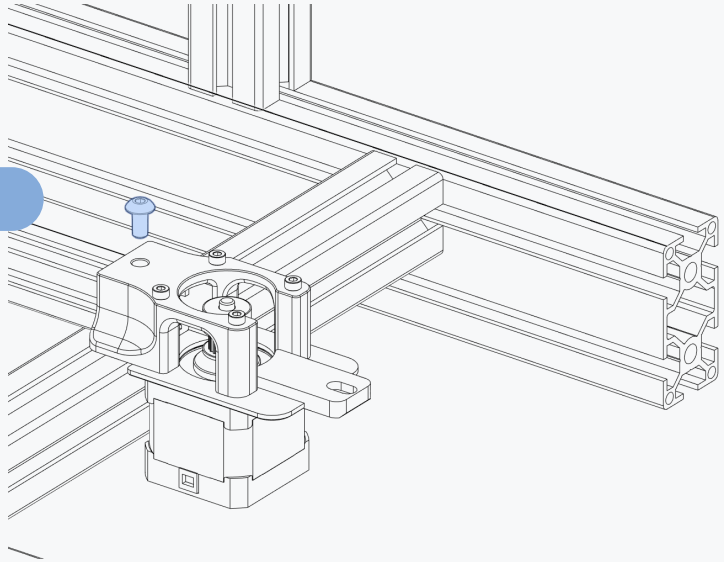


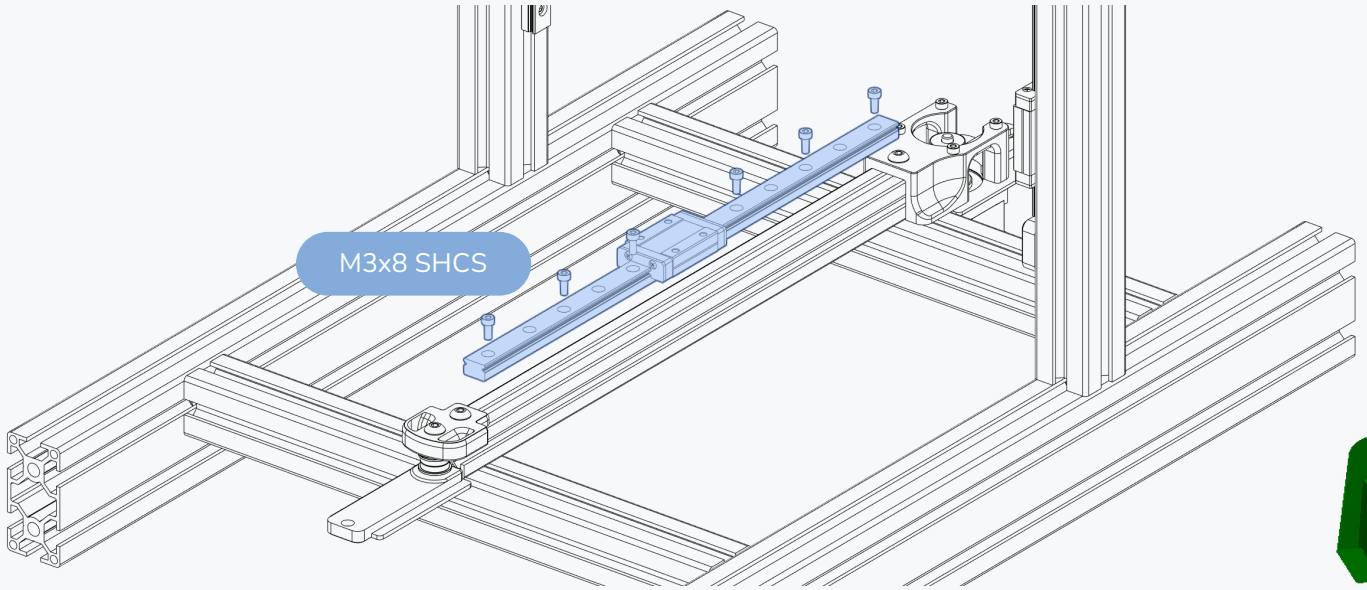


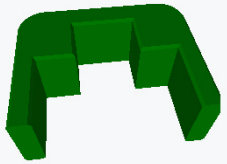
MOTOR ORIENTATION

Rotate the motor in such a way that the connector/wires are on the left side when looking at it from the back.

M5x10 BHCS

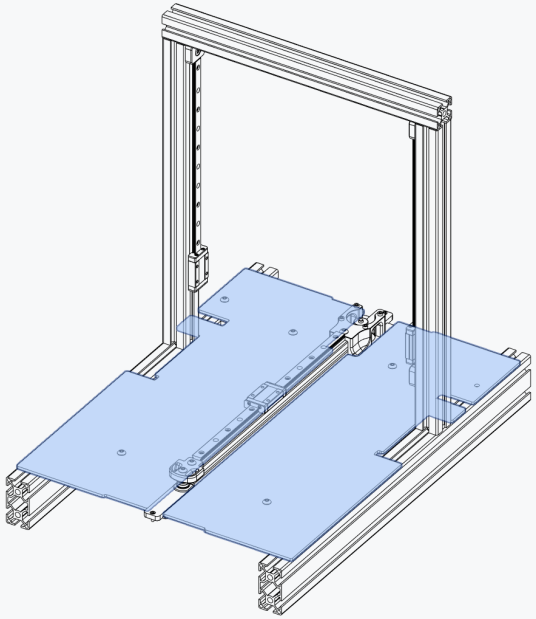






CENTERED RAIL INSTALLATION GUIDE

Use the guides to position the rail in the center of the extrusion prior to fastening the bolts.



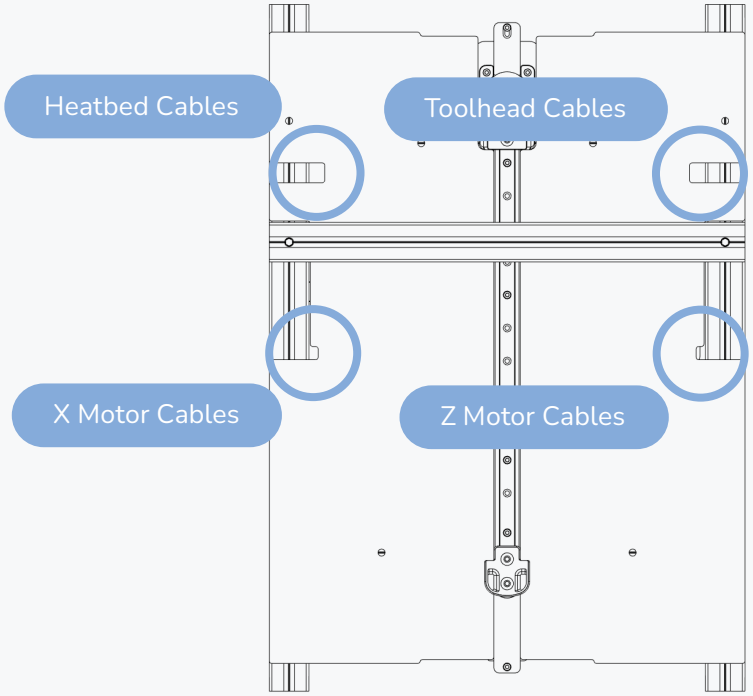
DECK PANELS?!

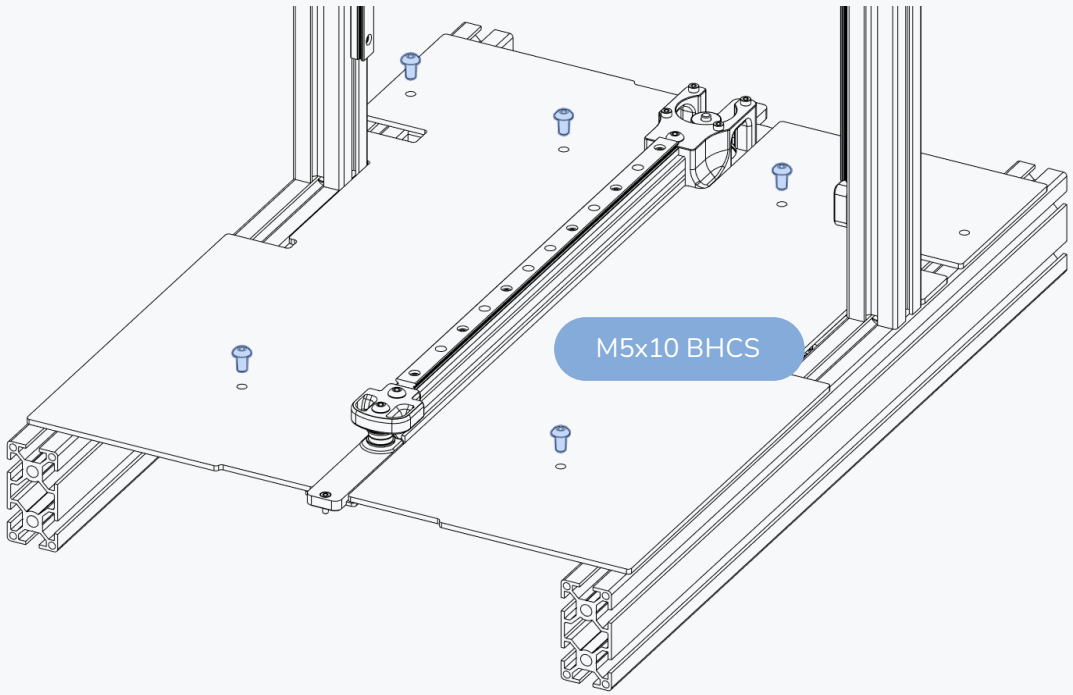
You can install the deck panels at this point if you like.
We didn't show them to make some captures easier.

PREMADE WIRE HARNESS

LDO MOTORS

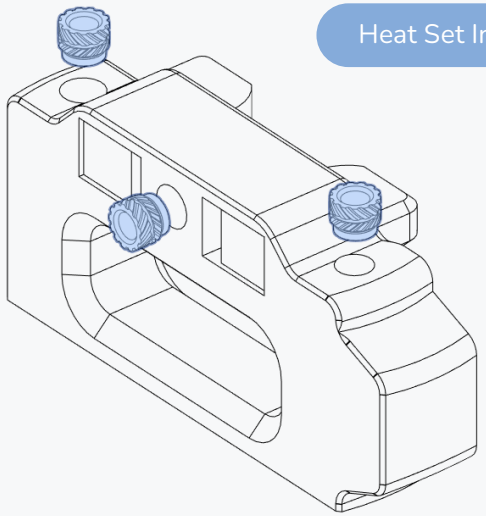
If you sourced a premade wire harness add the cables that pass through the deck panels now.




M5x10 BHCS

SENSORLESS HOMING FOR Y

Should you desire to use a microswitch as an endstop follow the instructions on the next page instead.



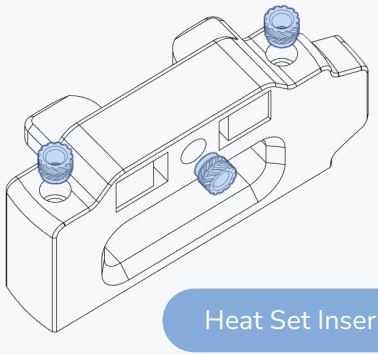
HEAT SET INSERTS

This design relies heavily on heat set inserts. Make sure you have the proper inserts (check the hardware reference for a close up picture).

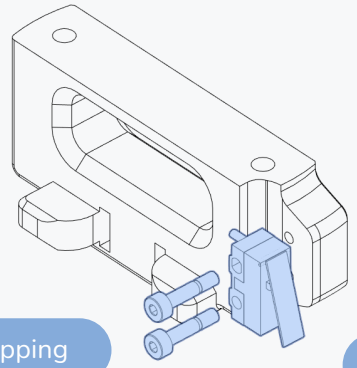
If you've never worked with heat set inserts before we recommend you watch the linked guide.

https://voron.link/m5ybt4d

Y BELT HOLDER - MICROSWITCH

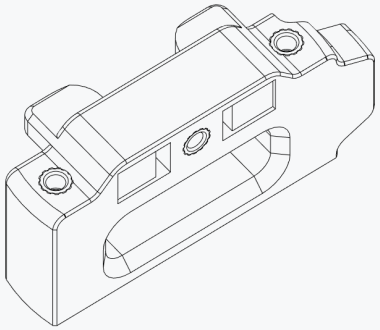


Heat Set Insert



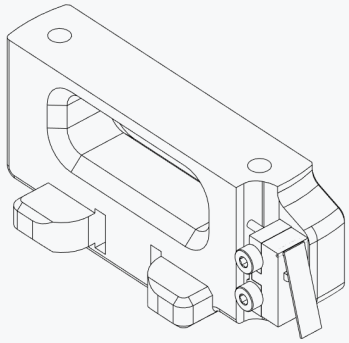
M2x10 Self Tapping

Microswitch

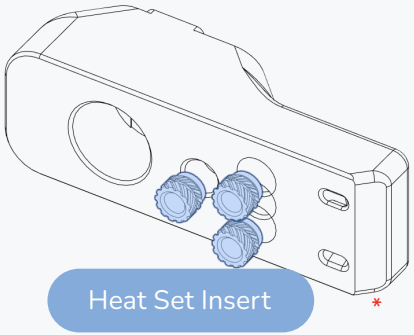


SOLDER WIRES

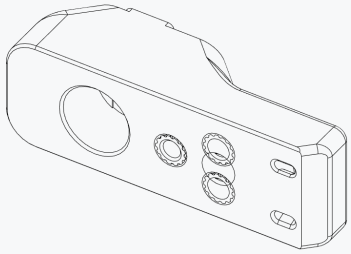
Solder approx. 100mm of wire to the mircoswitch before installing it.
See setup guide for details.

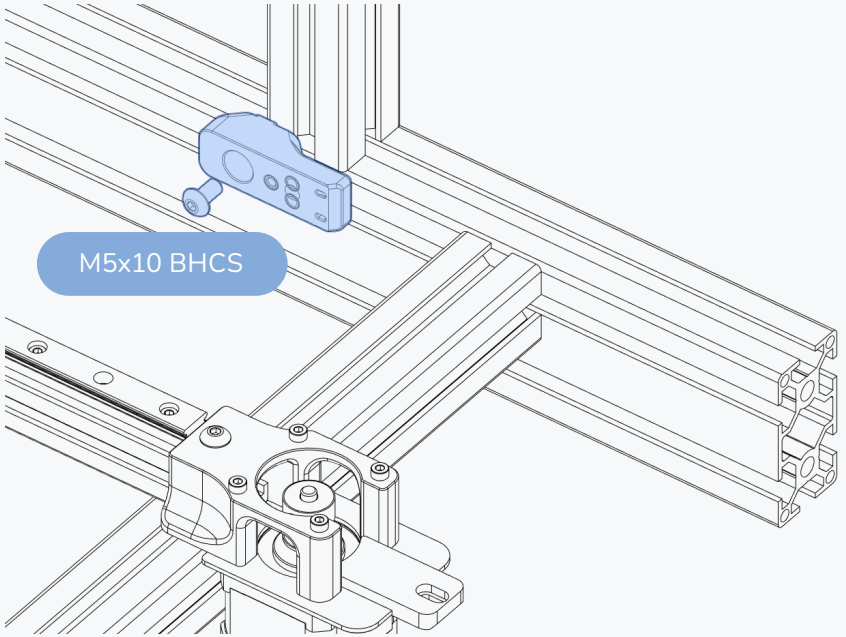


Y CABLE CHAIN MOUNT



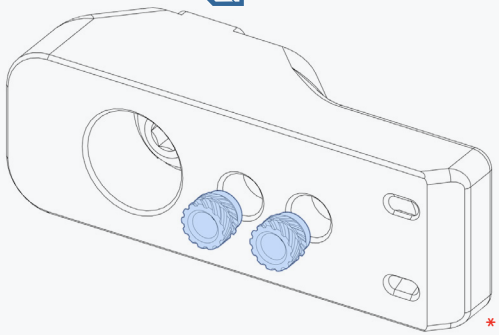
TRIANGLE PATTERN

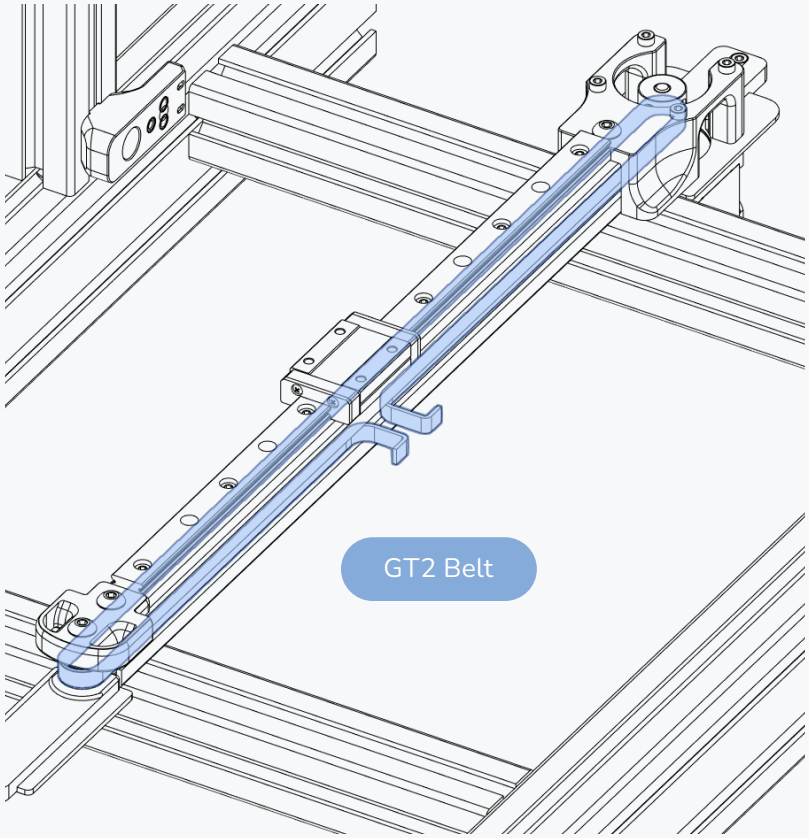


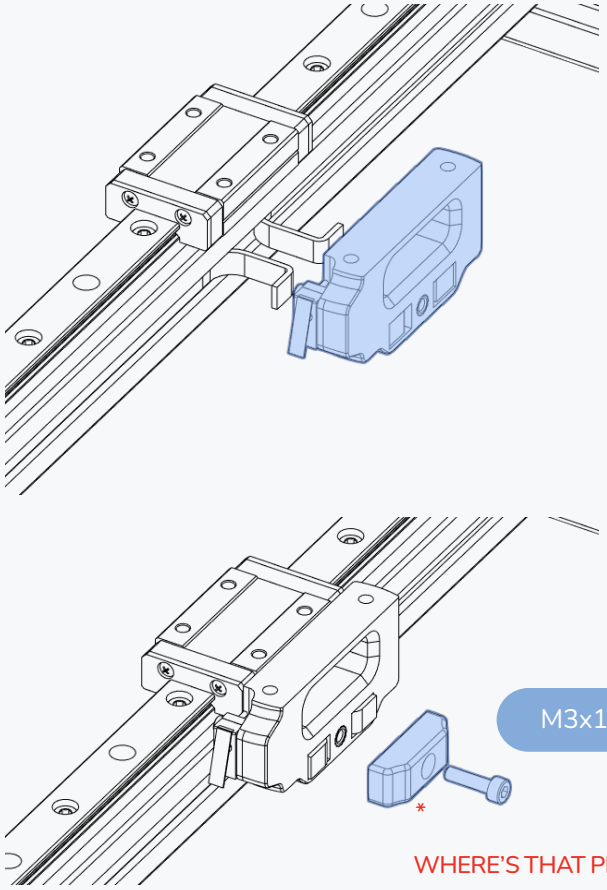


IGUS PATTERN

LDO MOTORS



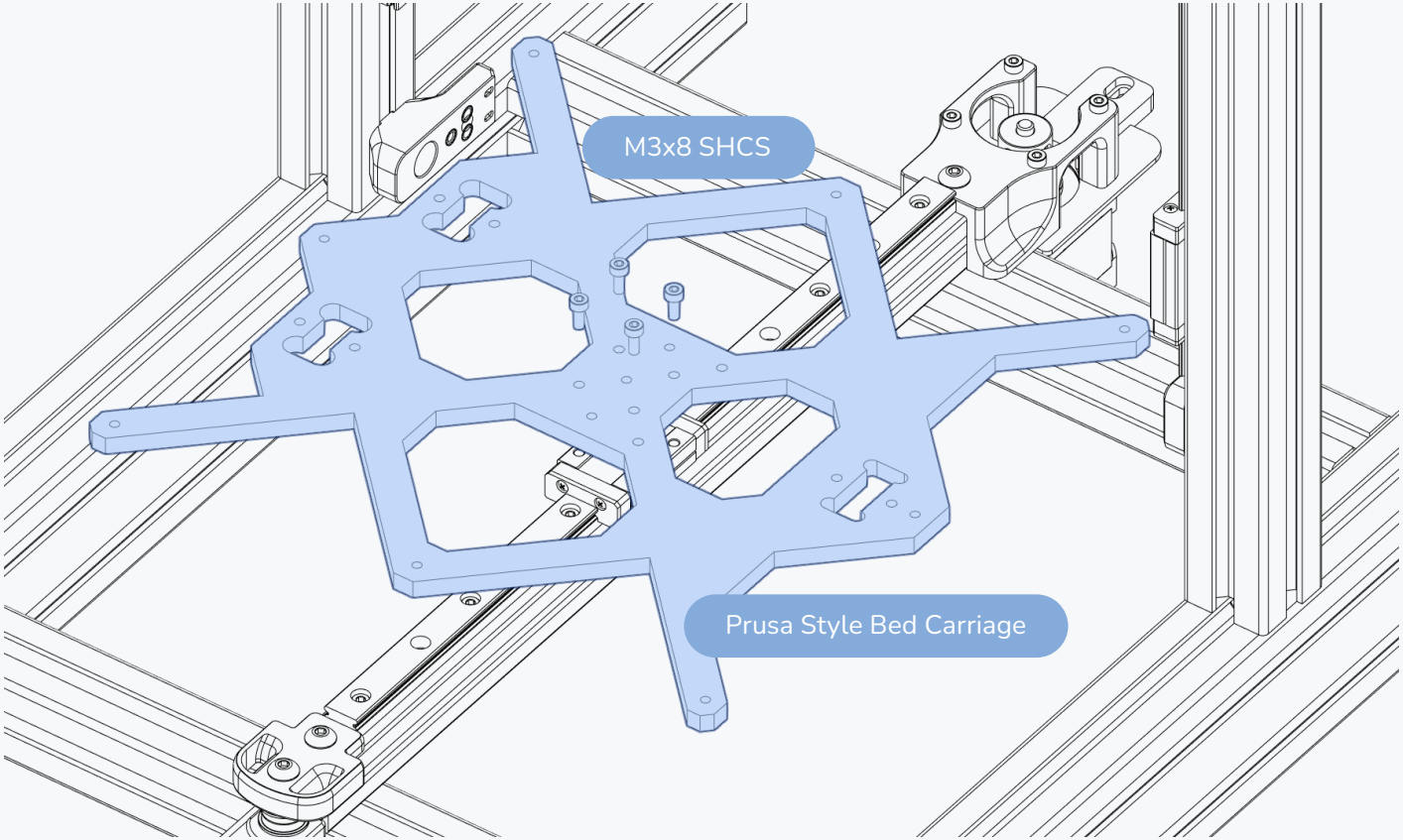


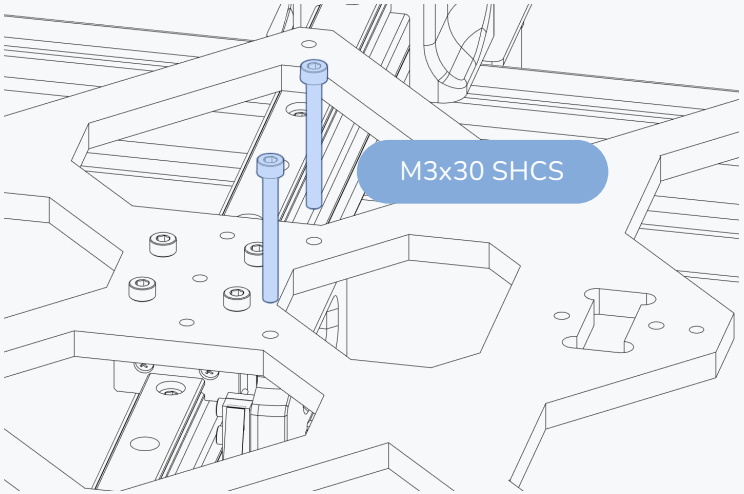


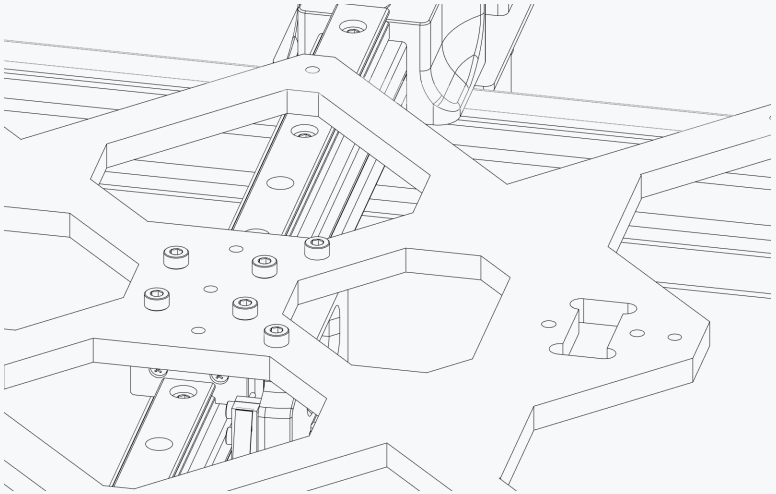
WHERE'S THAT PRINTED PART?

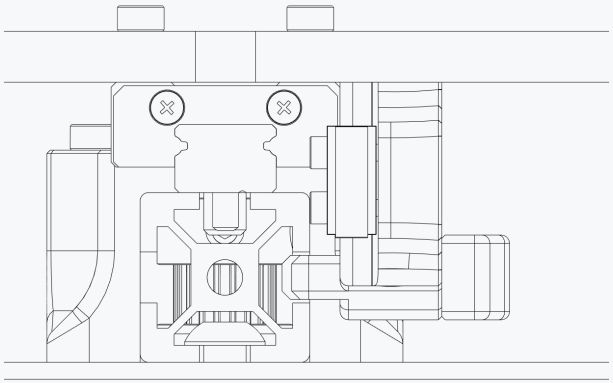
Look for asterix next to the part.

It indicates that this is an accent part.


M3x8 SHCS
Prusa Style Bed Carriage


M3x30 SHCS



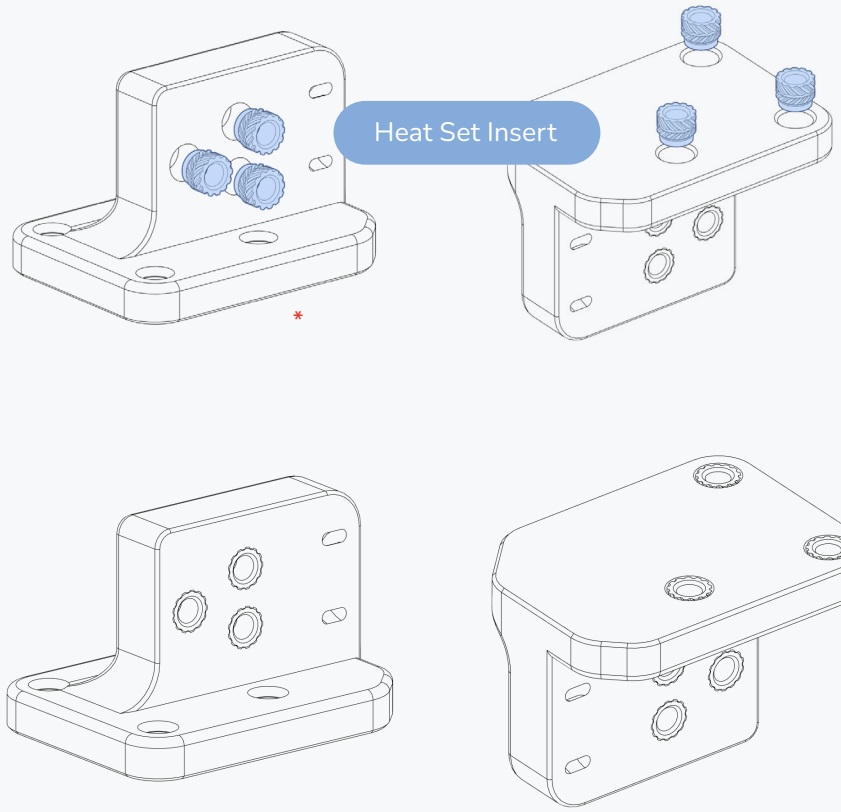


Y CABLE CHAIN MOUNT

CABLE CHAIN HOLE PATTERN

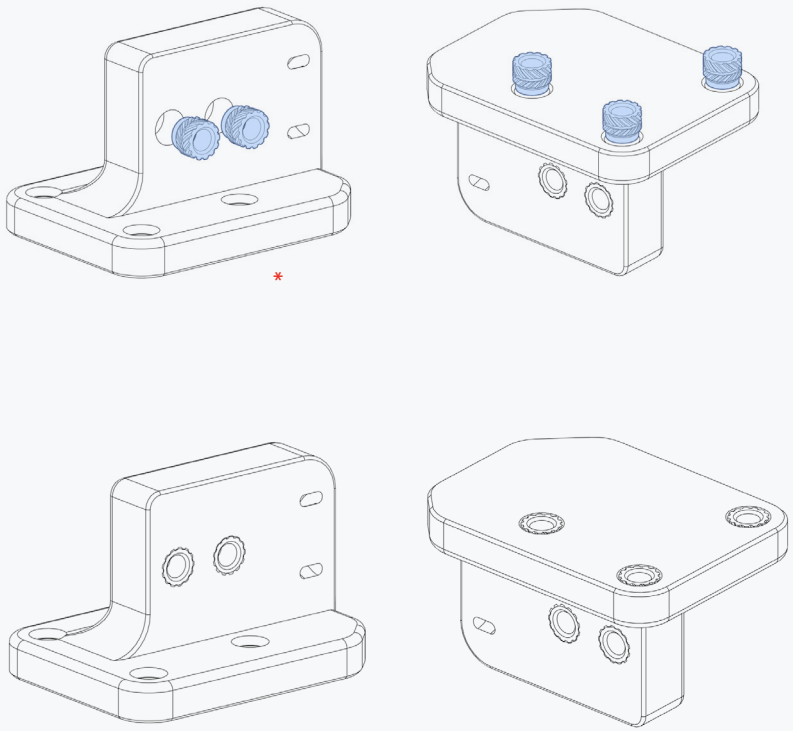
Cable chains ends have a couple of different hole patterns. We offer mounts for both the inline pattern used on IGUS chains, and the triangle pattern present on budget chains.

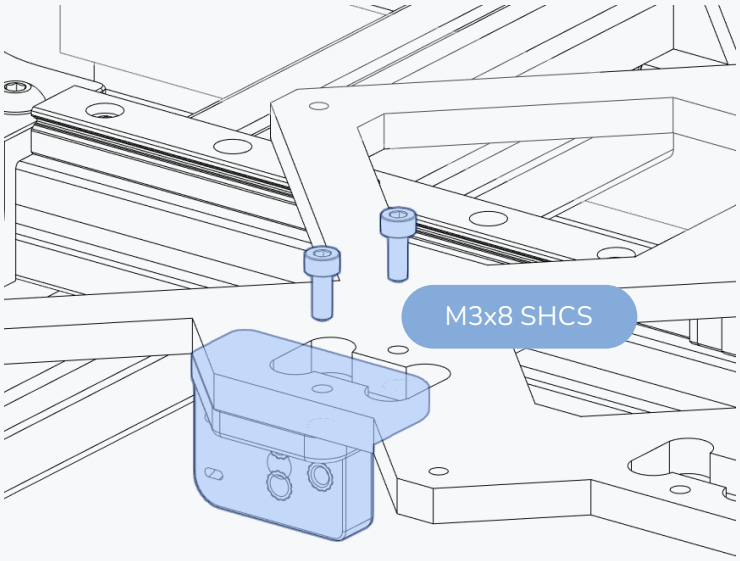
TRIANGLE PATTERN

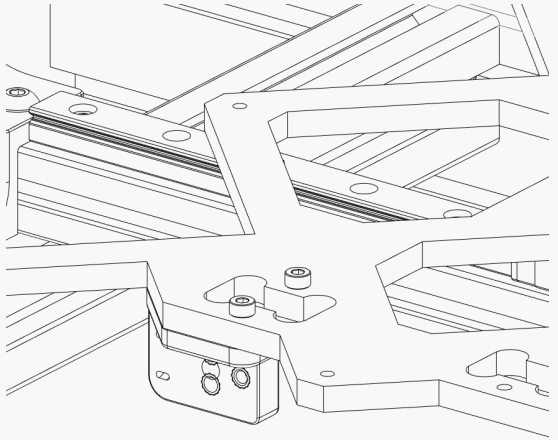


IGUS PATTERN

LDO MOTORS




M3x8 SHCS



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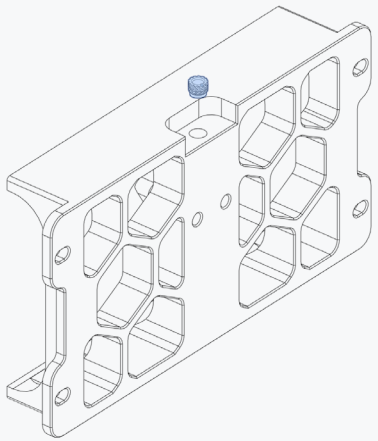
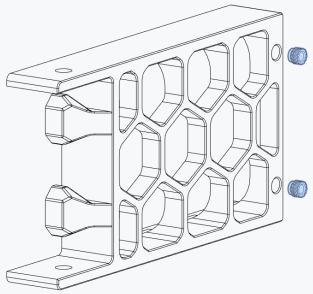
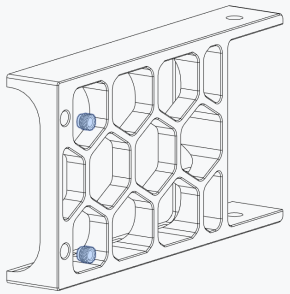
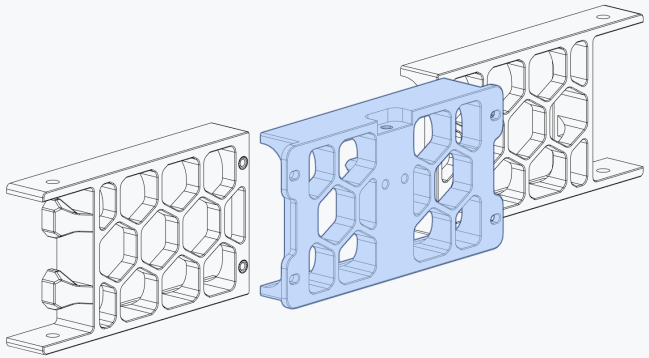


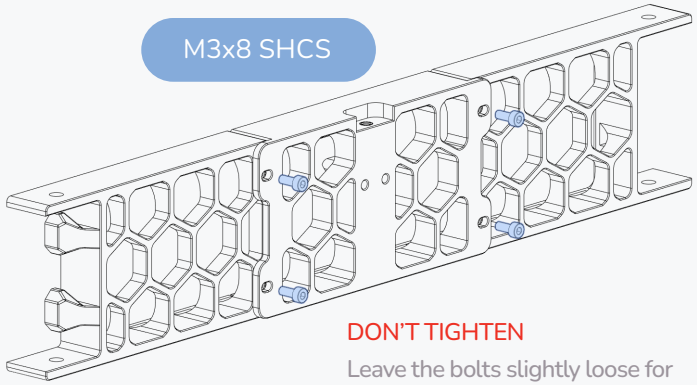
Heat Set Insert





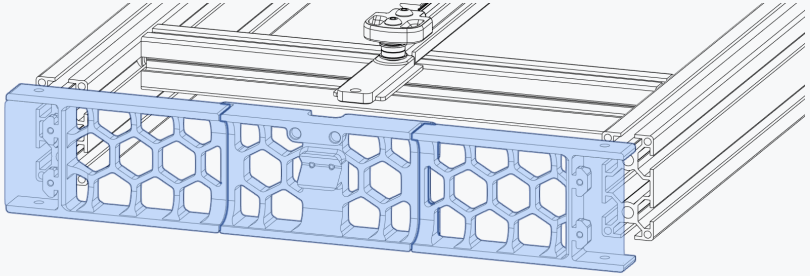


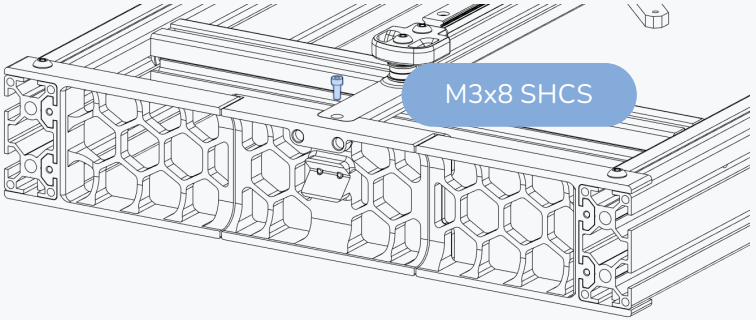
M3x8 SHCS

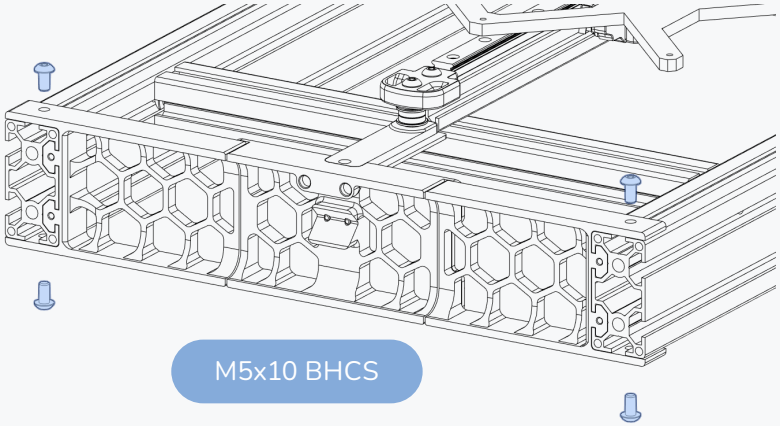


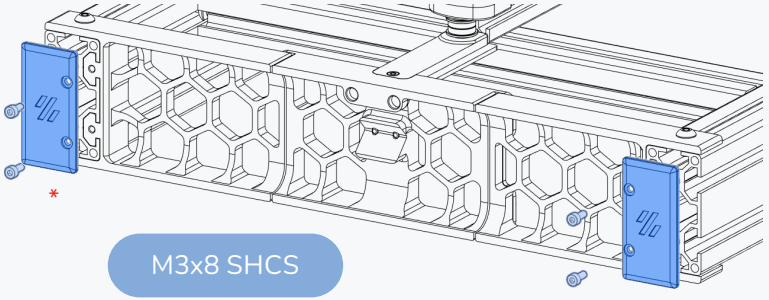
DON'T TIGHTEN

Leave the bolts slightly loose for the next step.



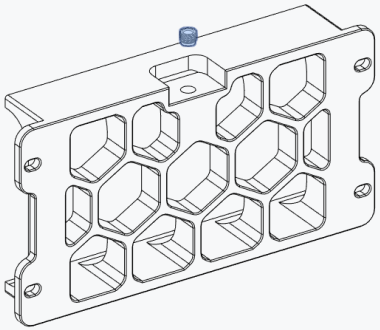




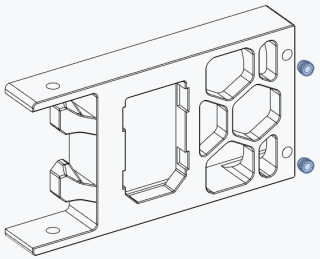


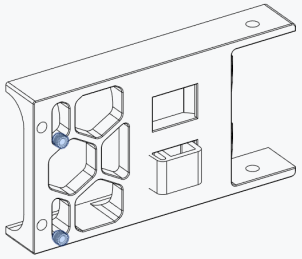
TIGHTEN BOLTS

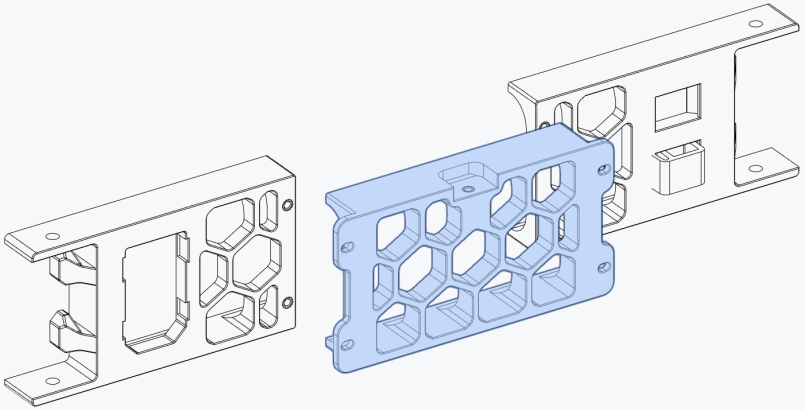
Tighten the bolts that connect the centre piece once installed on the frame.

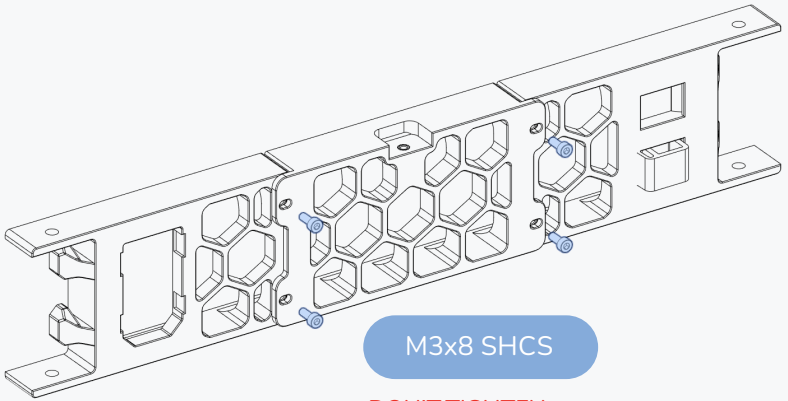


Heat Set Insert







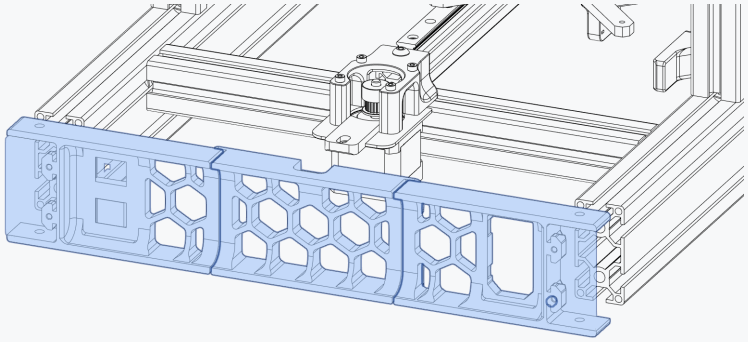


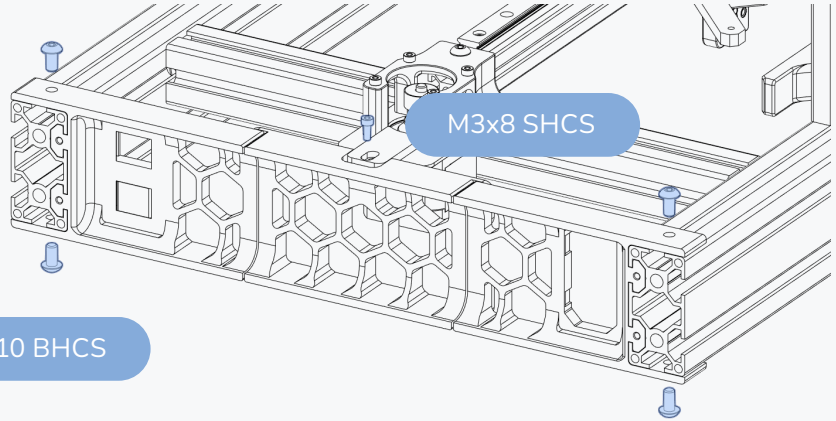
M3x8 SHCS

DON'T TIGHTEN

Leave the bolts slightly loose for the next step.

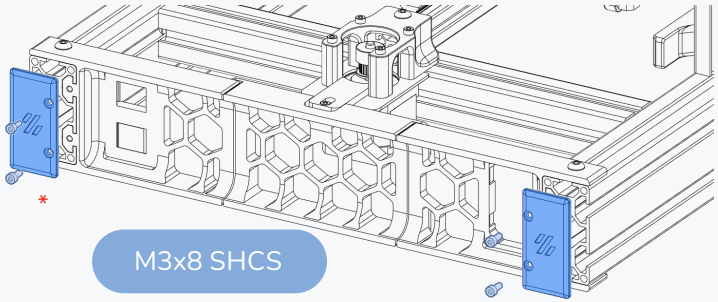
BACK SKIRT





TIGHTEN BOLTS

Tighten the bolts that connect the centre piece once installed on the frame.



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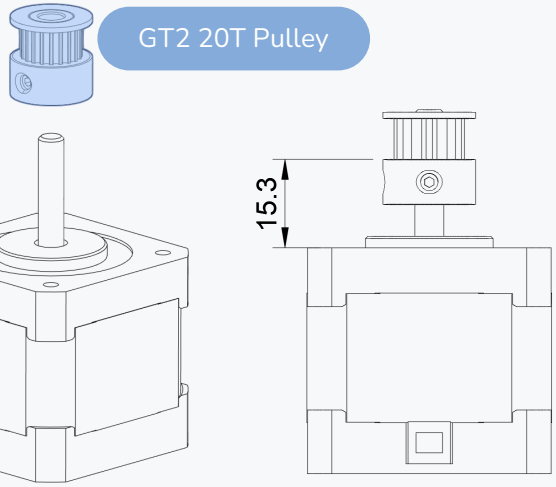
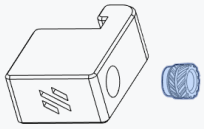
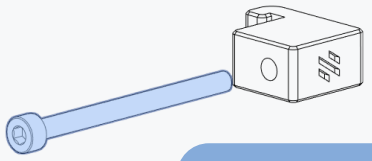
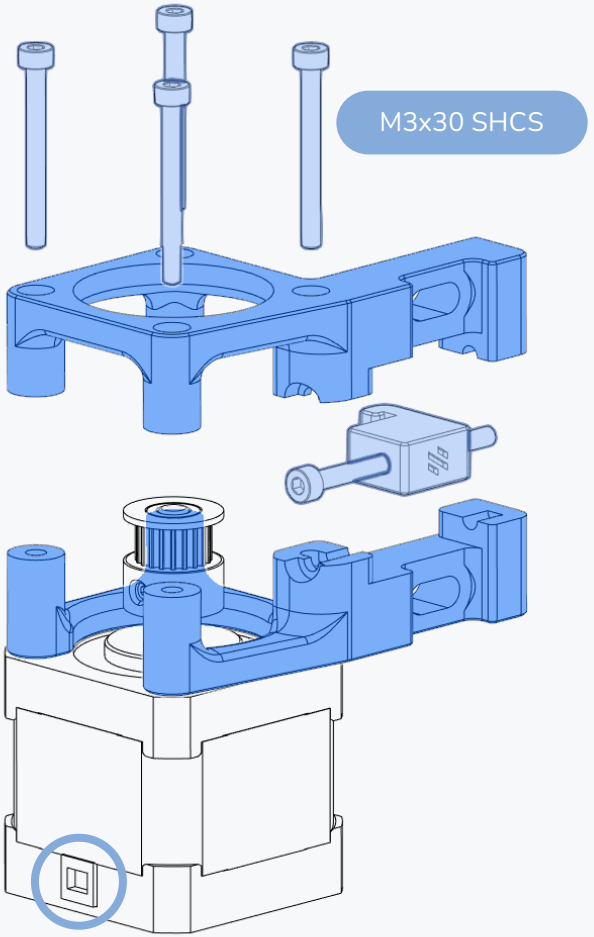


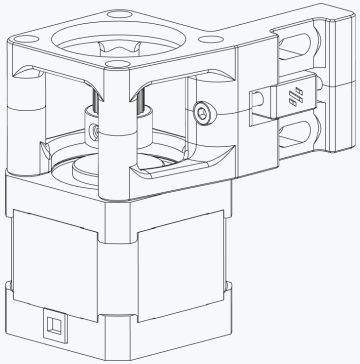


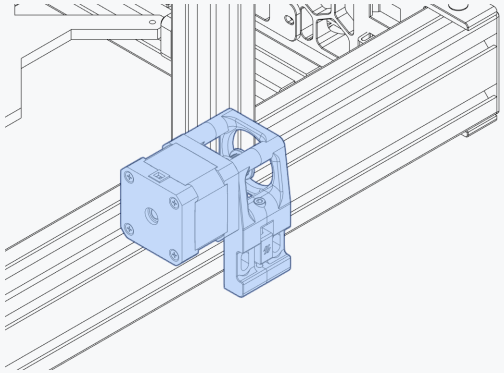


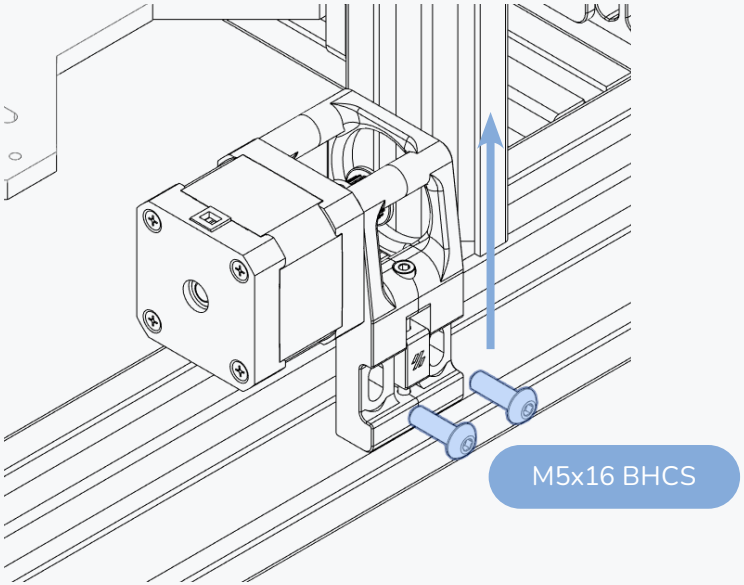


MOTOR ORIENTATION

Make sure the motor is oriented as shown.



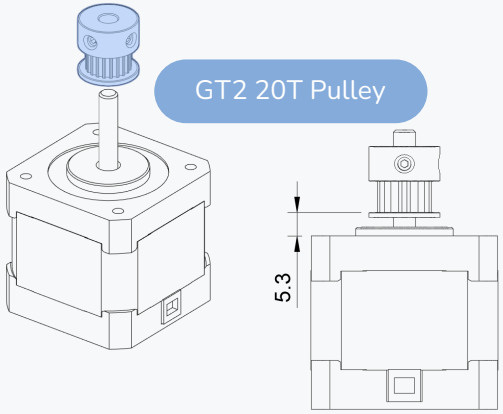


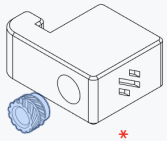


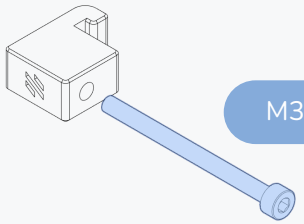
MOTOR MOUNT INSTALLATION

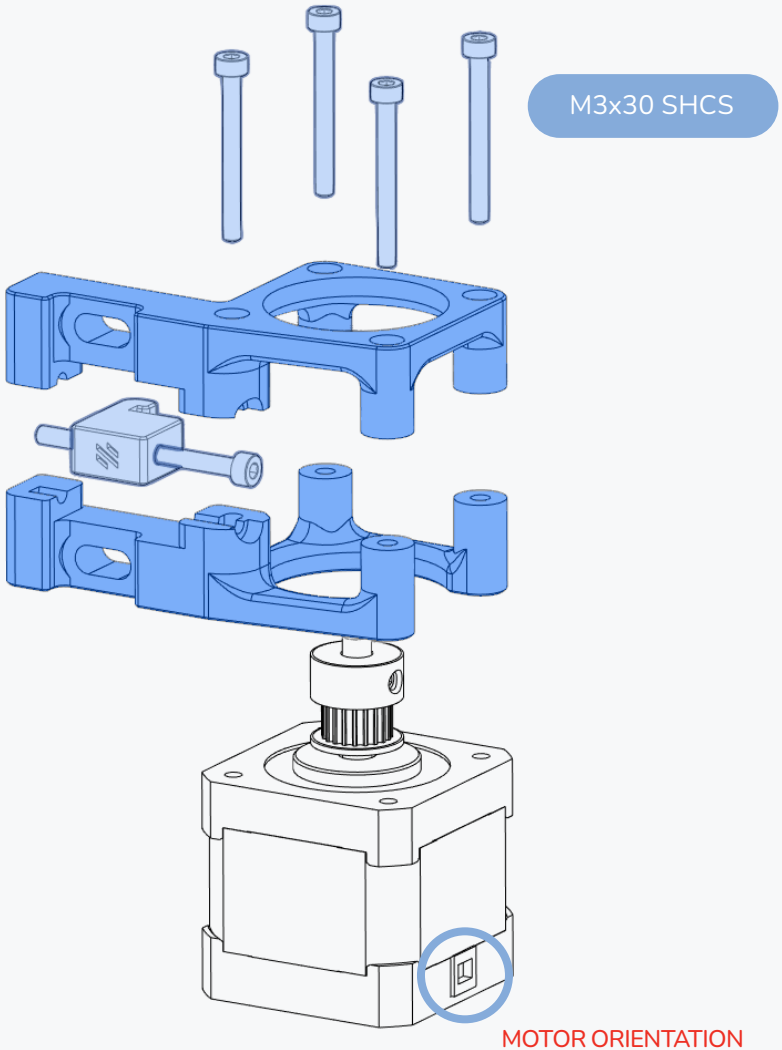
Push the motor mount as far up as the slots in the parts allow you to before fastening the bolts. You may need to turn the M3x40 counter clock wise to fully bottom out in the slot.

"X" MOTOR MOUNT



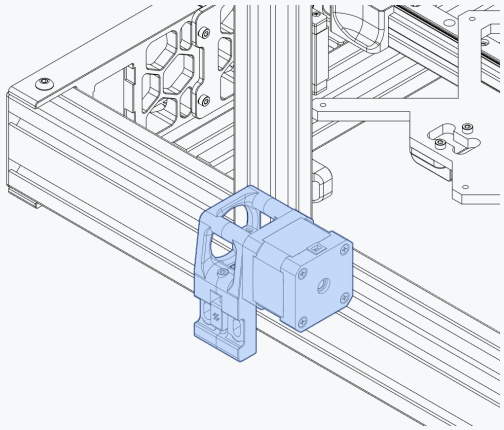


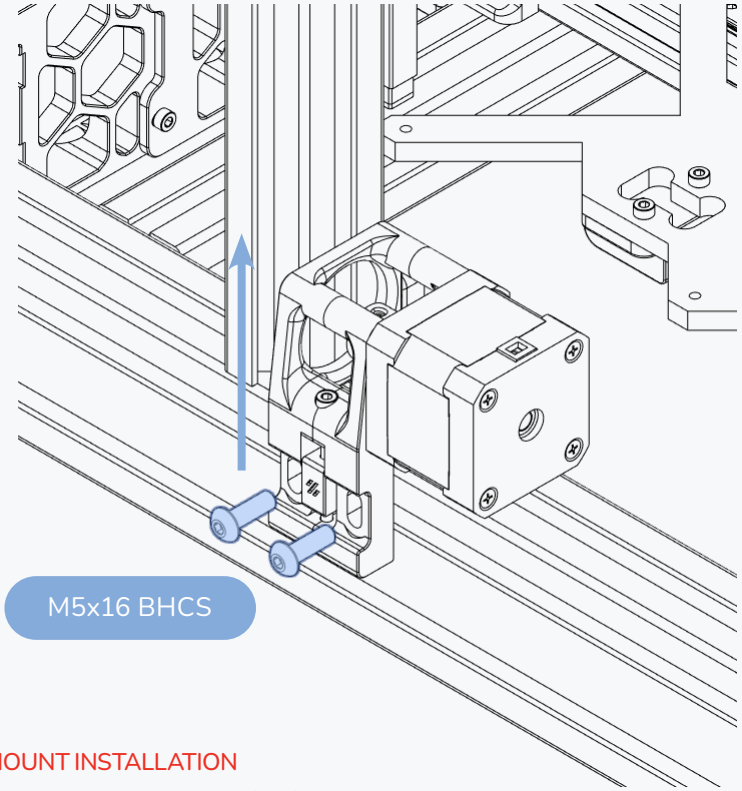




MOTOR ORIENTATION

Make sure the motor is oriented as shown.

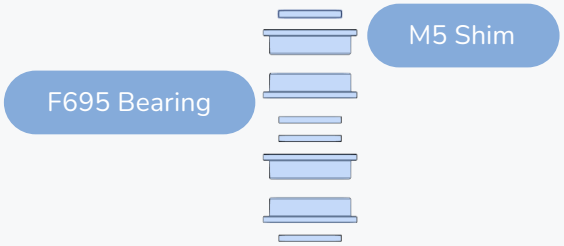


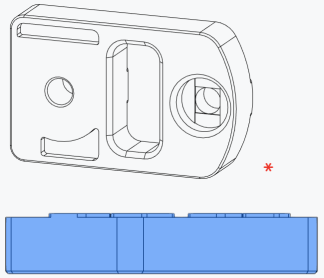


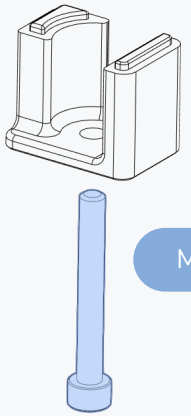
MOTOR MOUNT INSTALLATION

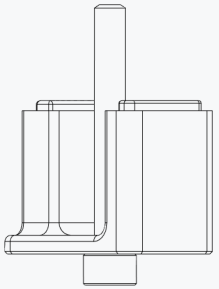
Push the motor mount as far up as the slots in the parts allow you to before fastening the bolts. You may need to turn the M3x40 counter clock wise to fully bottom out in the slot.

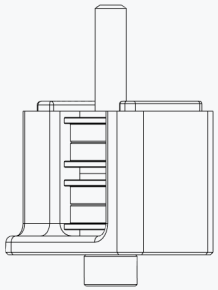
LEFT IDLER

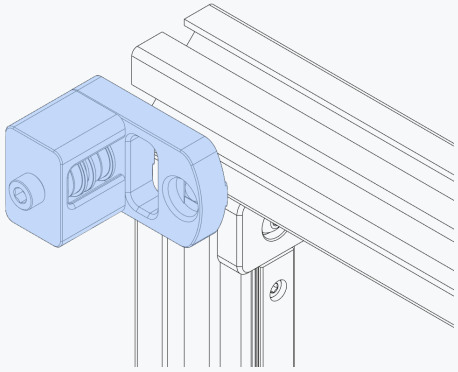


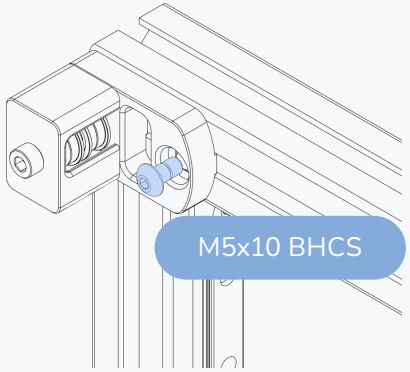


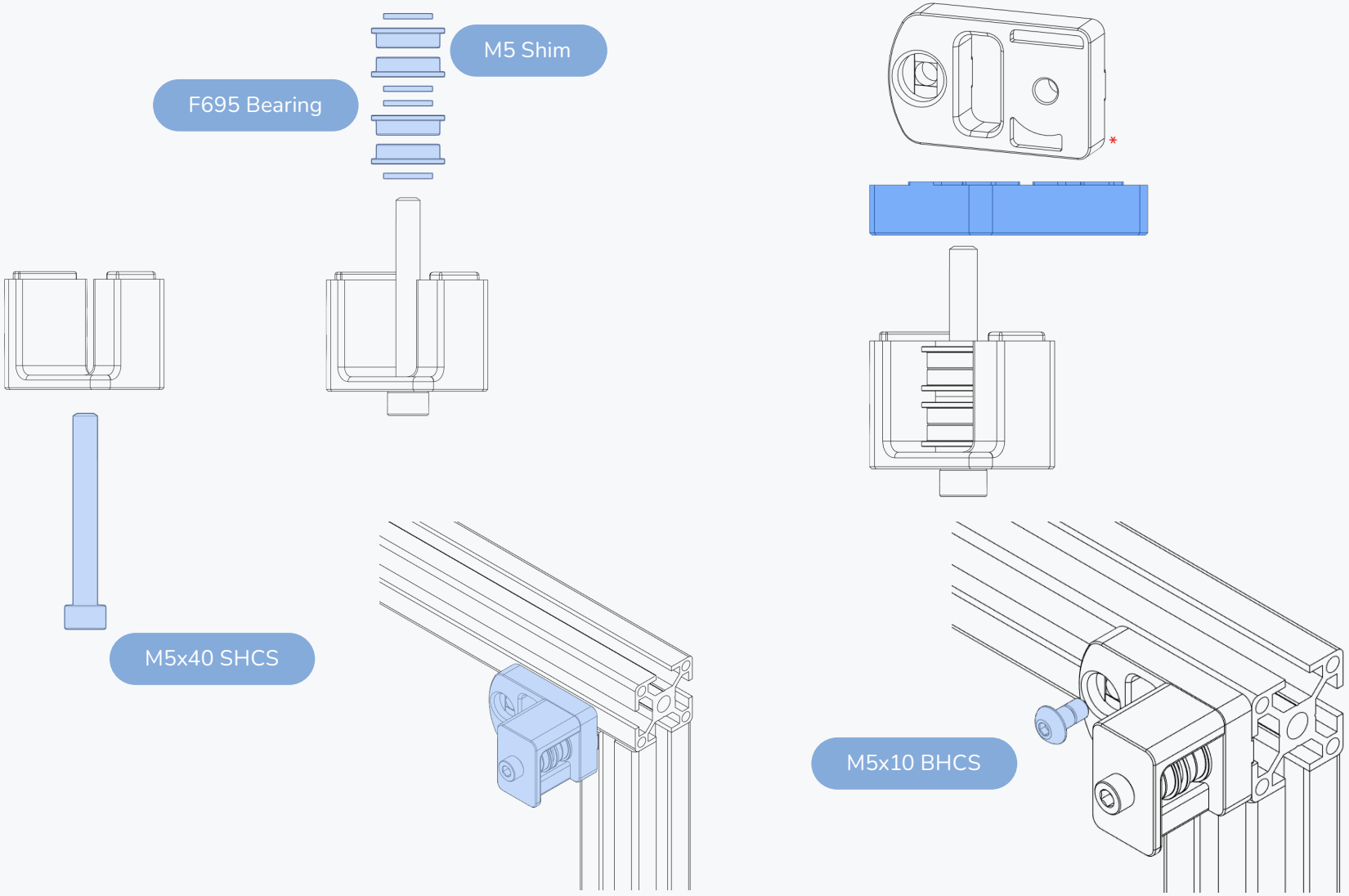






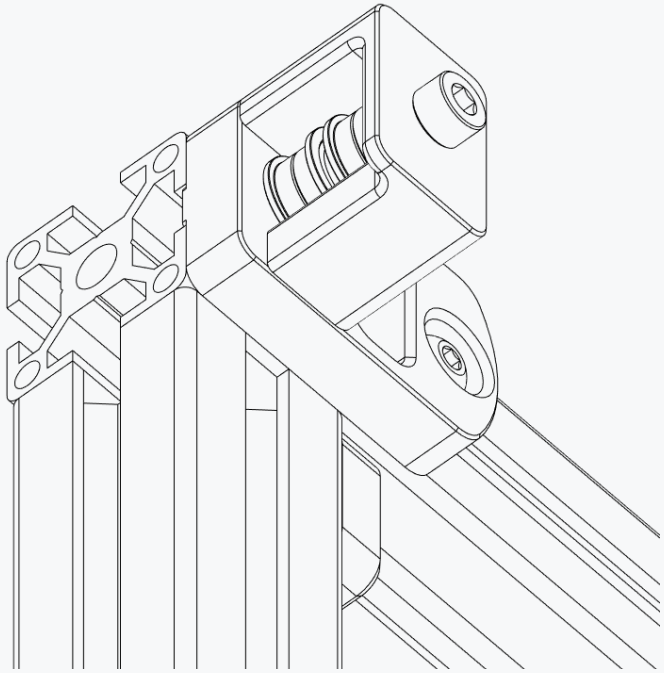


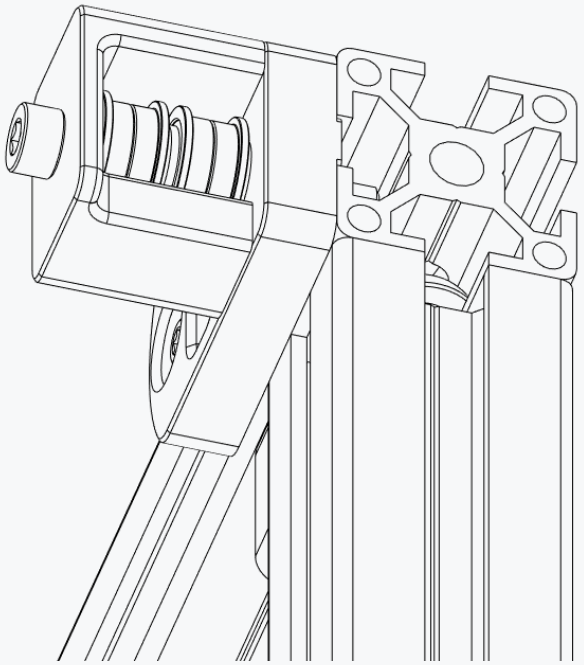



M5 Shim
F695 Bearing
M5x40 SHCS
*
M5x10 BHCS

CHECK ORIENTATION

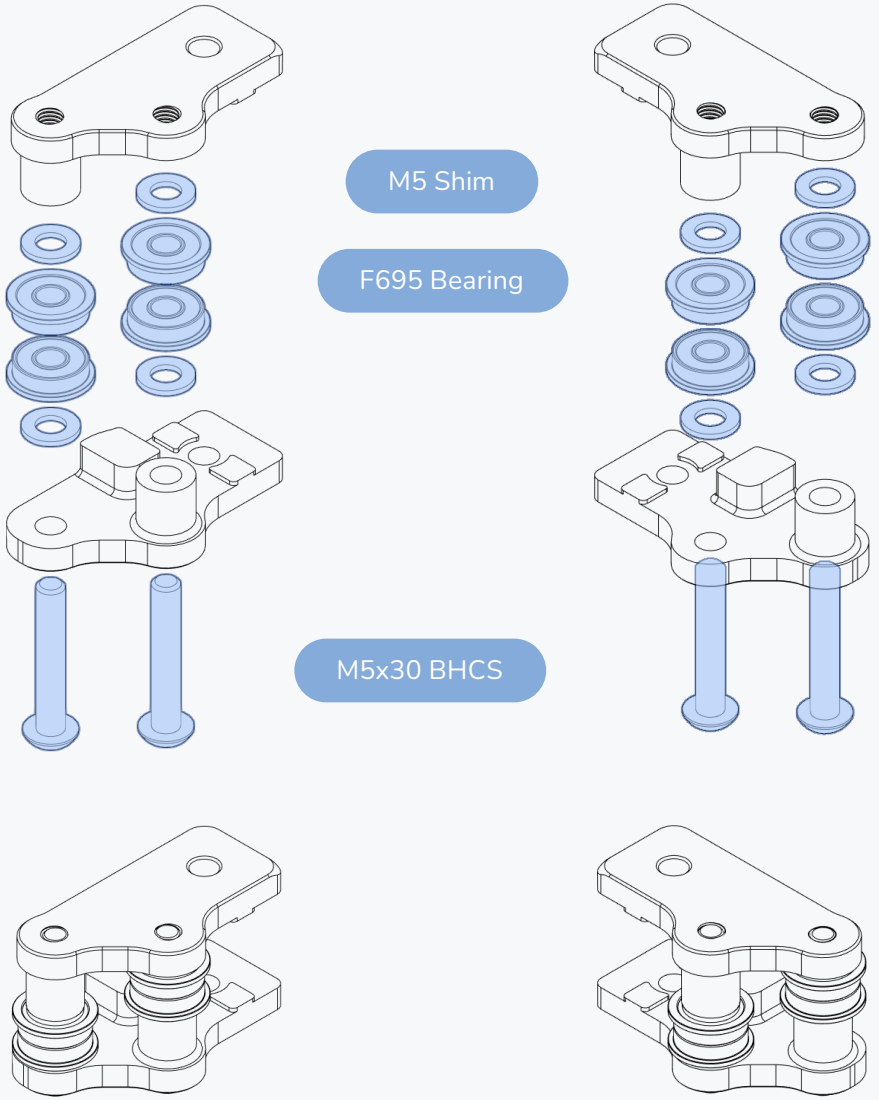
Make sure the cut-out is facing down.

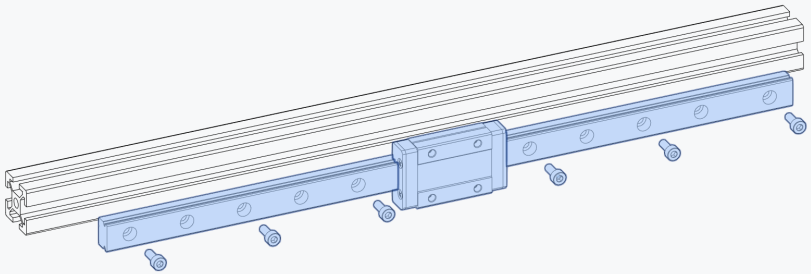




UPSIDE DOWN ASSEMBLY

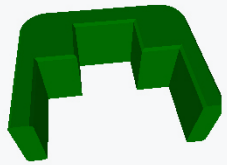
We'll flip these components a few times.
Pay attention to their final orientation





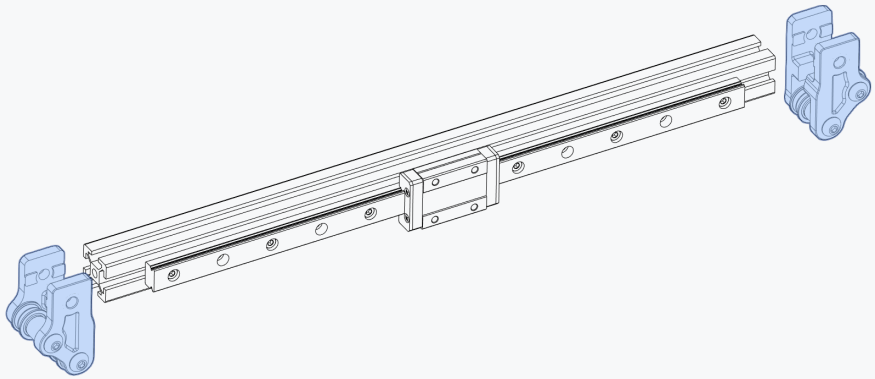
M3x8 SHCS

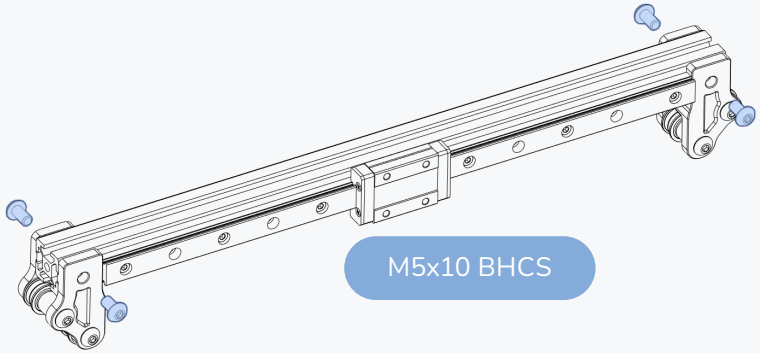
MGN12 Linear Rail



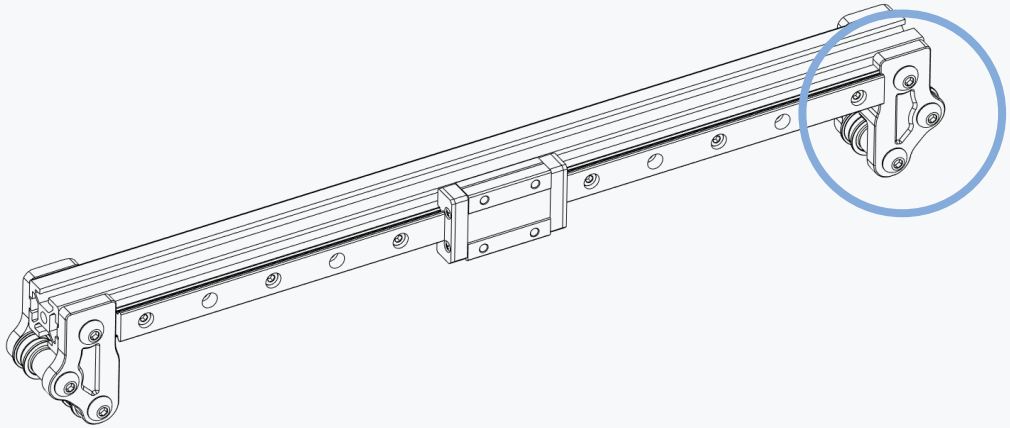
CENTERED RAIL INSTALLATION GUIDE

Use the guides to position the rail in the center of the extrusion prior to fastening the bolts.



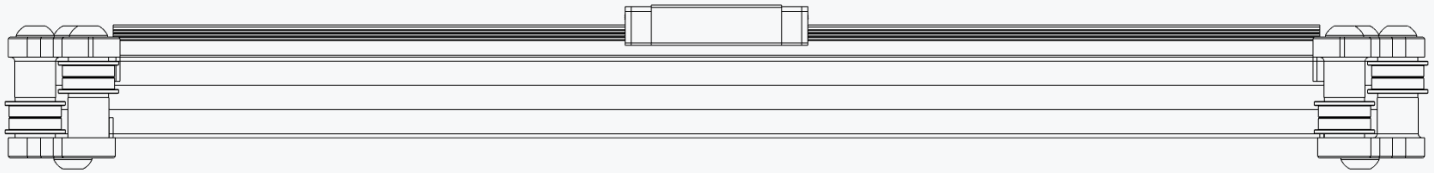


M5x10 BHCS

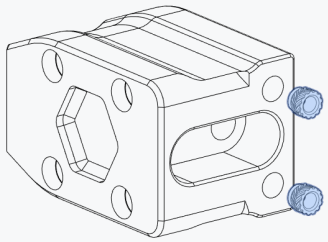


CHECK ORIENTATION

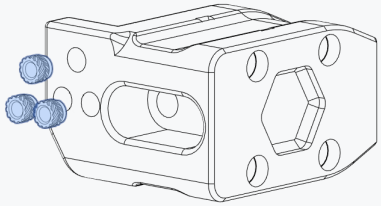
Make sure the XZ joints are oriented correctly. The reliefed/cutout section points towards the front and the idlers towards the bottom.

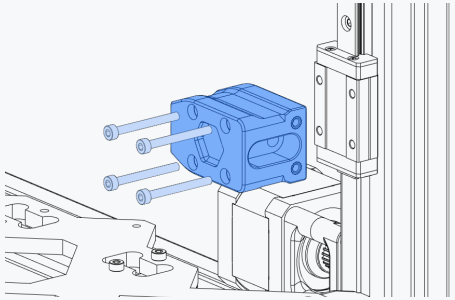


Z BEARING BLOCK

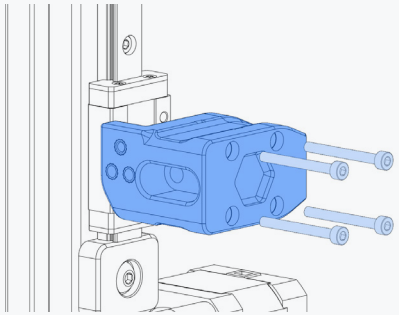


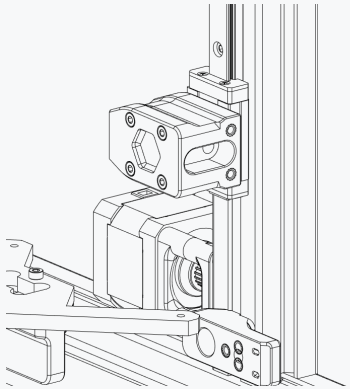
Heat Set Insert

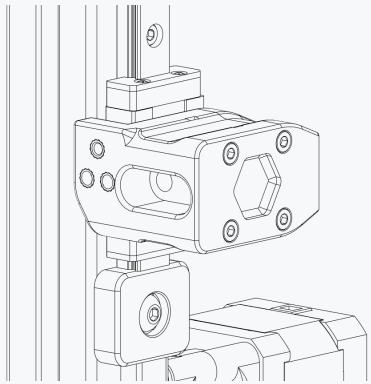


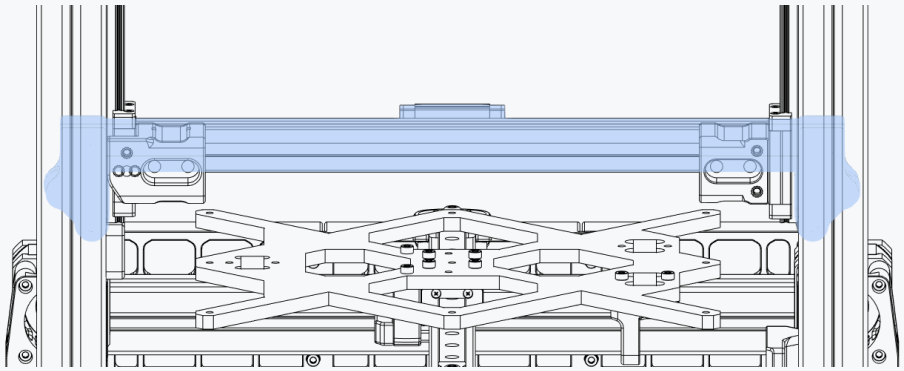


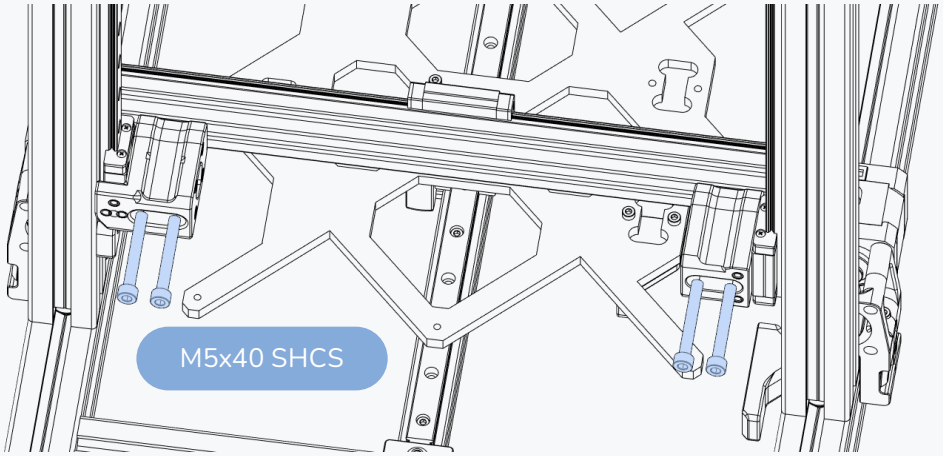
M3x30 SHCS











X AXIS INSTALLATION

Lightly tighten the M5x40 bolts and align the X axis horizontally by moving it all the way to the top. Make sure it is centered across the front.

Hold the X axis against the stoppers and fully tighten the bolts. Move the axis up and down and check for binding.

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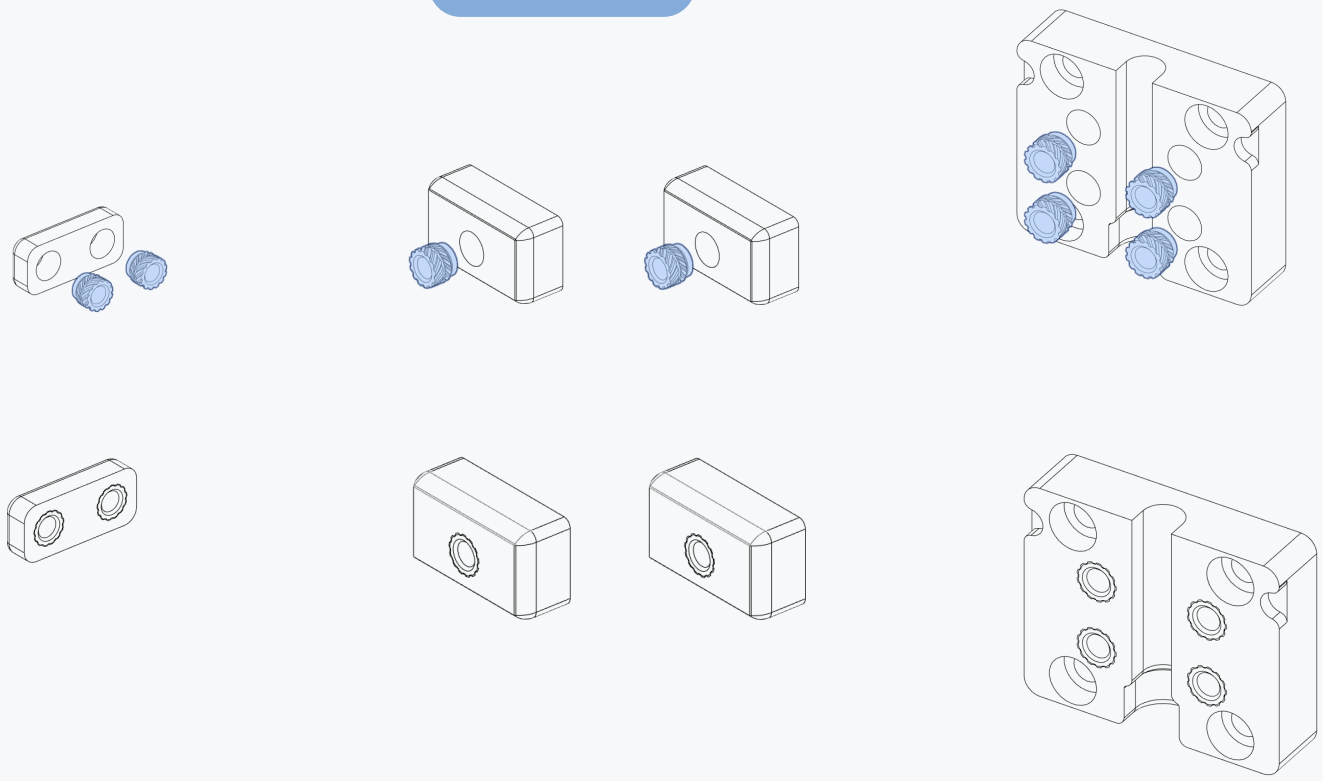
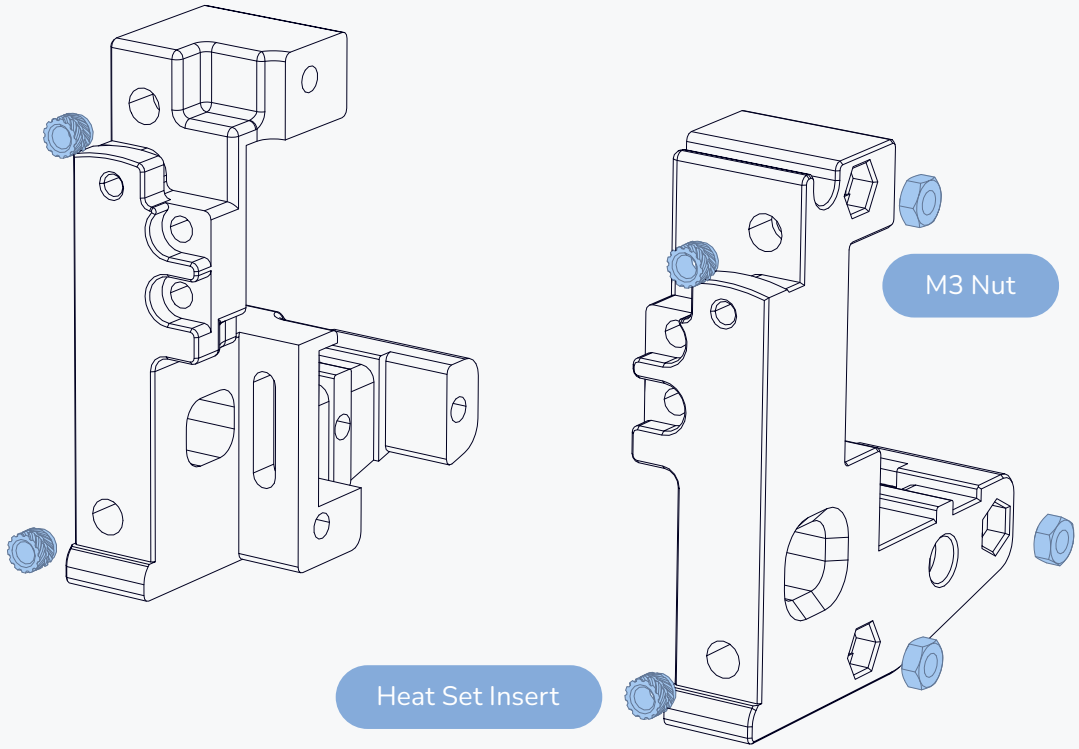
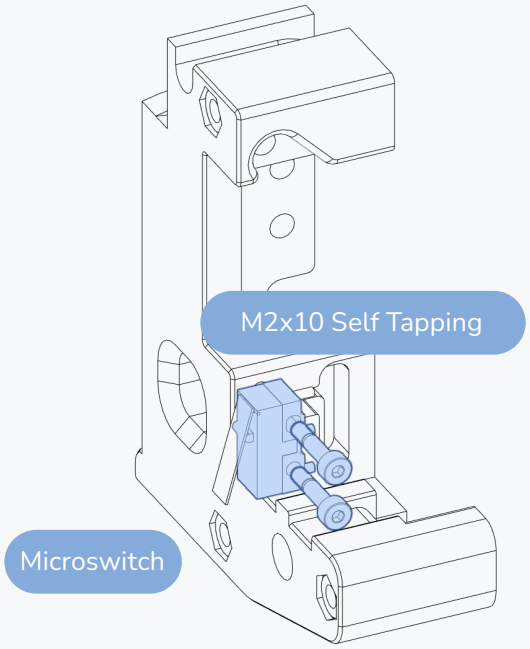
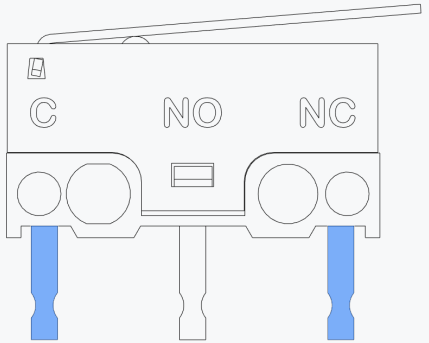
Heat Set Insert




M3 Nut
Heat Set Insert





END-STOP SWITCHES FOR X AND Y

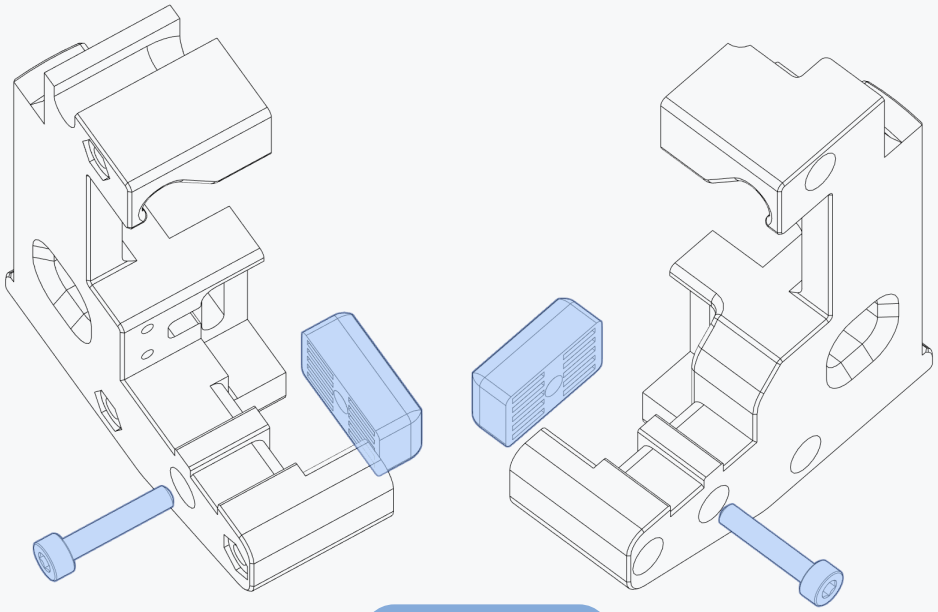
End-stops are wired in a “Normally Closed” configuration. On microswitches those are the 2 outer terminals indicated by C and NC.

Prepare the switches for X by soldering 150mm of wire to each of the outer terminals.

PRESOLDERED WIRES

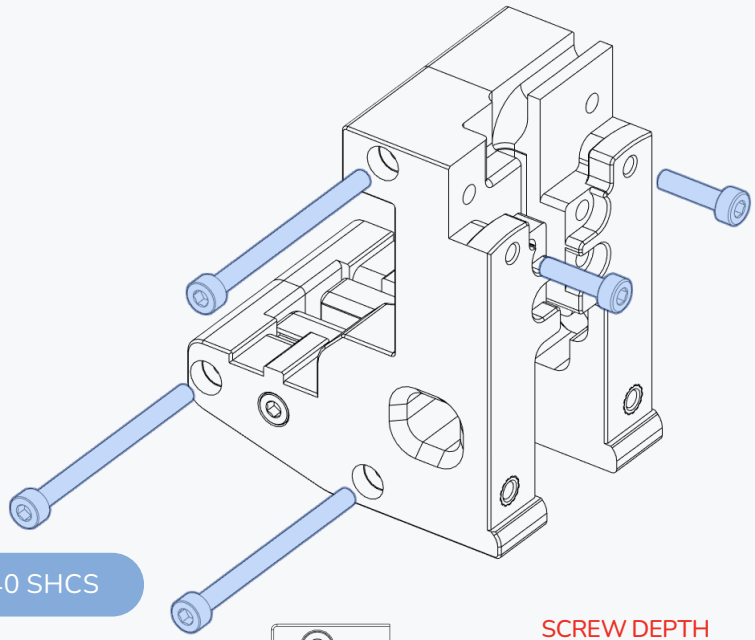
LDO MOTORS

Route the wires through the slot to the front of the carriage. Only attach the connector on the crimps after you routed them to the front.

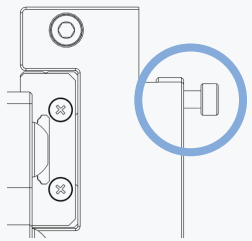


M3x16 SHCS

M3x12 SHCS

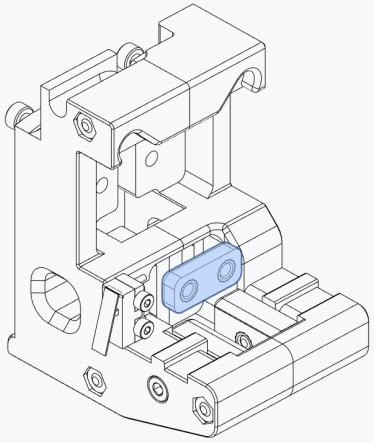


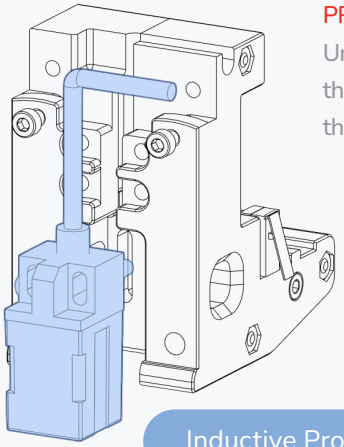
M3x40 SHCS



SCREW DEPTH

Don't tighten the bolts all the way. Leave 3-4mm of exposed thread.





Inductive Probe

PROBE WIRES

Unless your probe comes preterminated cut the probes wires to about 15cm. Don't cut them flush with the carriage frame.

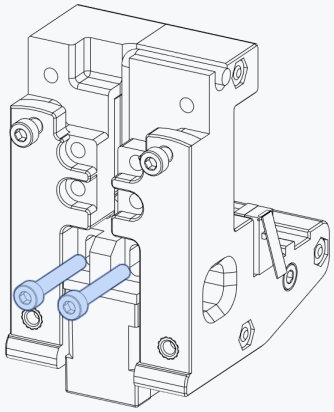
OTHER PROBE TYPES

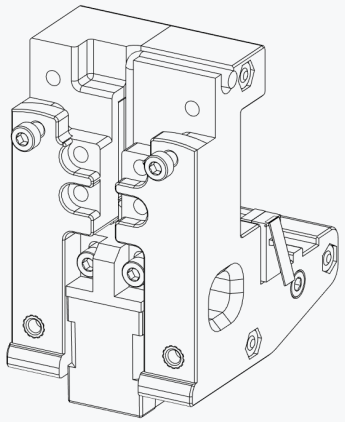
LDO MOTORS

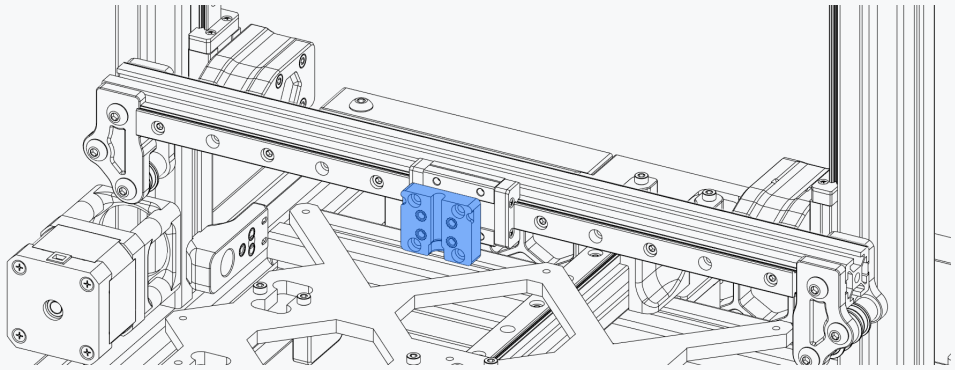
The picture shows a PL-08 probe. The recommended Omron TL-Q5MC probe will fit in the same space but requires M3x25 SHCS.

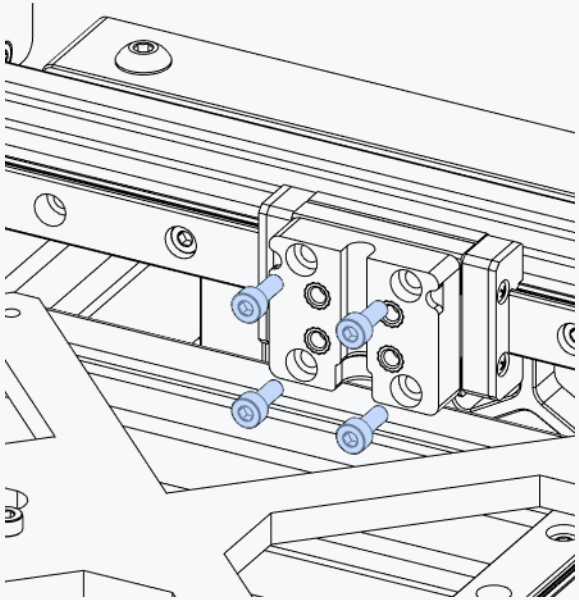
Other probes with a similar form factor and characteristics might work as well.

M3x20 SHCS

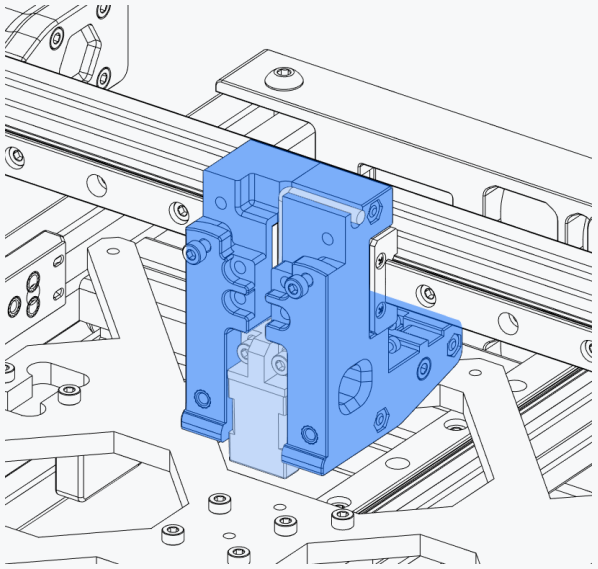




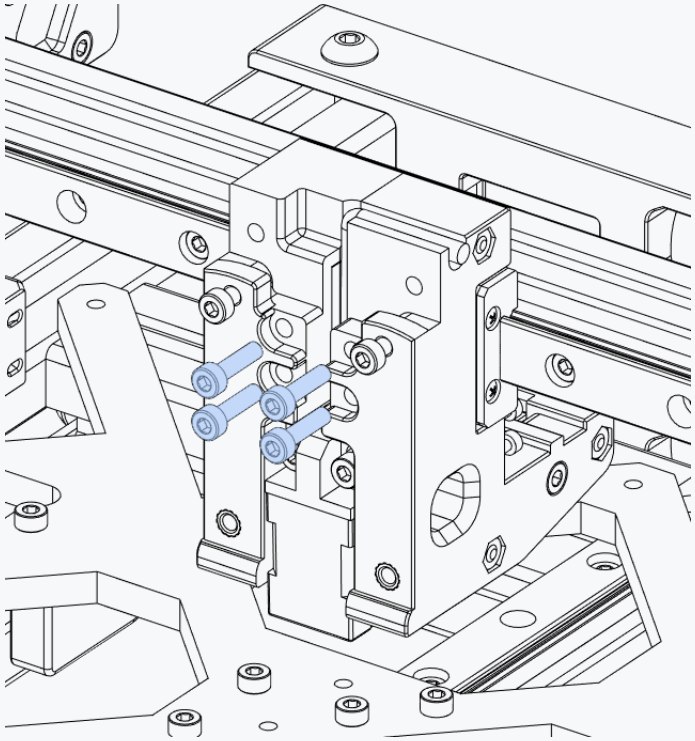




M3x8 SHCS

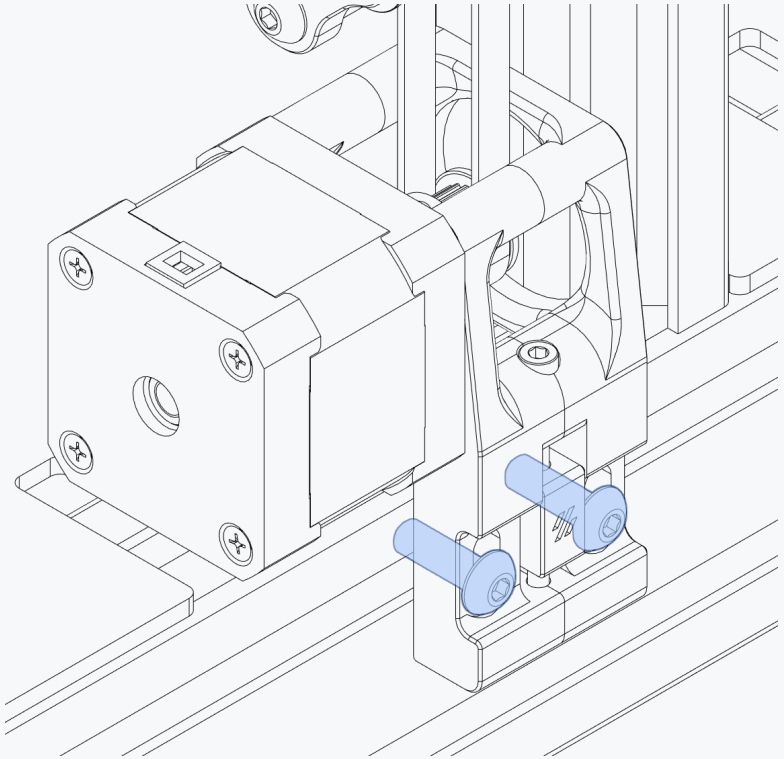


M3x12 SHCS



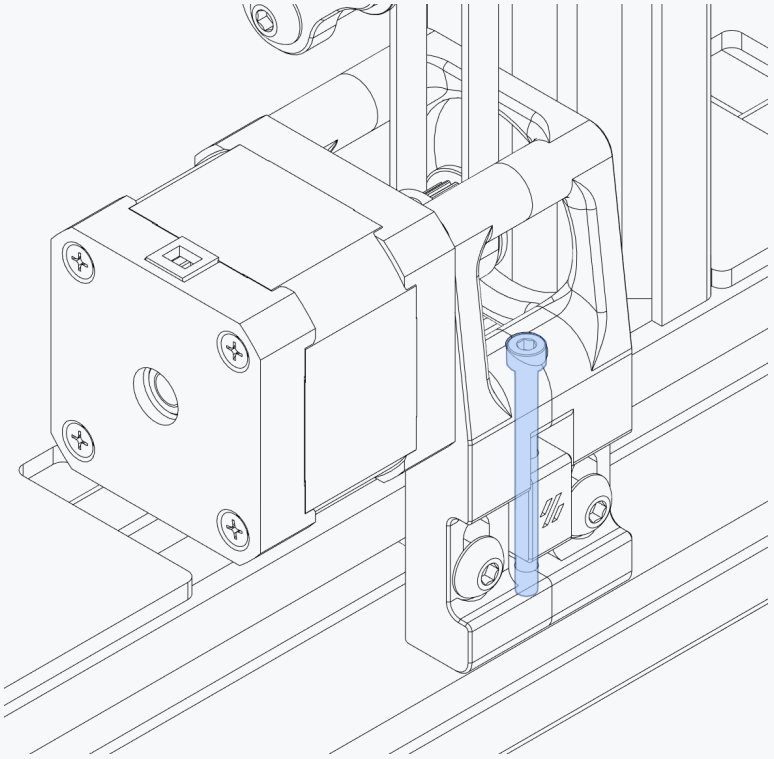
BELT TENSIONER FOR “X” AND “Z”

The motor mounts for the “X” and the “Z” motors have build in tensioners.



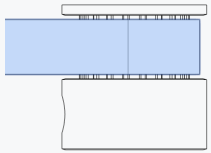
LOCK SCREWS

Slightly loosen these screws before adjusting the belt tension. Fasten them again after adjusting the tension.



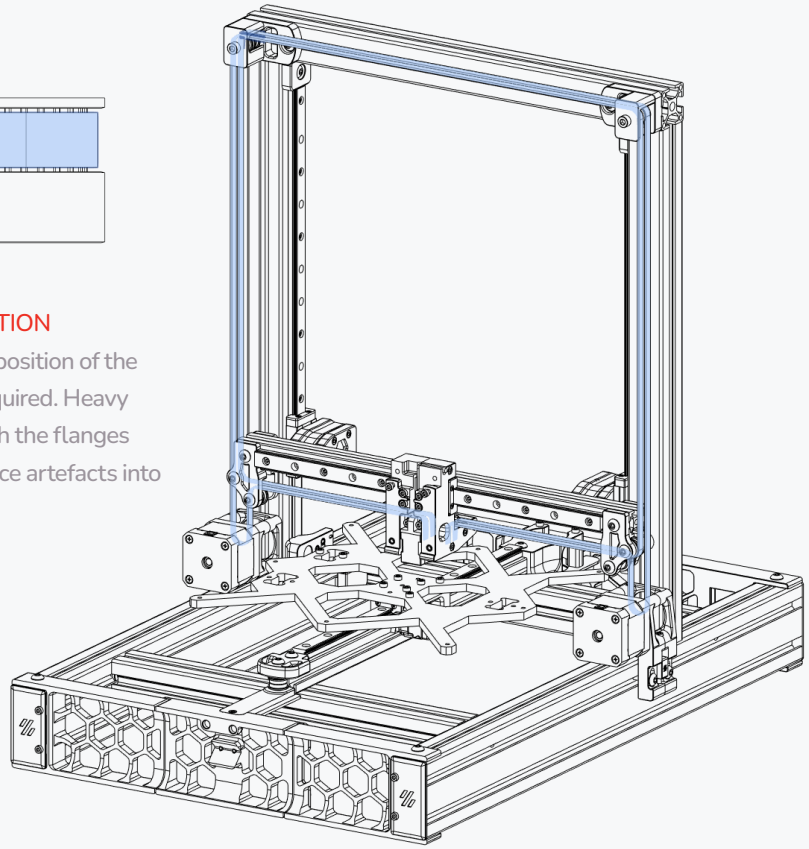
ADJUSTMENT SCREW

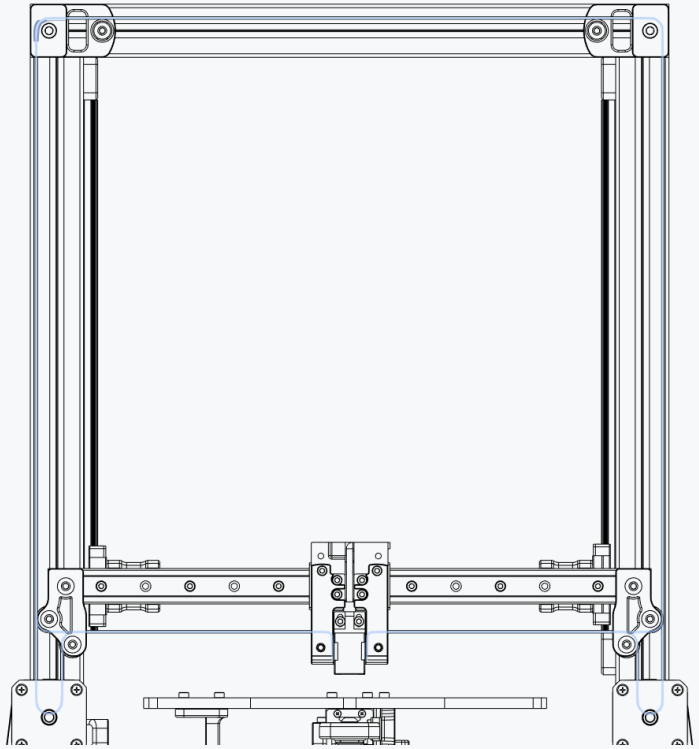
Turn the screw clockwise to pull a higher tension on the belt.



BELT POSITION

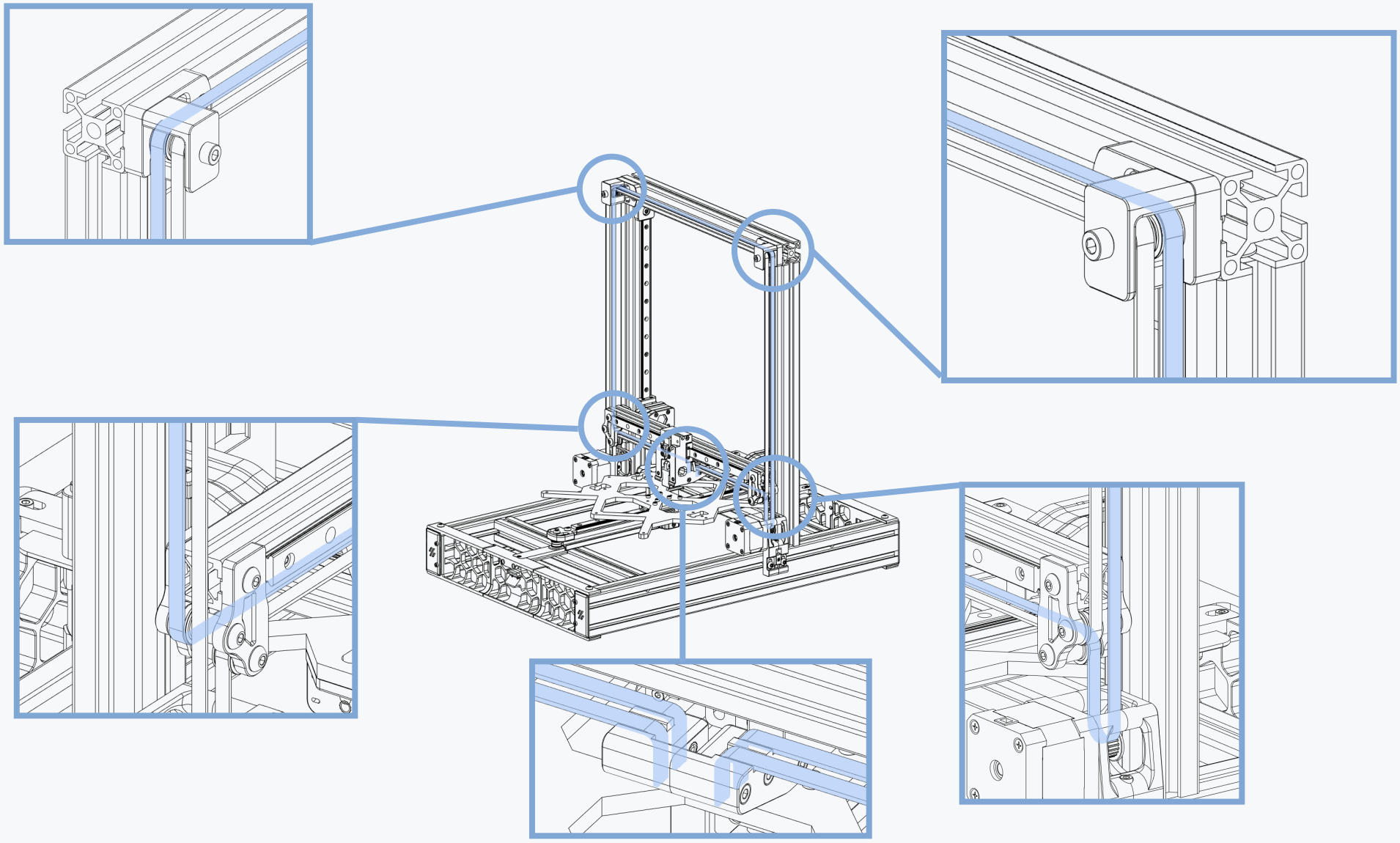
Adjust the position of the pulley if required. Heavy contact with the flanges will introduce artefacts into your prints.

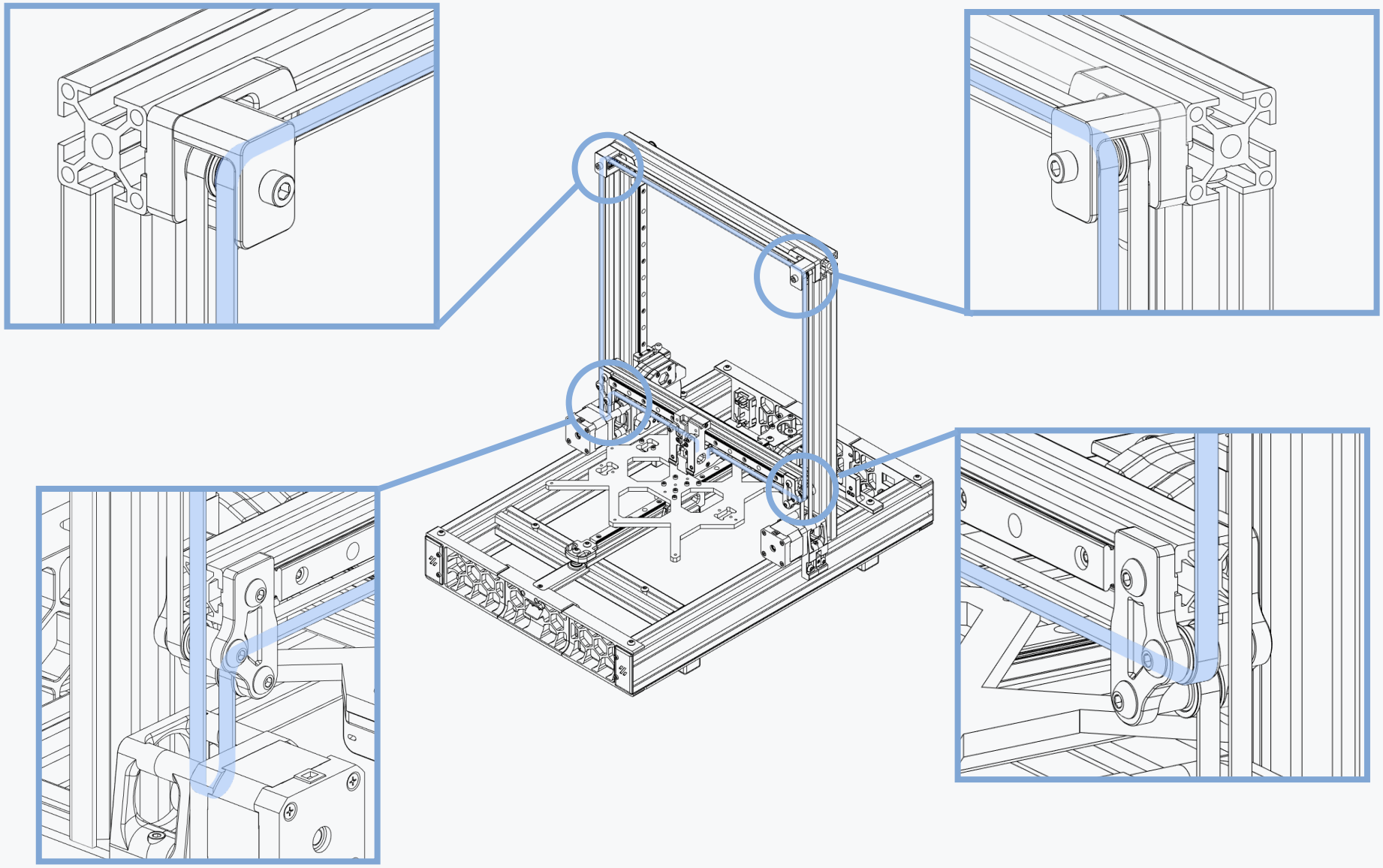


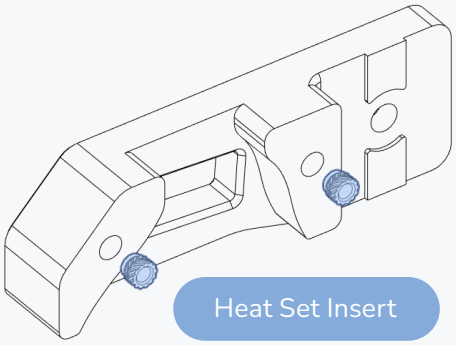


BELT PATH

Take your time and make sure your belt routing is as intended. Belts should never rub on plastic parts.



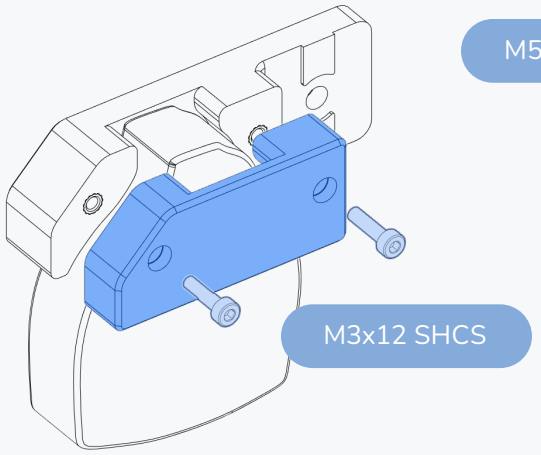


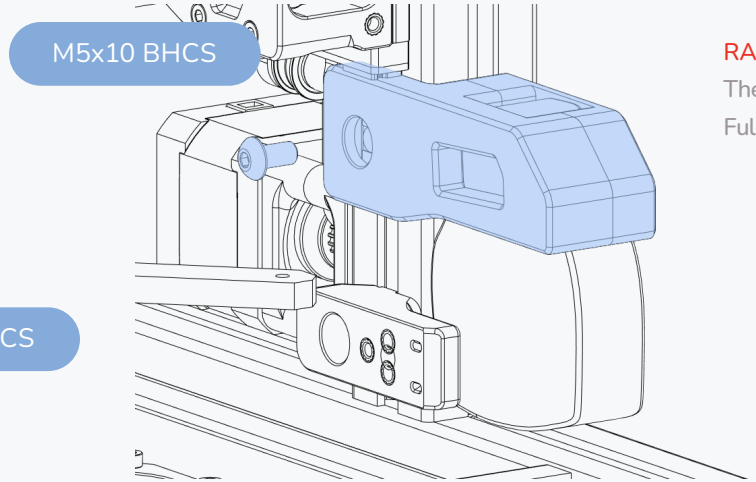


WEAR EYE PROTECTION

The Key-Bak will violently retract if you lose grip of the end piece.

Wear eye protection or a face shield until it is securely mounted.

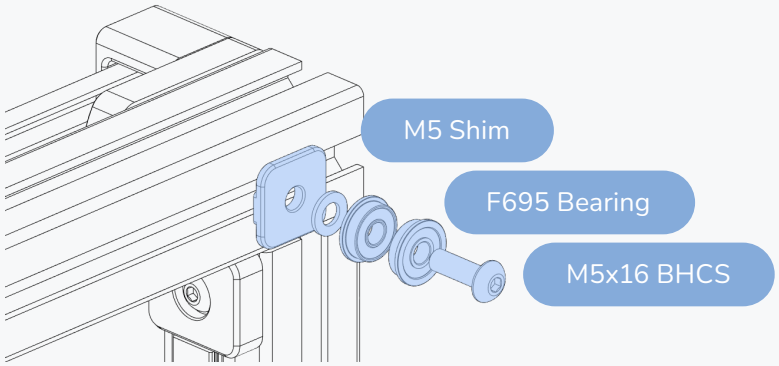


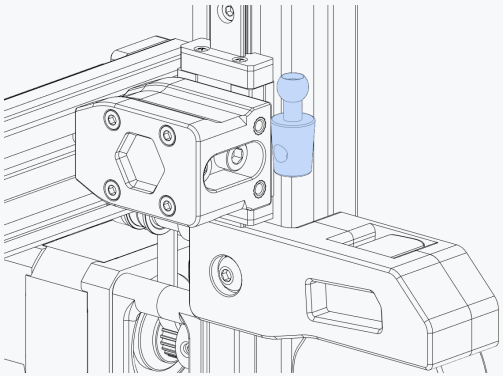


RAIL CARRIAGE

The rail carriage is now secure.
Fully fasten the bottom most bolt.

COUNTERWEIGHT IDLER

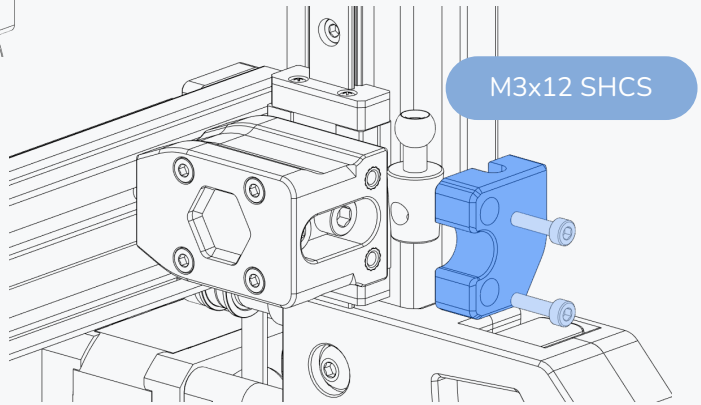




MOUNTING THE KEYBACK END PIECE

Pull the end piece of the Key-Bak and mount it as shown.

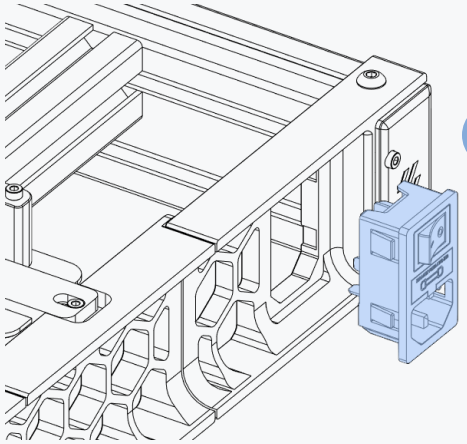
Extend more of the cord and hook it in the idler mounted above.



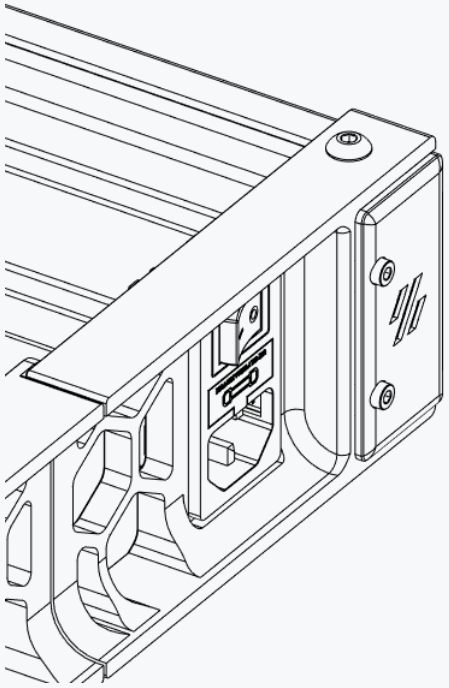
UNIVERSAL TOOLHEAD

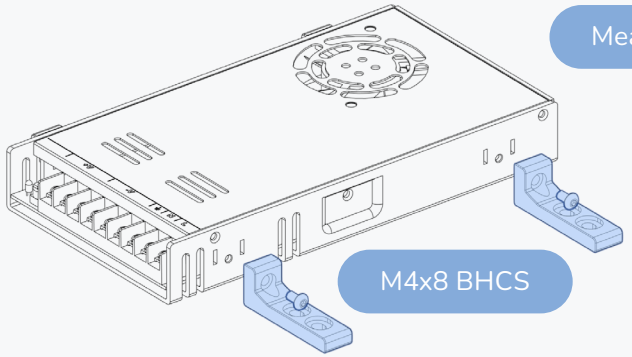
This printer uses the StealthBurner toolhead, which is compatible with several of the printers in the Voron lineup.
To keep things organized, StealthBurner's files are maintained separately. Follow the StealthBurner assembly manual to build your toolhead, and return here to proceed.

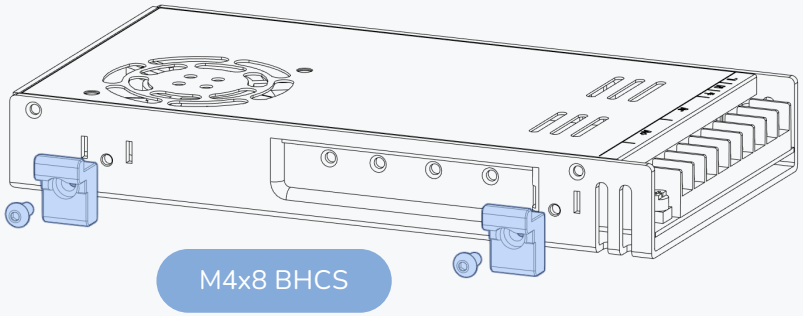
https://voron.link/6hbi9n3

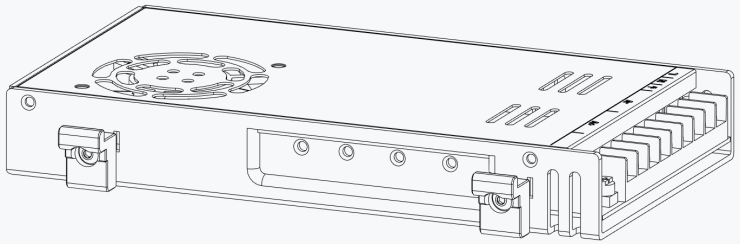


Mains Inlet

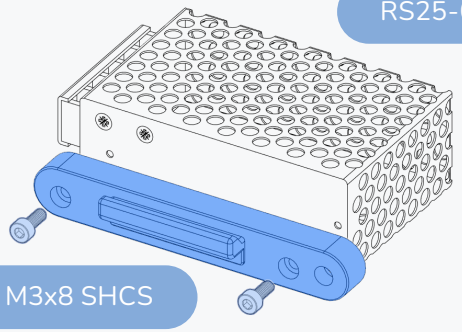



Meanwell LRS350-24
M4x8 BHCS

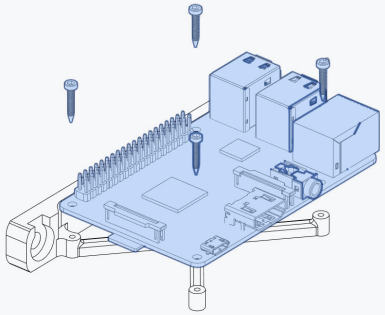

M4x8 BHCS



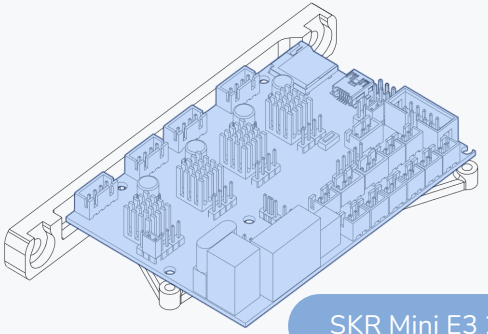
RS25-05 PSU



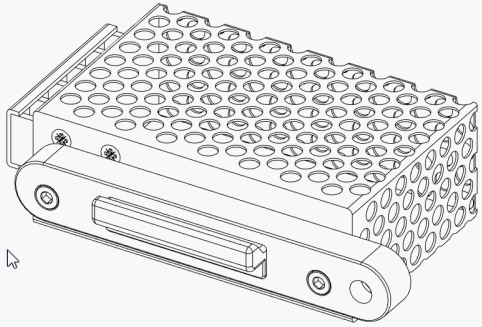
M2x10 Self Tapping



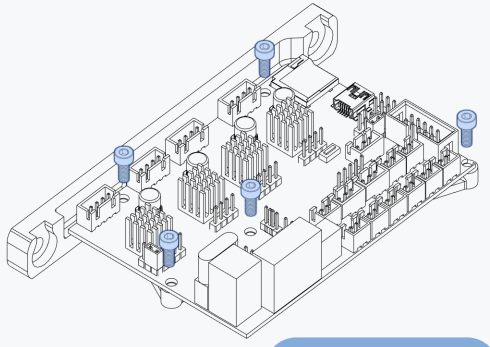
M3x8 SHCS



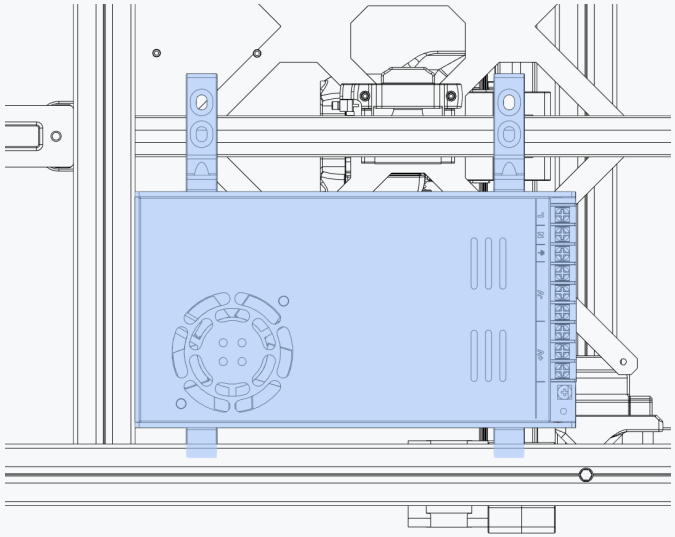
SKR Mini E3 2.0

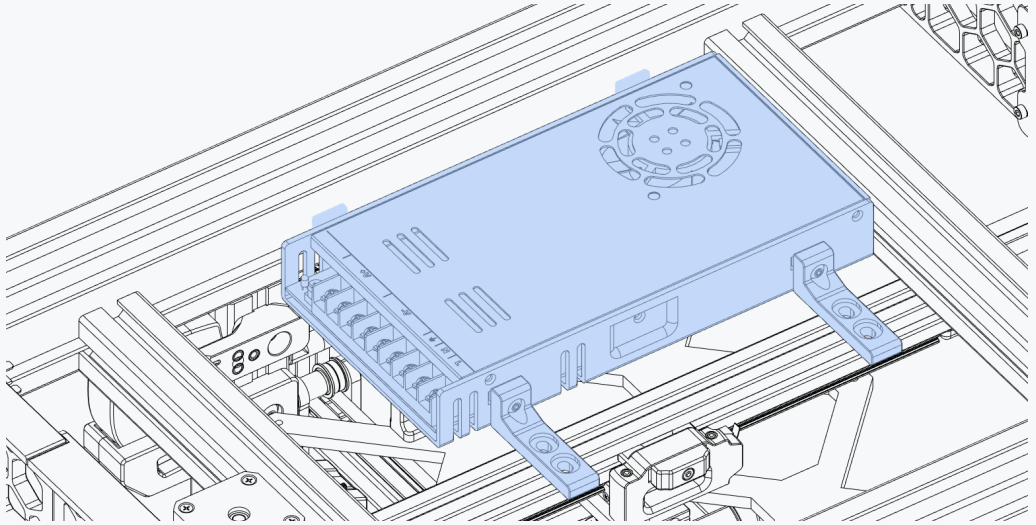


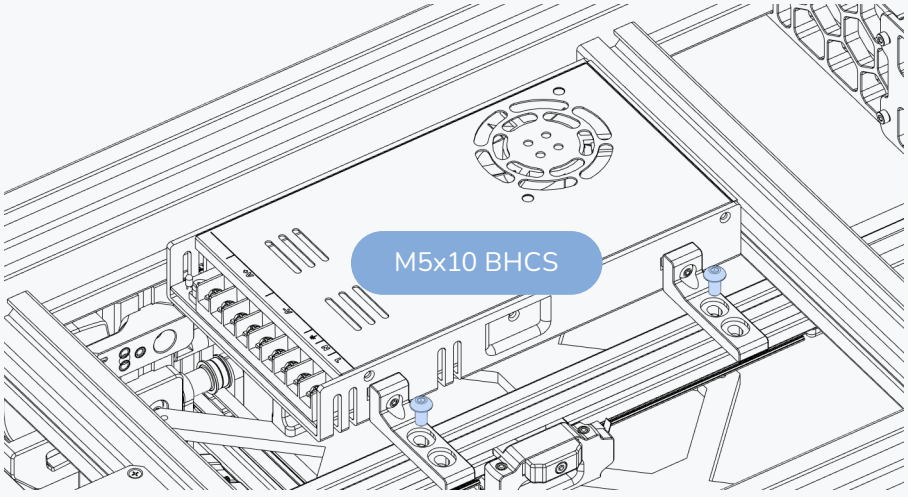
Dunar's Cursor

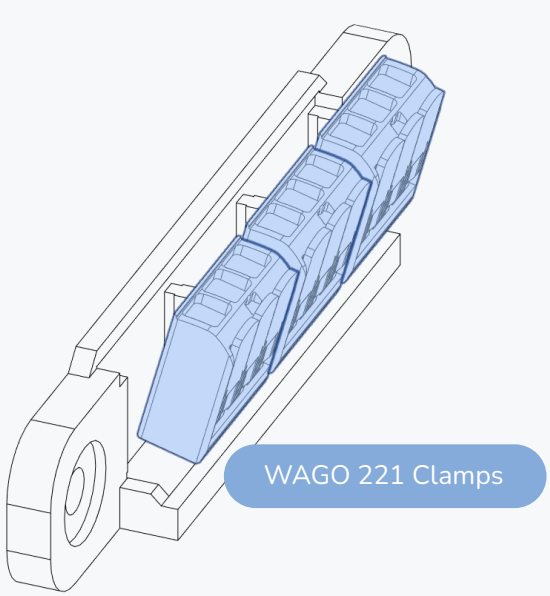


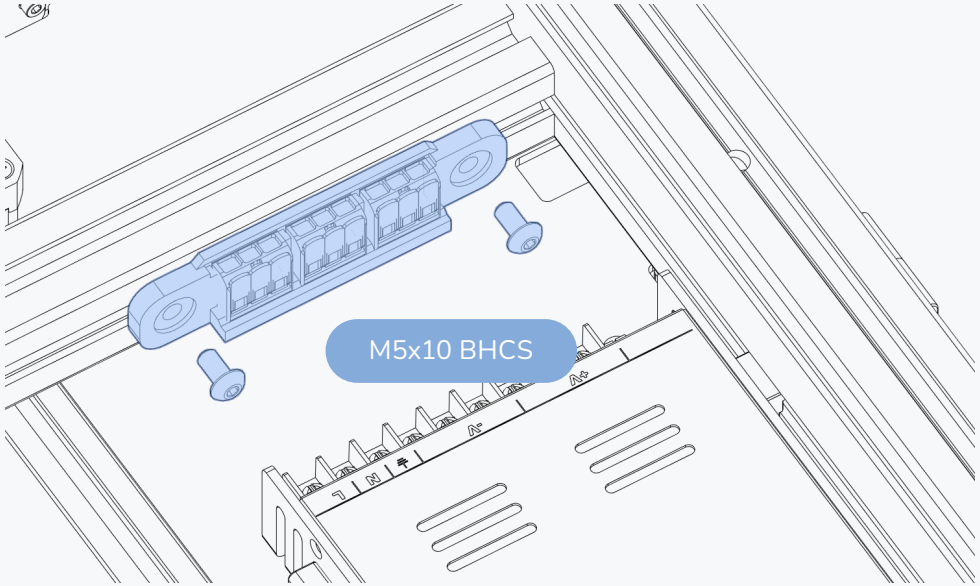
M3x8 SHCS






M5x10 BHCS

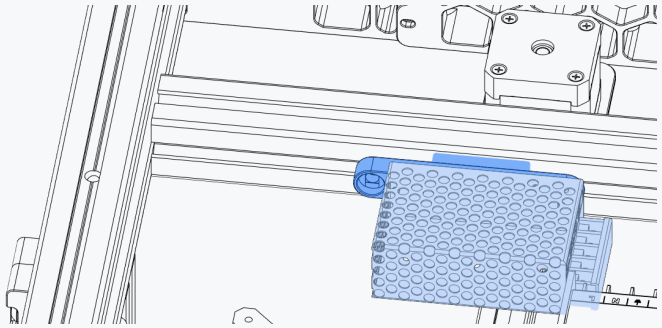


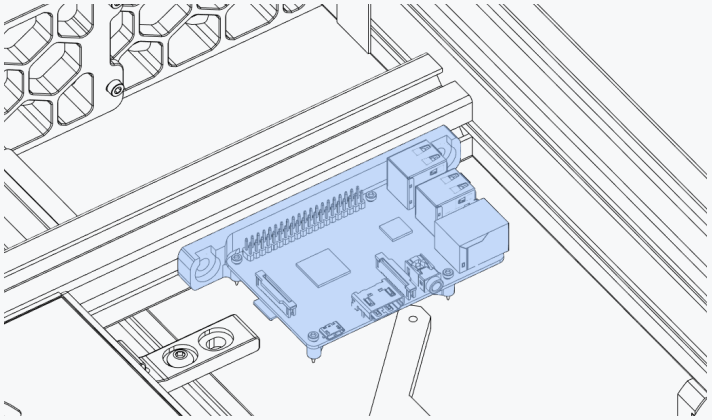


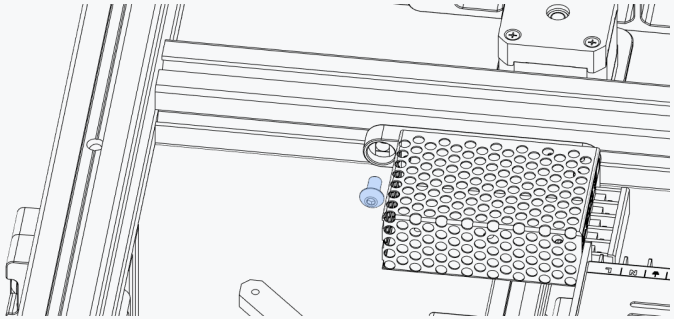
OPTION: WAGO TERMINAL STRIP

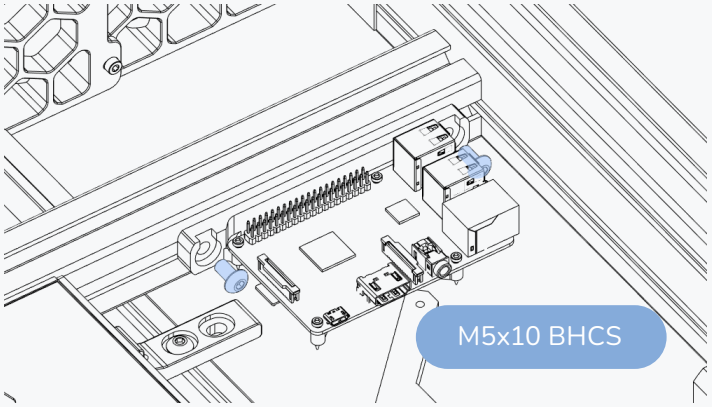
LDO MOTORS

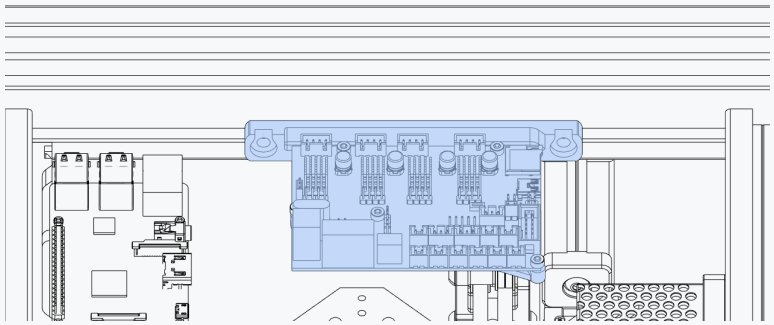
WAGO 221 series clamps can be used to simplify the mains wiring.

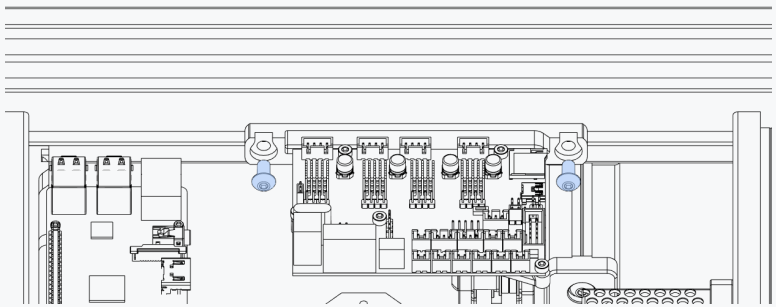




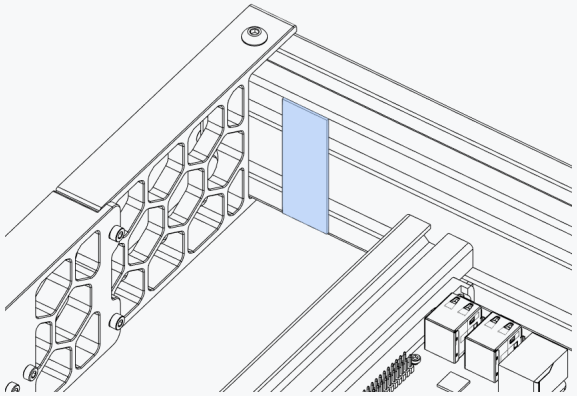

M5x10 BHCS

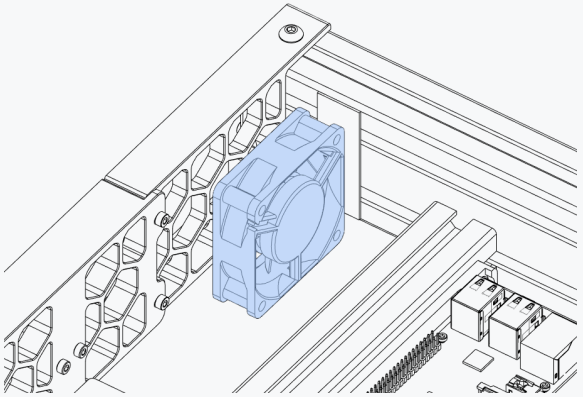

M5x10 BHCS





M5x10 BHCS

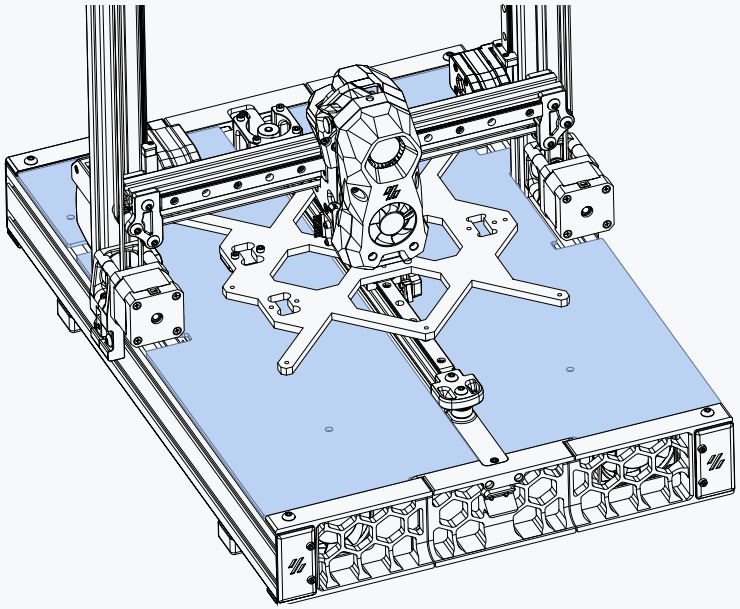




6020 Fan

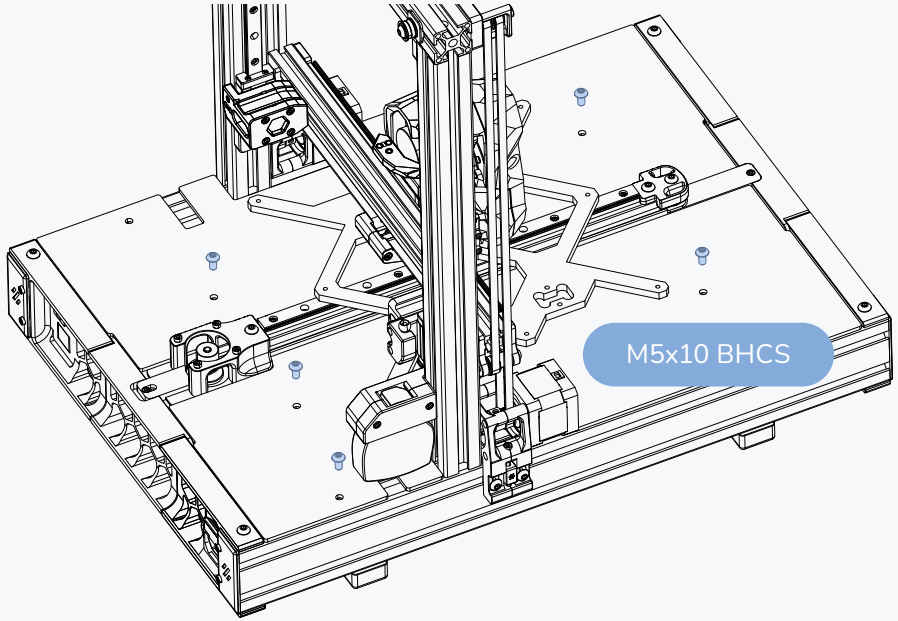
APPLY VHB TAPE

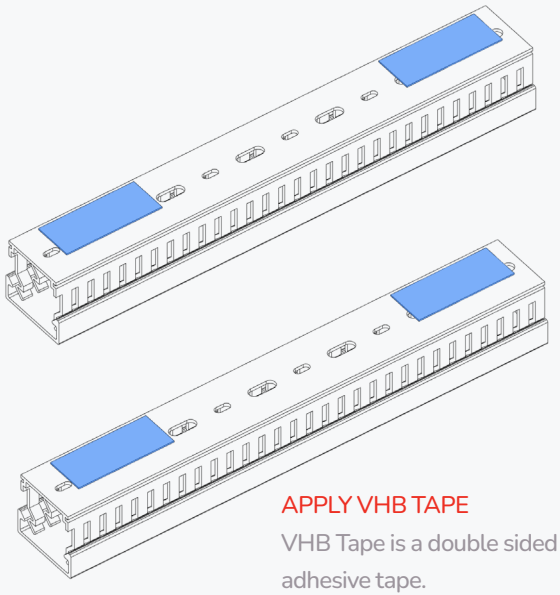
3M VHB tape is a double sided adhesive tape. Other vendors have similar products that you can use as a substitute.



DECK PANELS

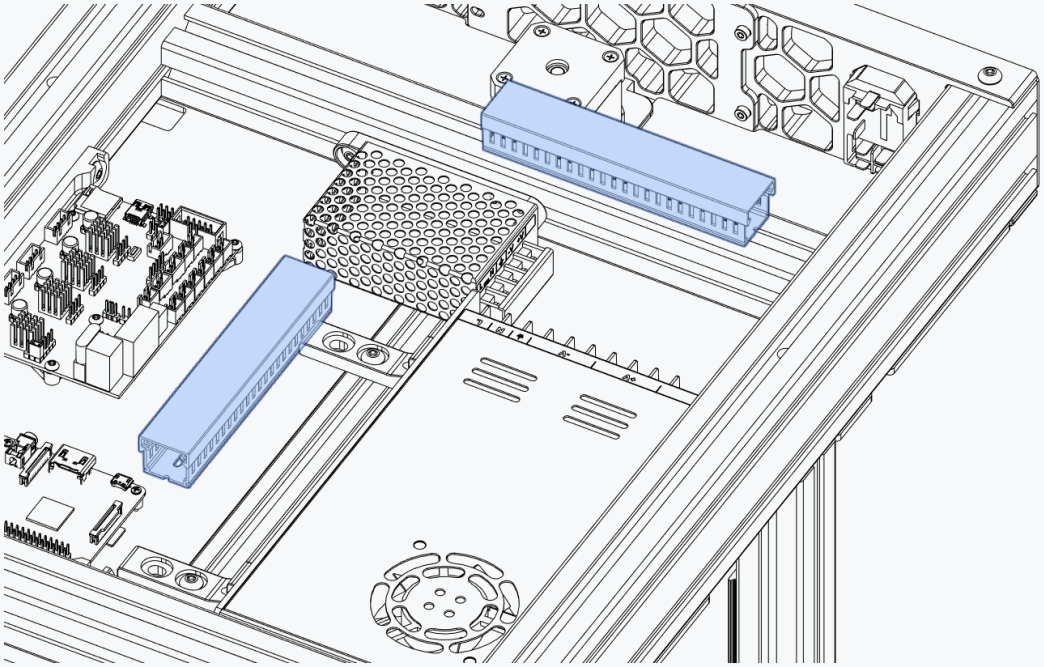
Unless you installed the panels earlier undo the 6 bolts on the bed carriage and remove the bed carriage to install the panels.





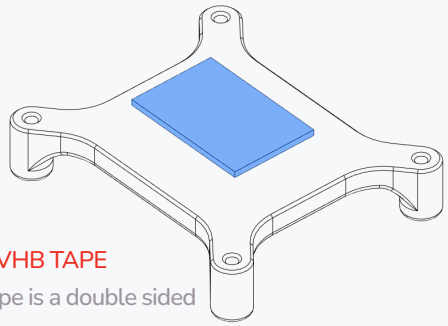
APPLY VHB TAPE

VHB Tape is a double sided adhesive tape.



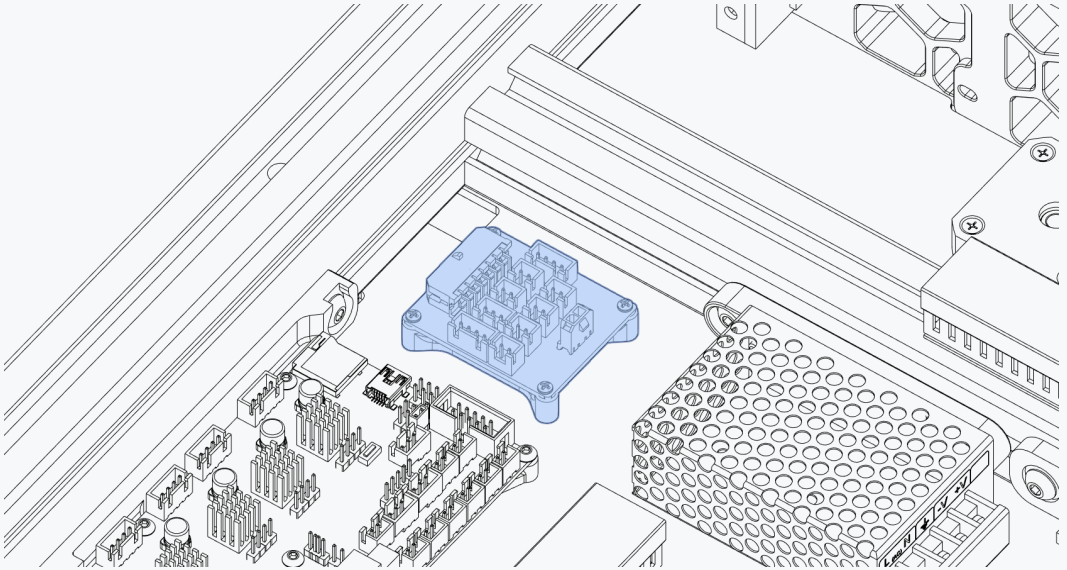
OPTION: CABLE RACEWAYS LDO MOTORS

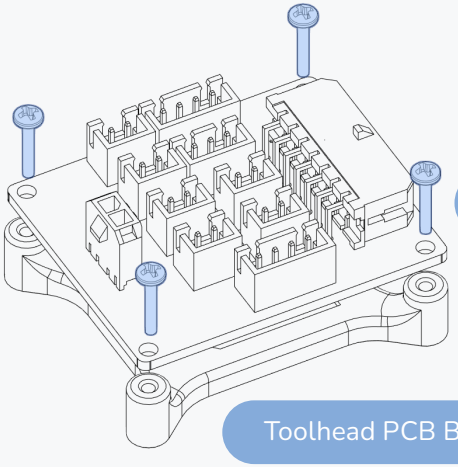
Raceways help in keeping the electronics compartment organized and clutter free.



APPLY VHB TAPE

VHB Tape is a double sided adhesive tape.





M2x10 Self Tapping

Toolhead PCB Breakout Board

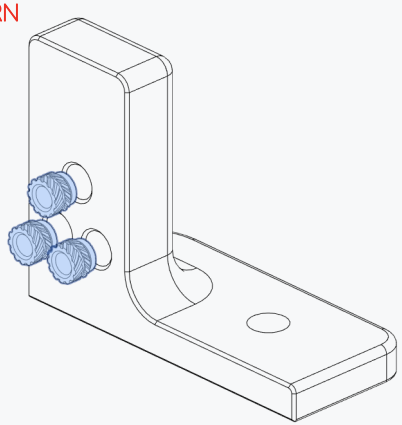
OPTION: TOOLHEAD PCB BREAKOUT BOARD

LDO MOTORS

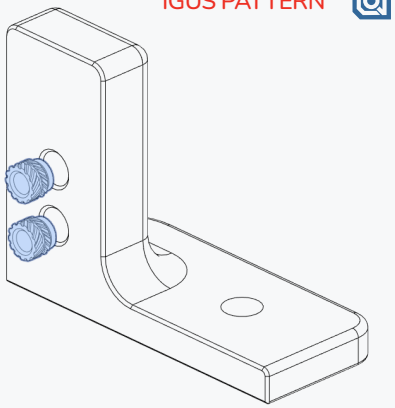
Some premade wiring harnesses come with a breakout board for the electronics compartment. If yours did install it now.

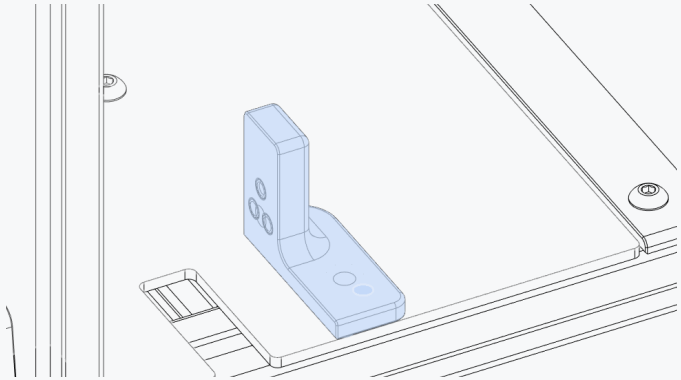
TRIANGLE PATTERN

Heat Set Insert

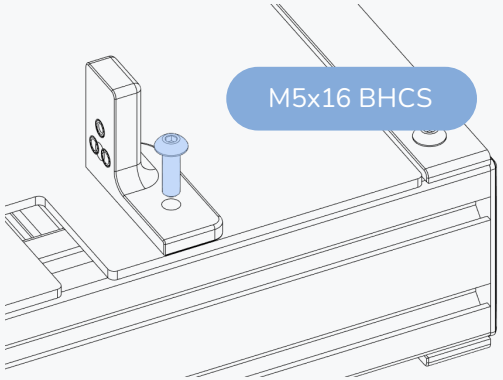


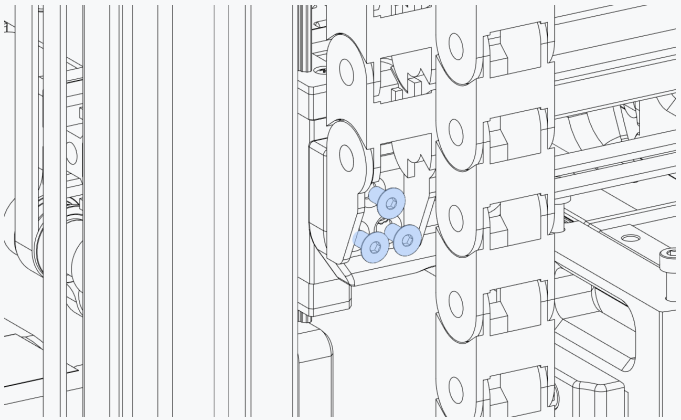
IGUS PATTERN



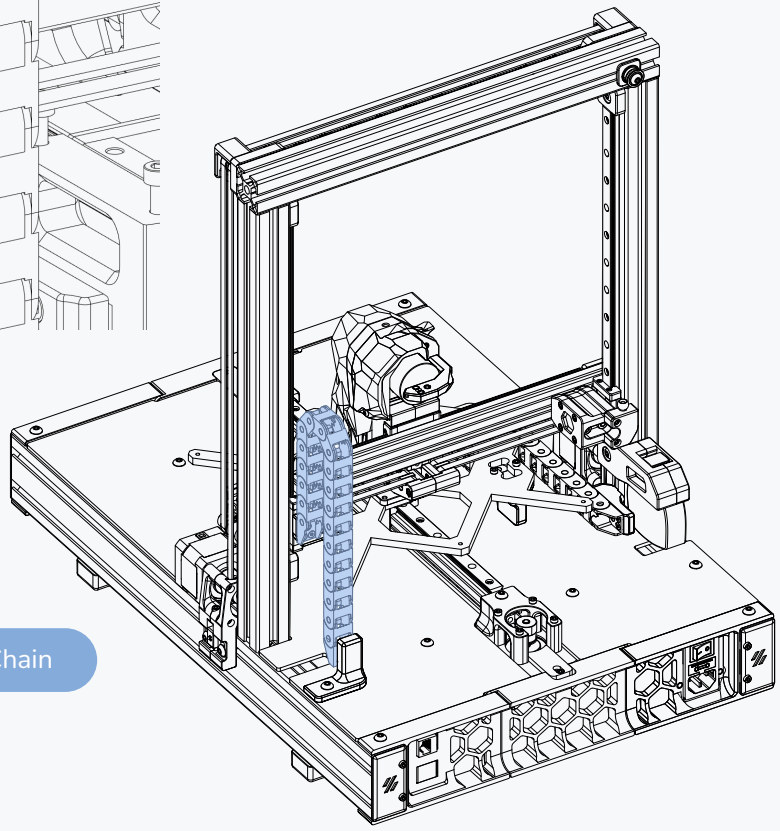


M5x16 BHCS

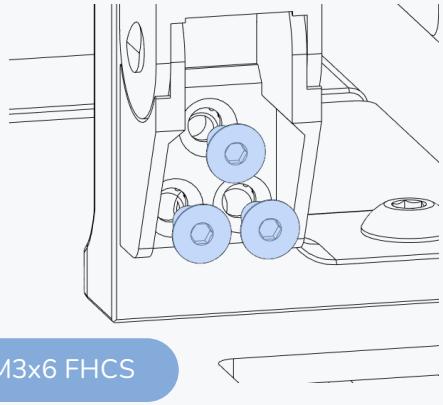




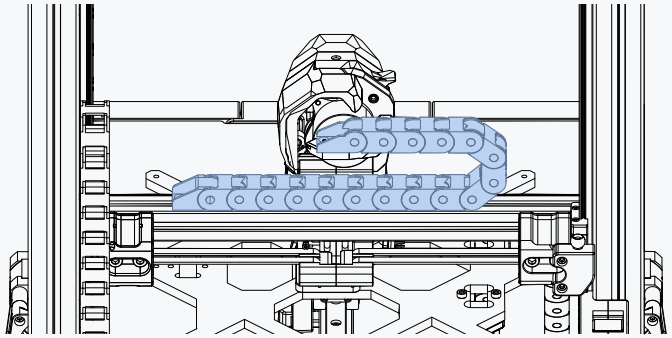
M3x6 FHCS



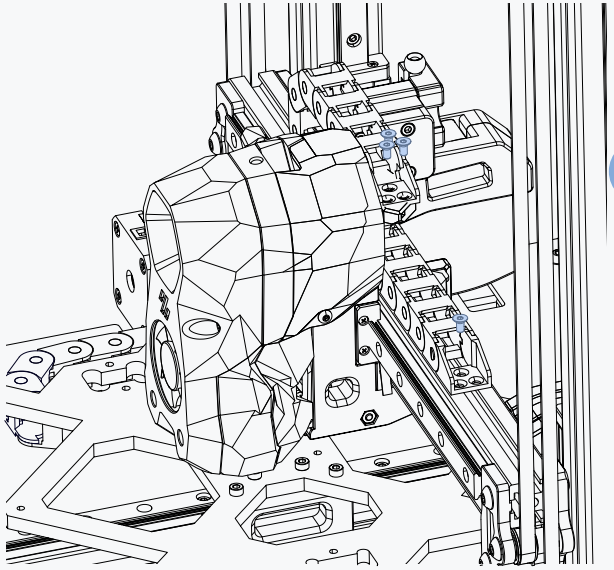
10x11 Cable Chain



M3x6 FHCS

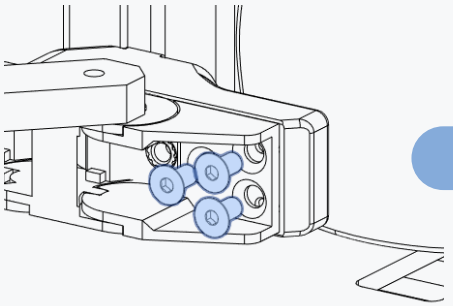


10x11 Cable Chain

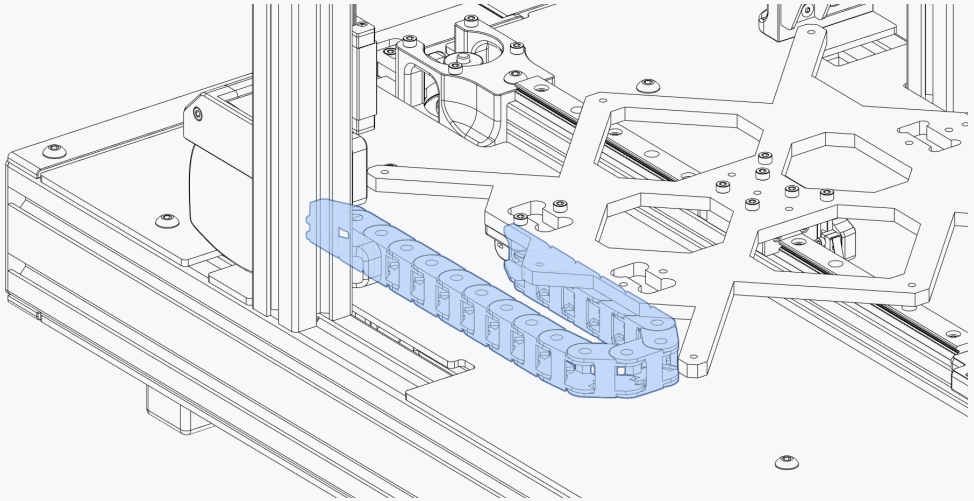


M3x6 FHCS

Y CABLE CHAIN

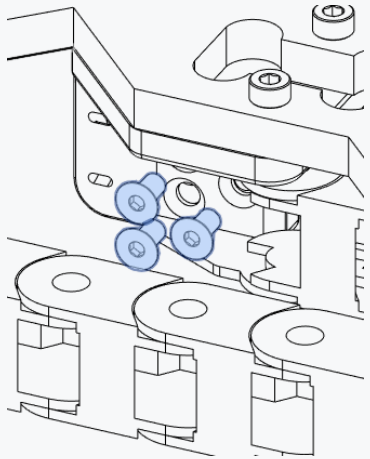


M3x6 FHCS

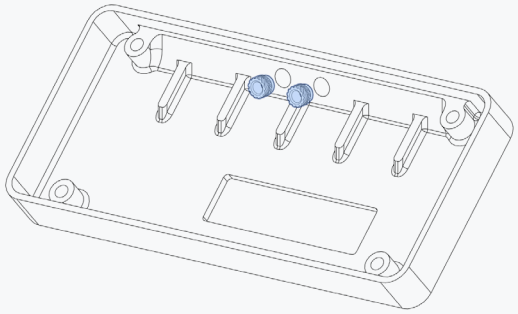


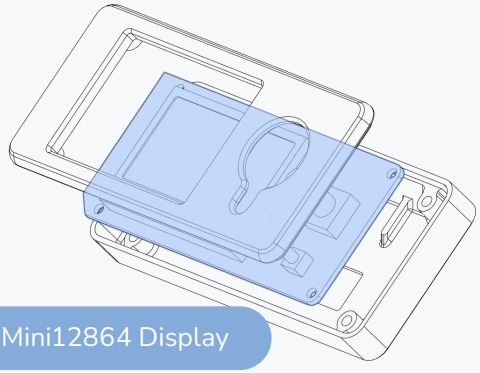
10x11 Cable Chain

M3x6 FHCS

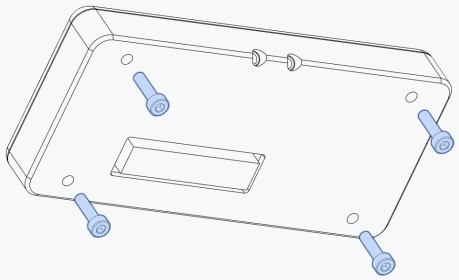


Heat Set Insert

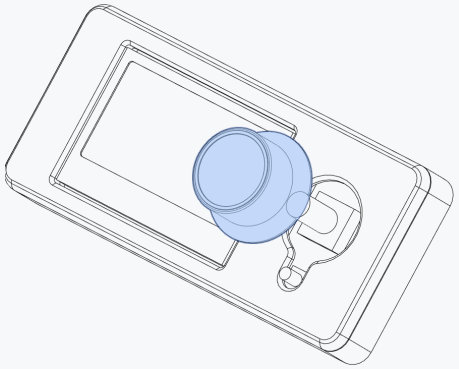


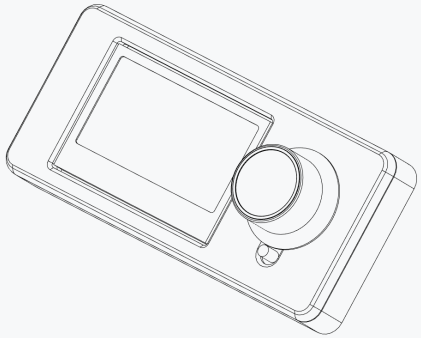


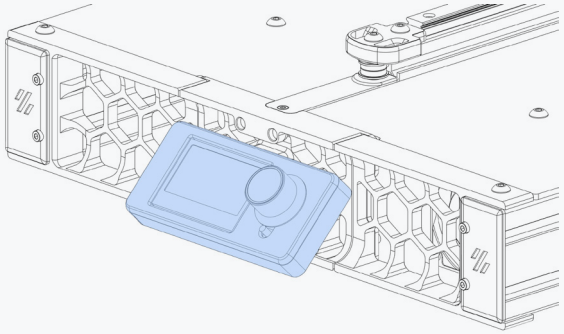
Mini12864 Display



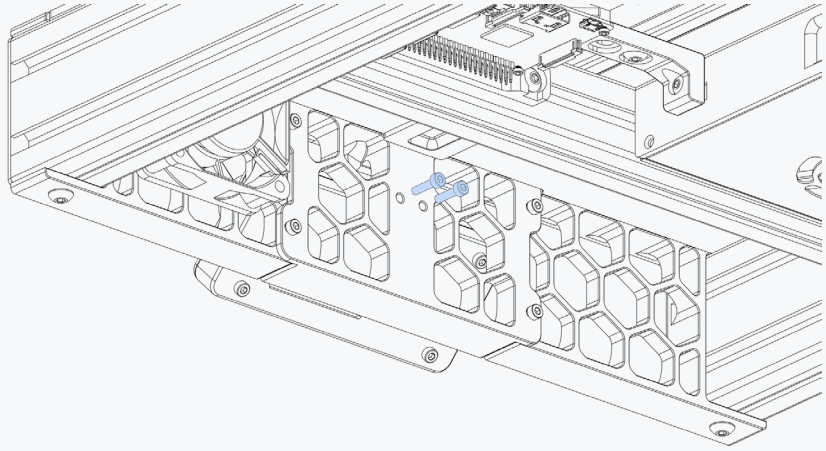
M3x12 SHCS







M3x16 SHCS



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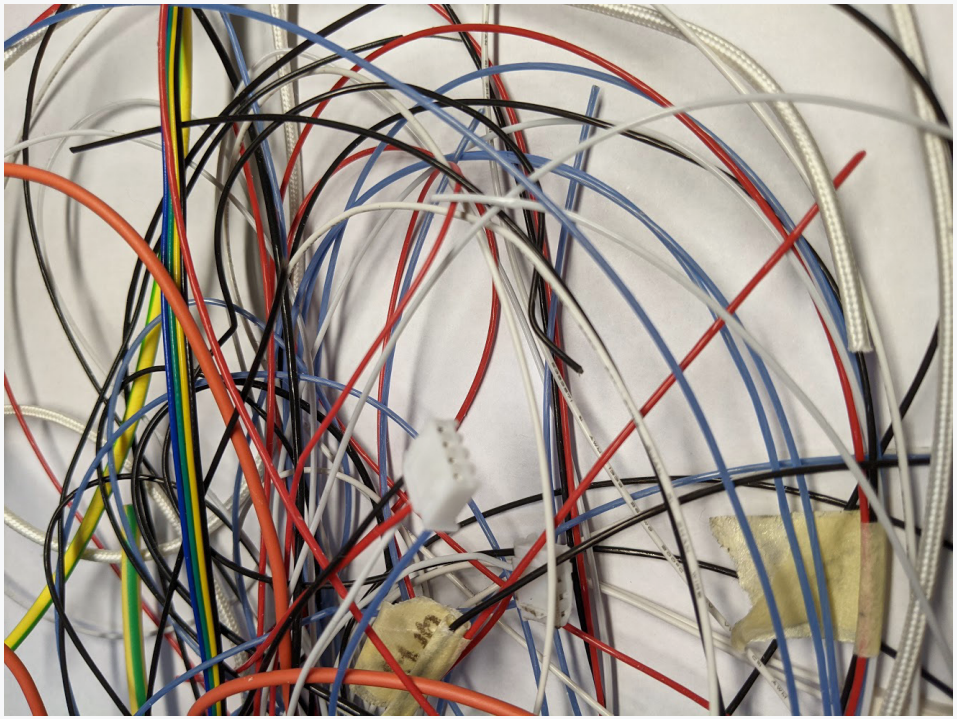
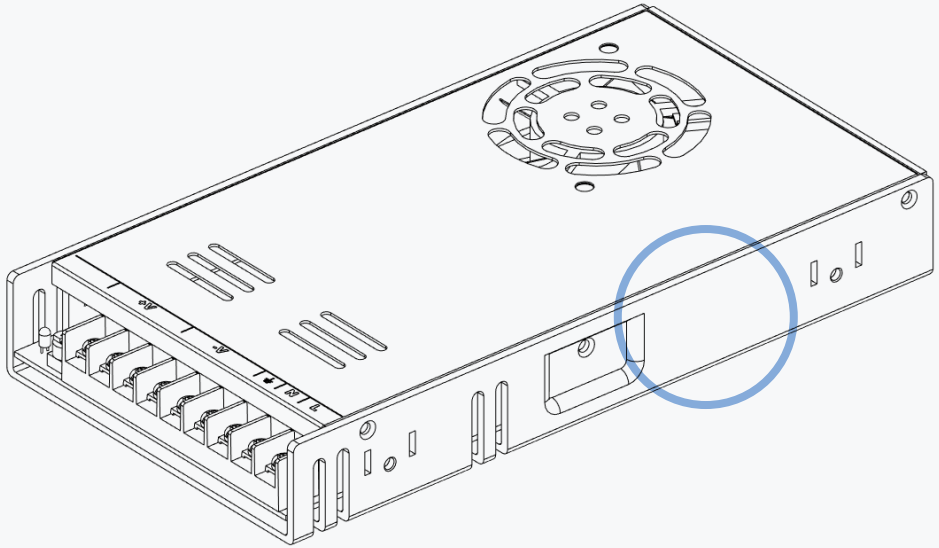
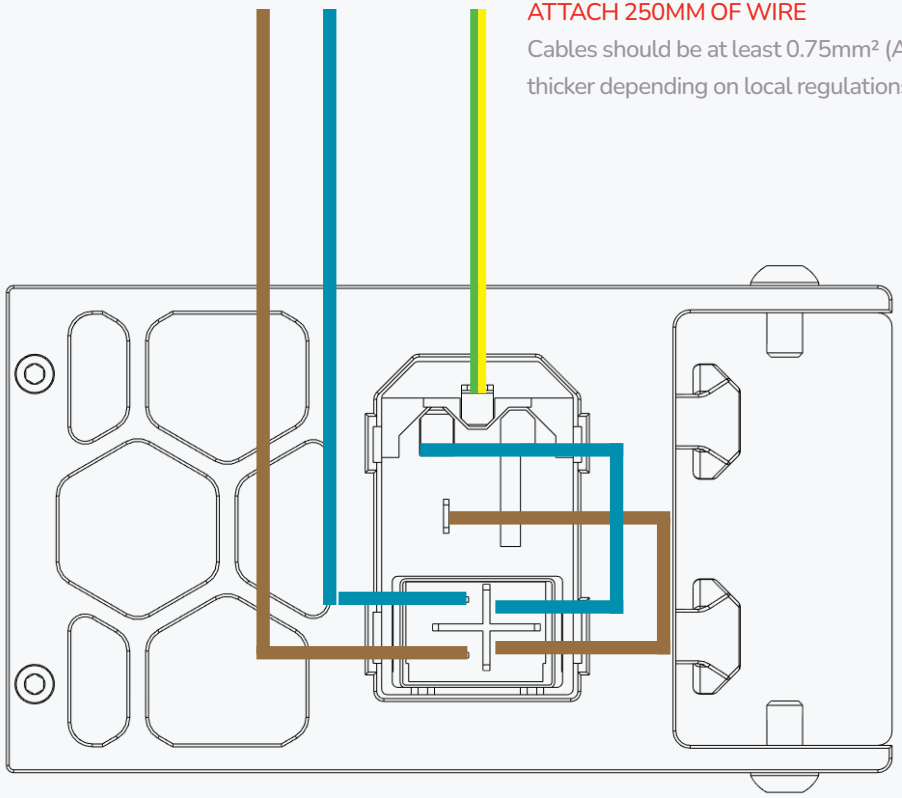




INPUT VOLTAGE SWITCH

Check the input voltage switch of the power supply. It is located in the highlighted area behind the metal mesh.

Make sure the selection matches your local mains voltage. Refer to the Mean Well LRS-350 datasheet for possible settings.



ATTACH 250MM OF WIRE

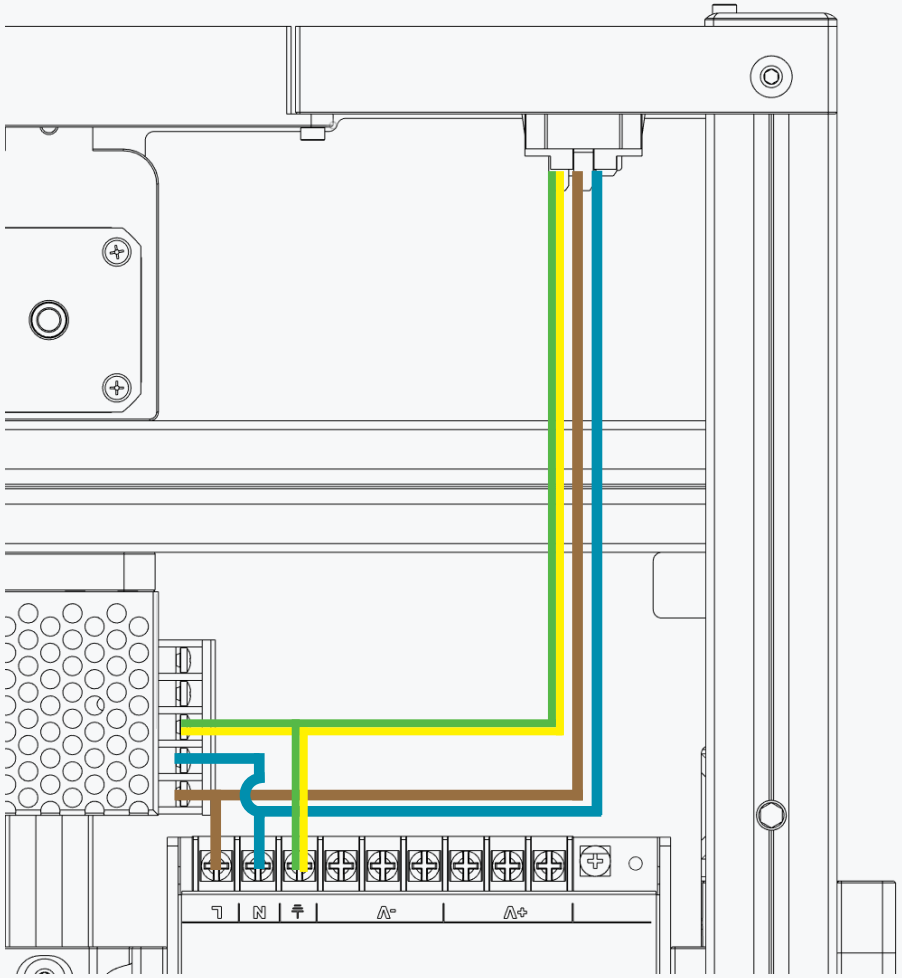
Cables should be at least 0.75mm² (AWG18) or thicker depending on local regulations.

MAINS INLET WIRING

We show the wiring in the IEC colour scheme. Depending on your region the colour scheme and wiring standards will differ.

Mains wiring should only be done by qualified personnel trained in local regulations and safety standards.

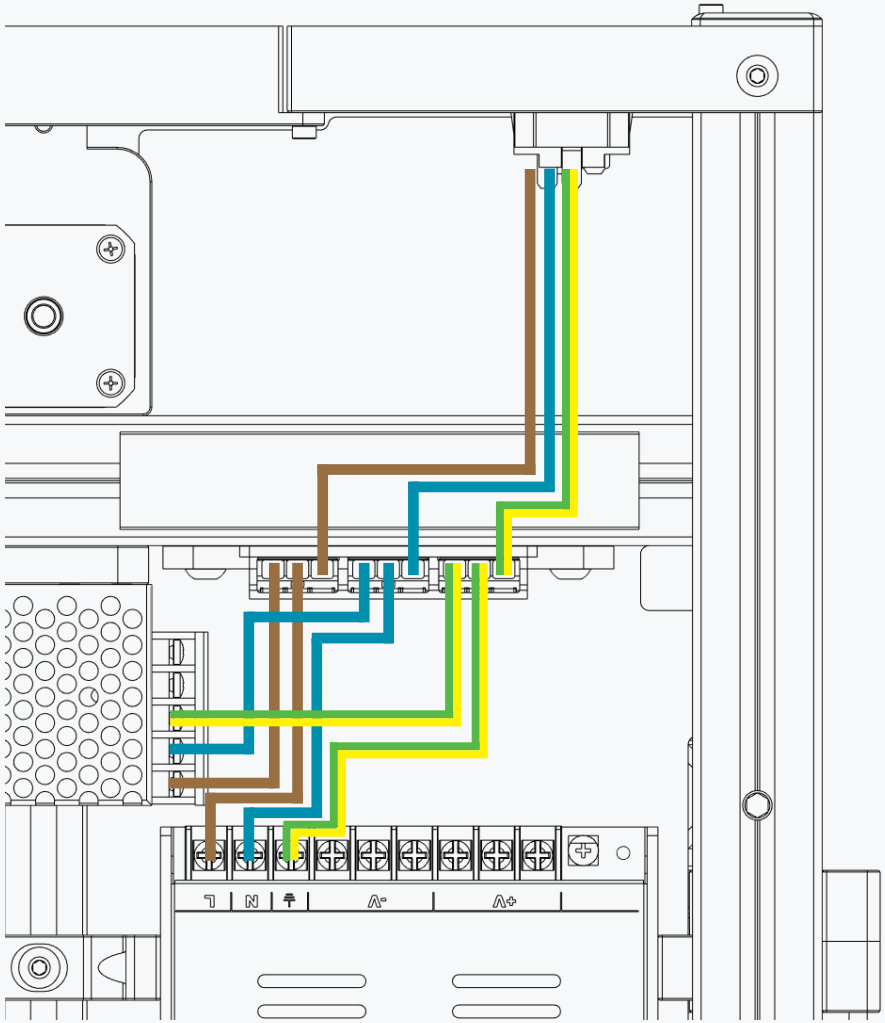
Failure to observe those could result in bodily harm.



MAINS WIRING CONTINUED

Secure the wires with cable clips / cable tie anchors.

Observe your local regulations in regards to grounding the frame/other components.

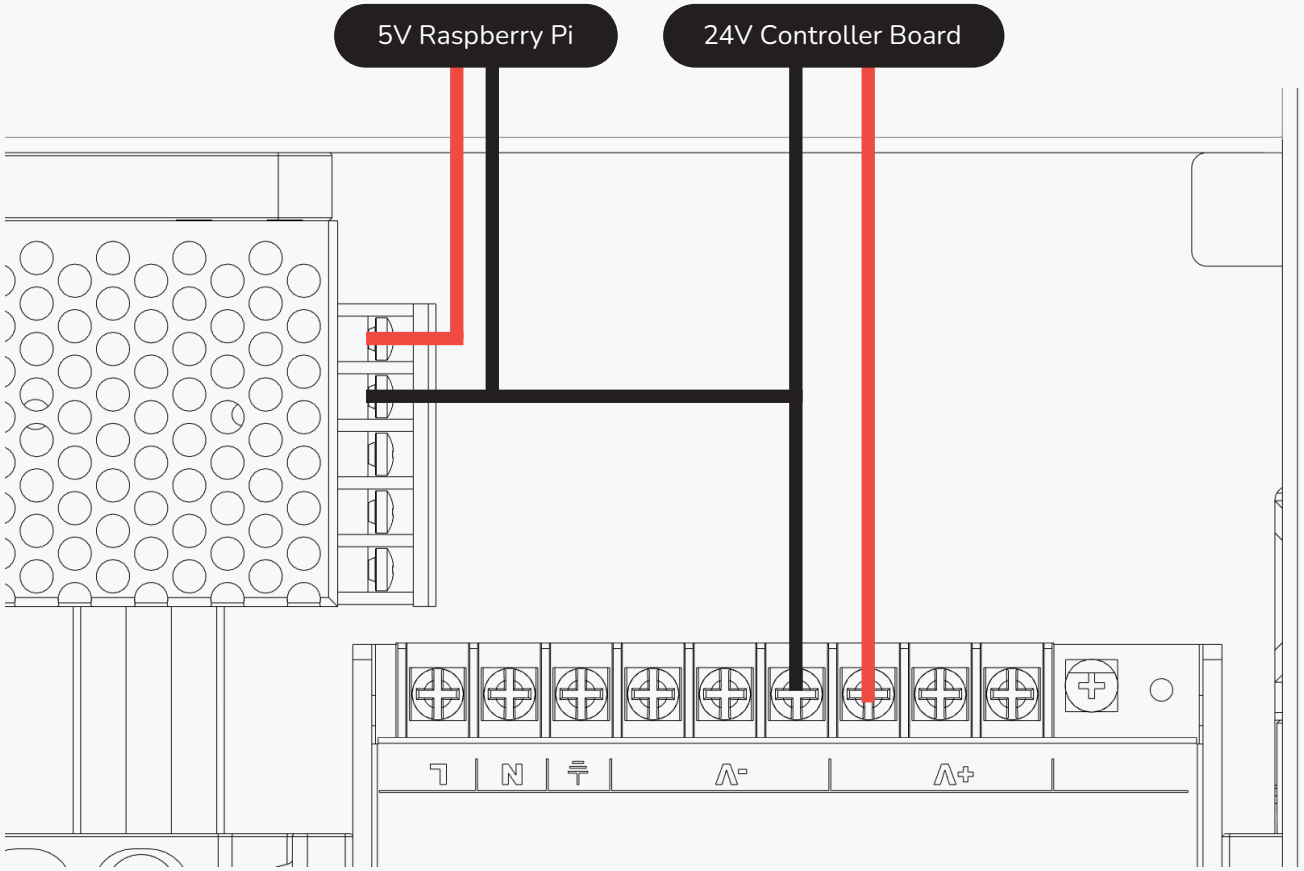


MAINS WIRING CONTINUED

LDO MOTORS

Some vendors provide preprovisioned mains wiring components along.

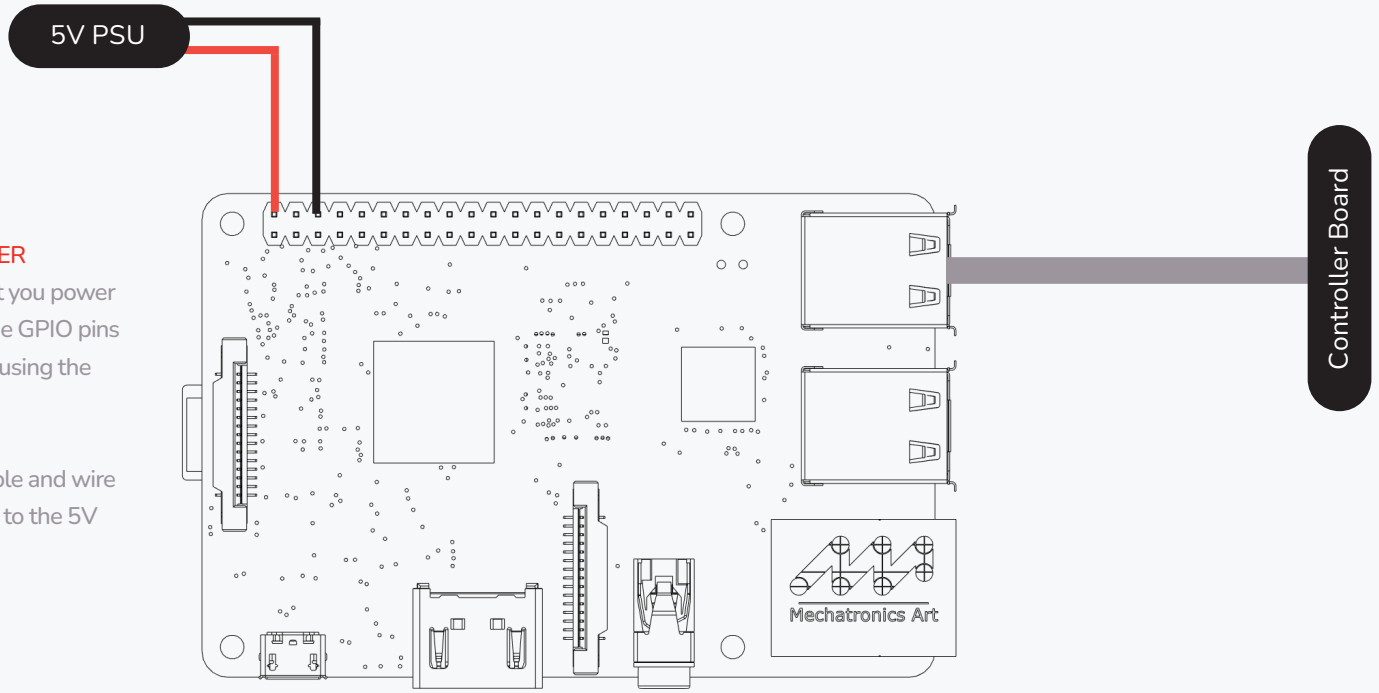
Observe your local regulations in regards to grounding the frame/other components.



CABLE CROSS SECTION

Cables to the controller board should be 2.5mm² (AWG14) or larger.

0.5mm² (AWG20) is sufficient for the connection to the Raspberry Pi.



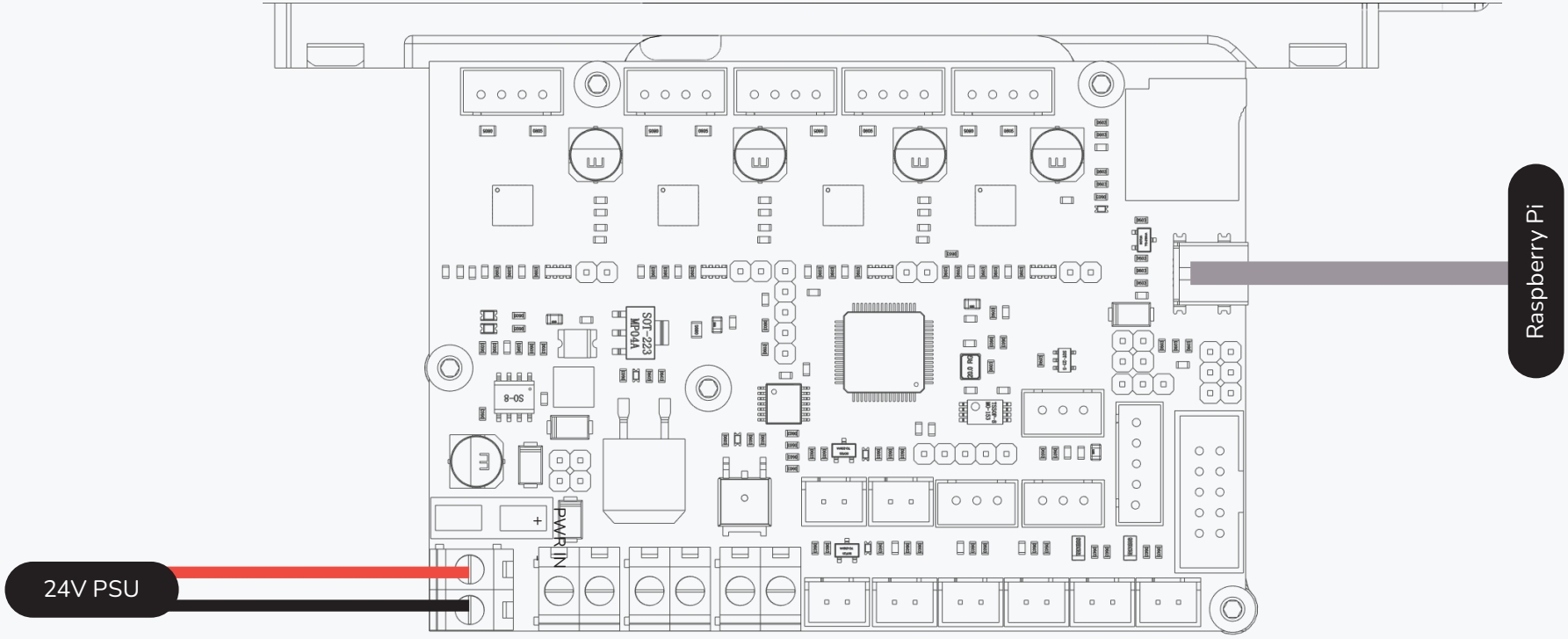
RASPBERRY PI POWER

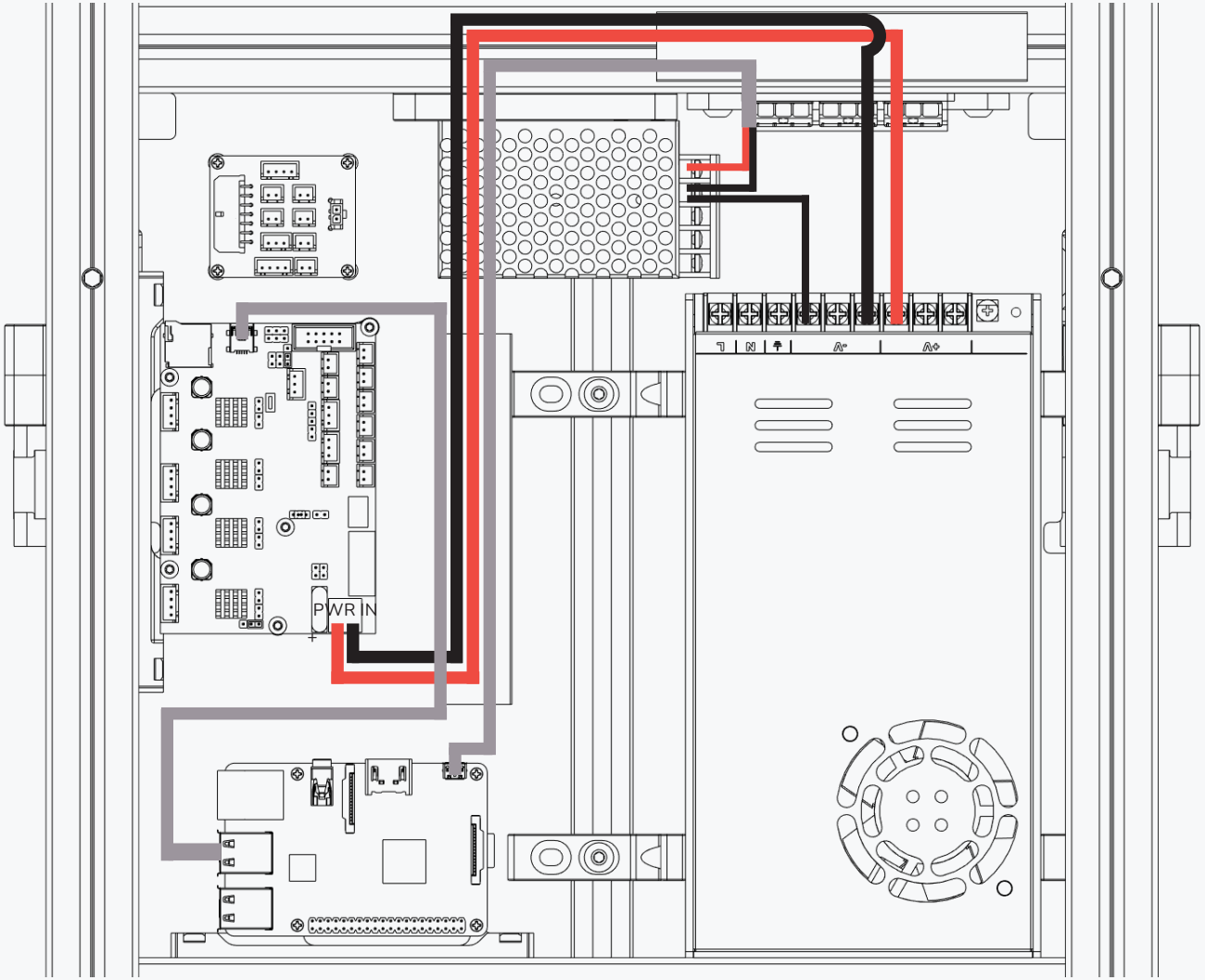
While we suggest that you power the Raspberry Pi via the GPIO pins you may also power it using the "Power-In" USB port.

Cut a suitable USB cable and wire the + and ground lines to the 5V PSU instead.

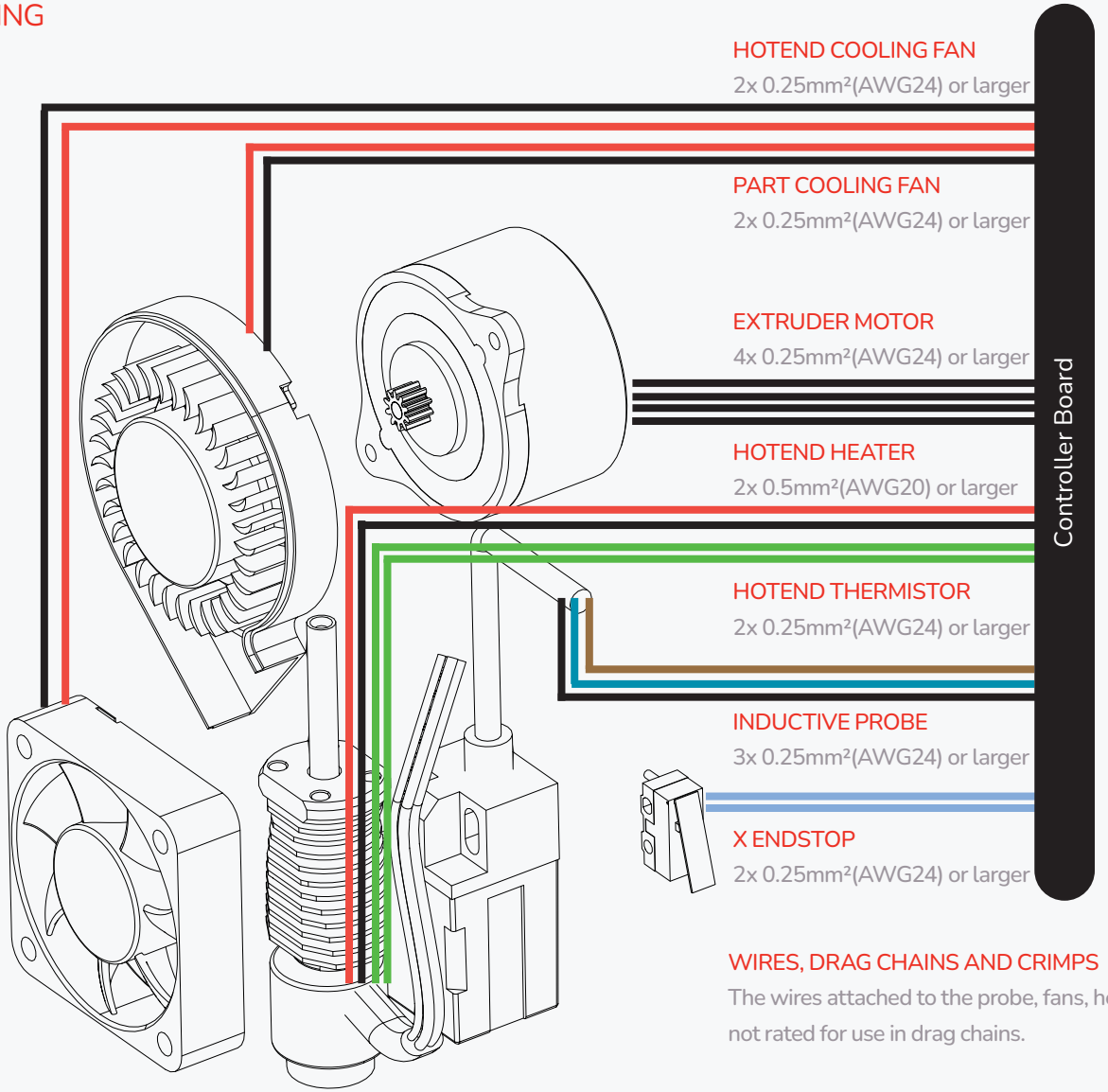
CONTROLLER BOARD

The assembly manual will outline the wiring for a BTT SKR Mini E3 2.0. You can find additional documentation and alternative configurations on docs.vorondesign.com




PWR IN

TOOLHEAD WIRING



LDO MOTORS

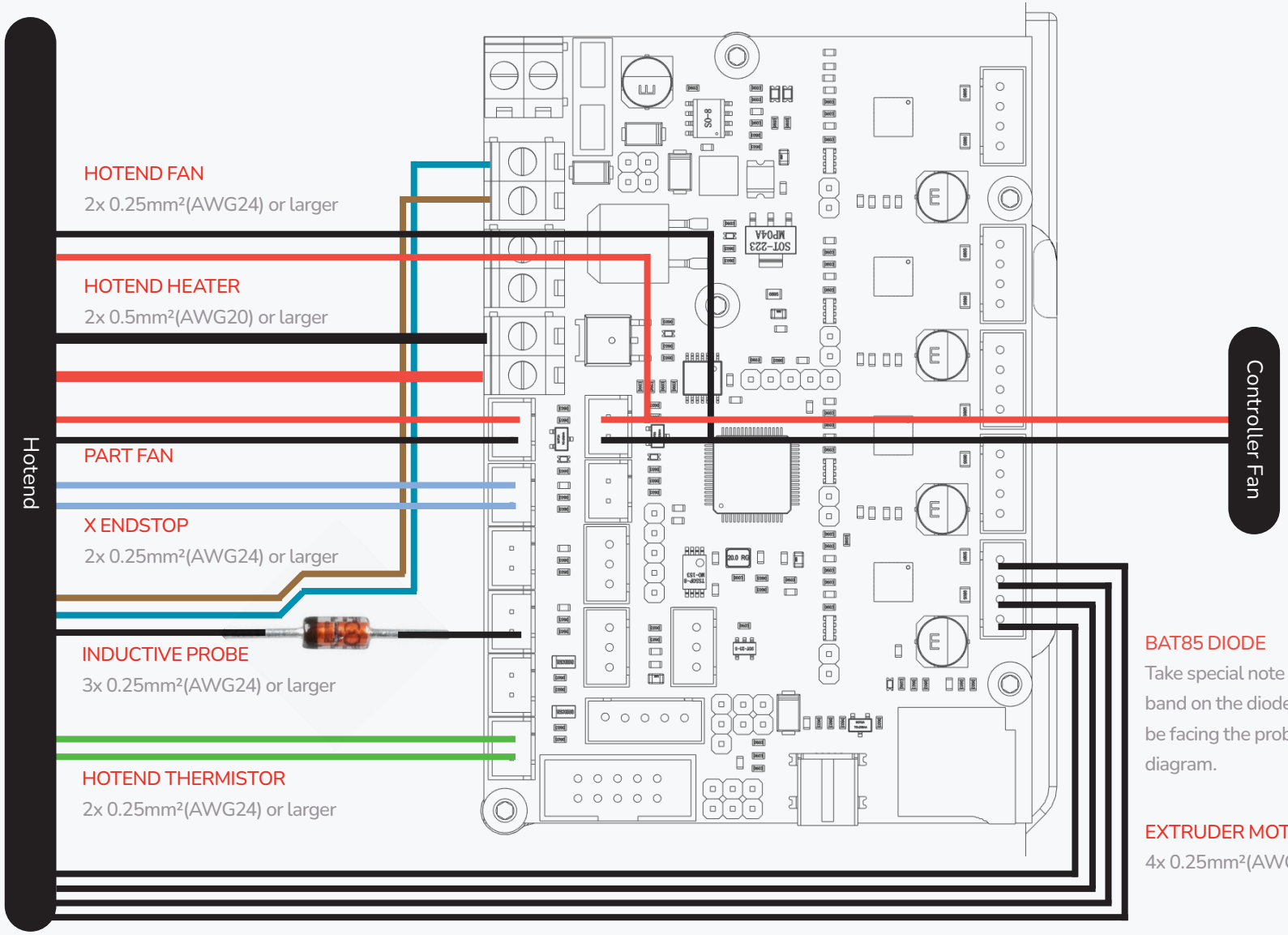
OPTION: TOOLHEAD PCB

If you are planing to use a toolhead PCB consult the Board manufacturer for wiring instructions

WIRES, DRAG CHAINS AND CRIMPS

The wires attached to the probe, fans, heater, etc. are usually not rated for use in drag chains.

Add crimp connectors at the toolhead and run suitable wire down the drag chains. Refer to the sourcing guide for options.

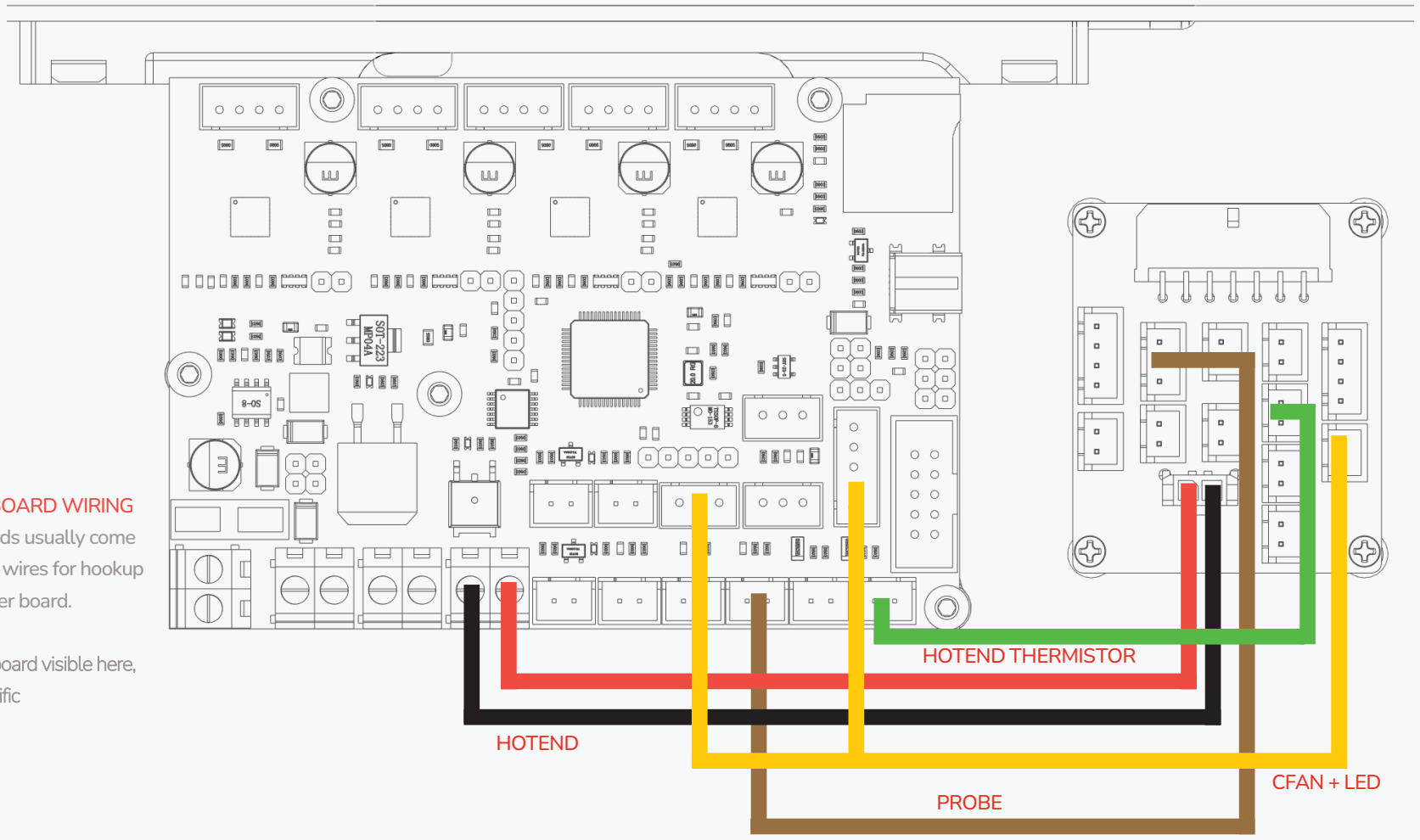


BAT85 DIODE

Take special note of the black band on the diode. It should be facing the probe as per the diagram.

EXTRUDER MOTOR

4x 0.25mm²(AWG24) or larger



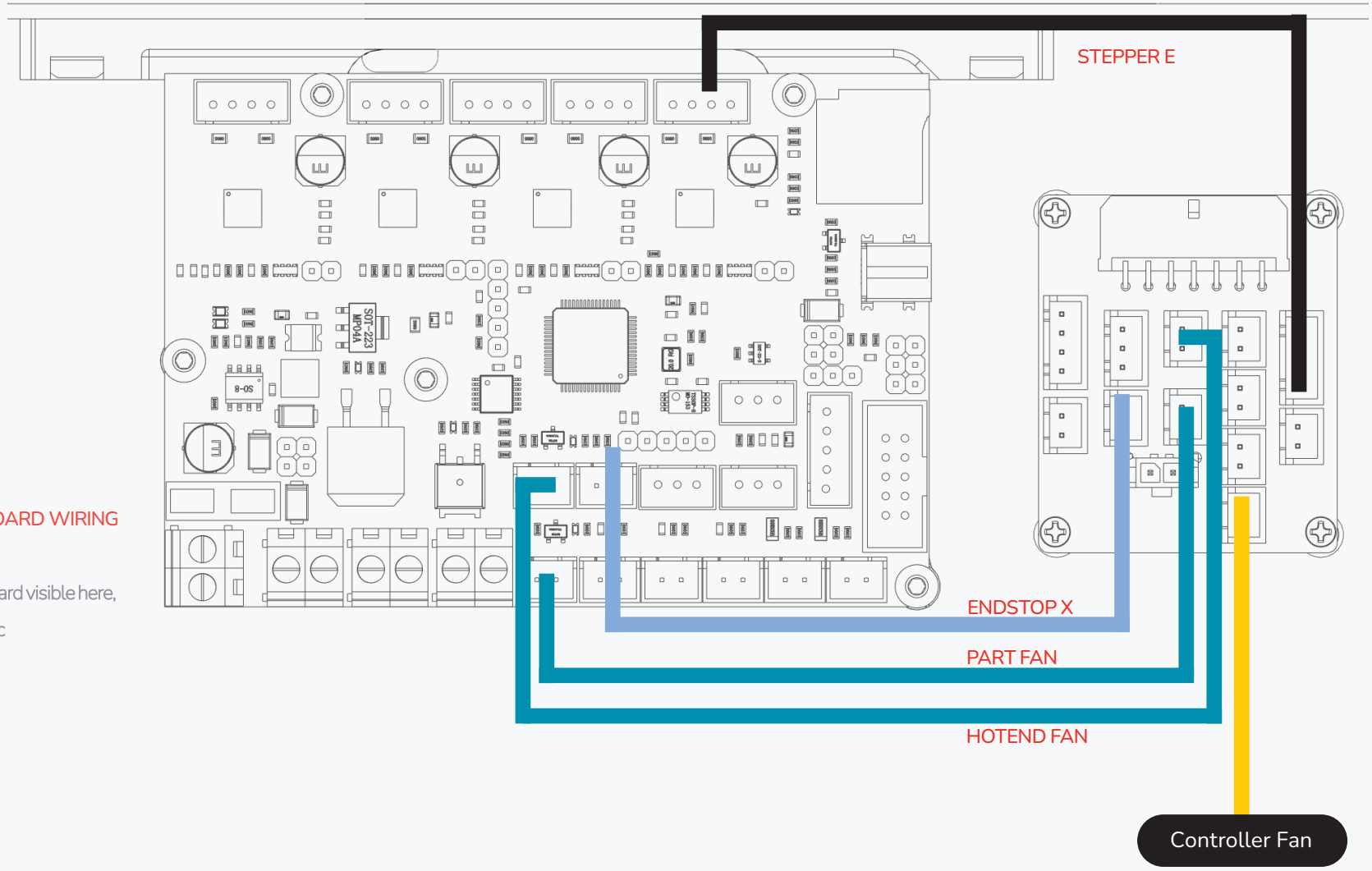
BREAKOUT BOARD WIRING

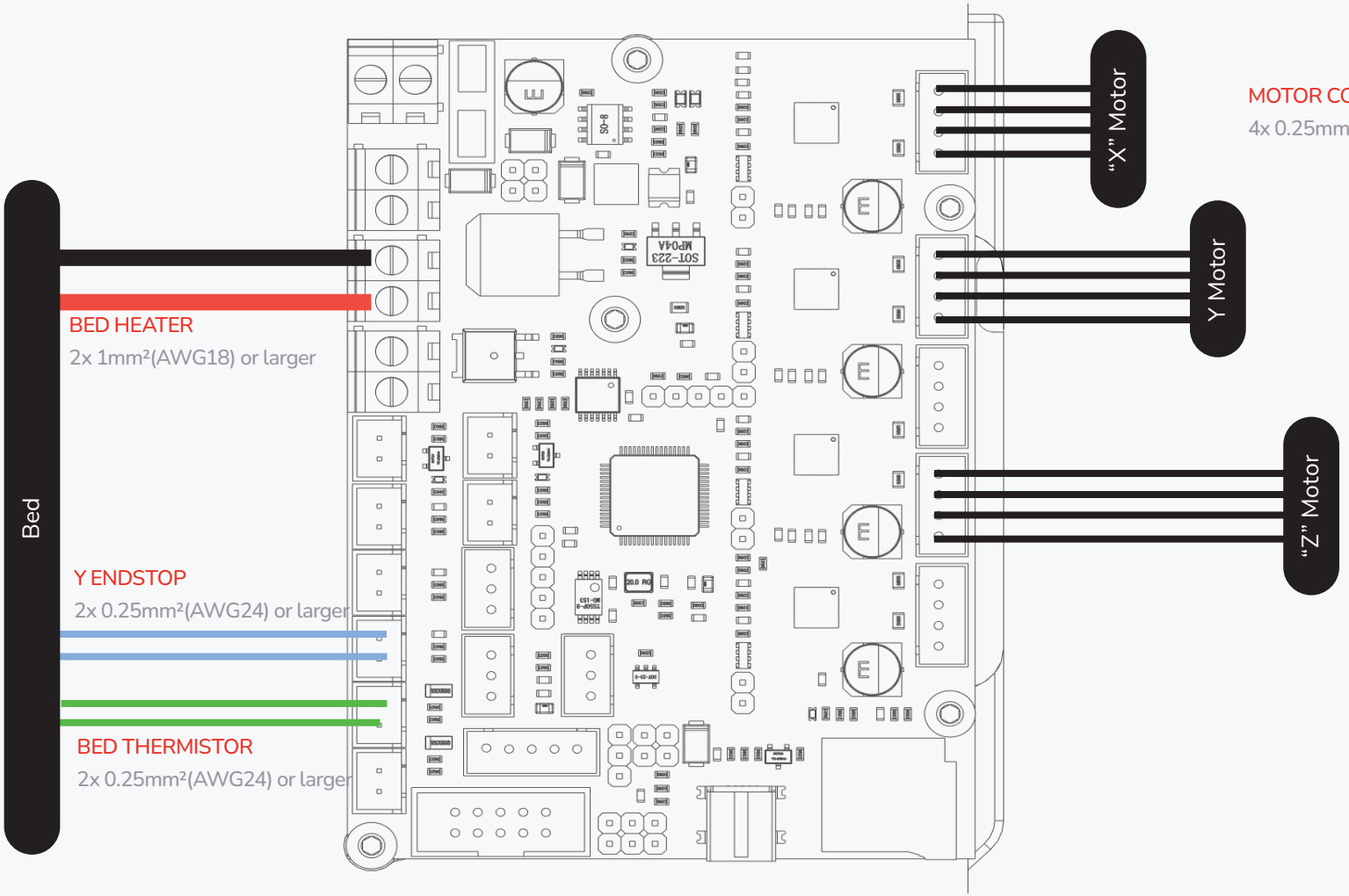
Breakout boards usually come with premade wires for hookup to the controller board.

The breakout board visible here, is LDO kit specific

BREAKOUT BOARD WIRING

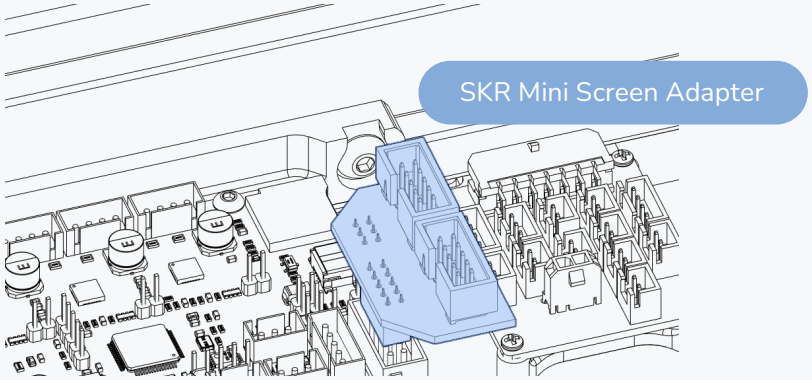
The breakout board visible here, is LDO kit specific

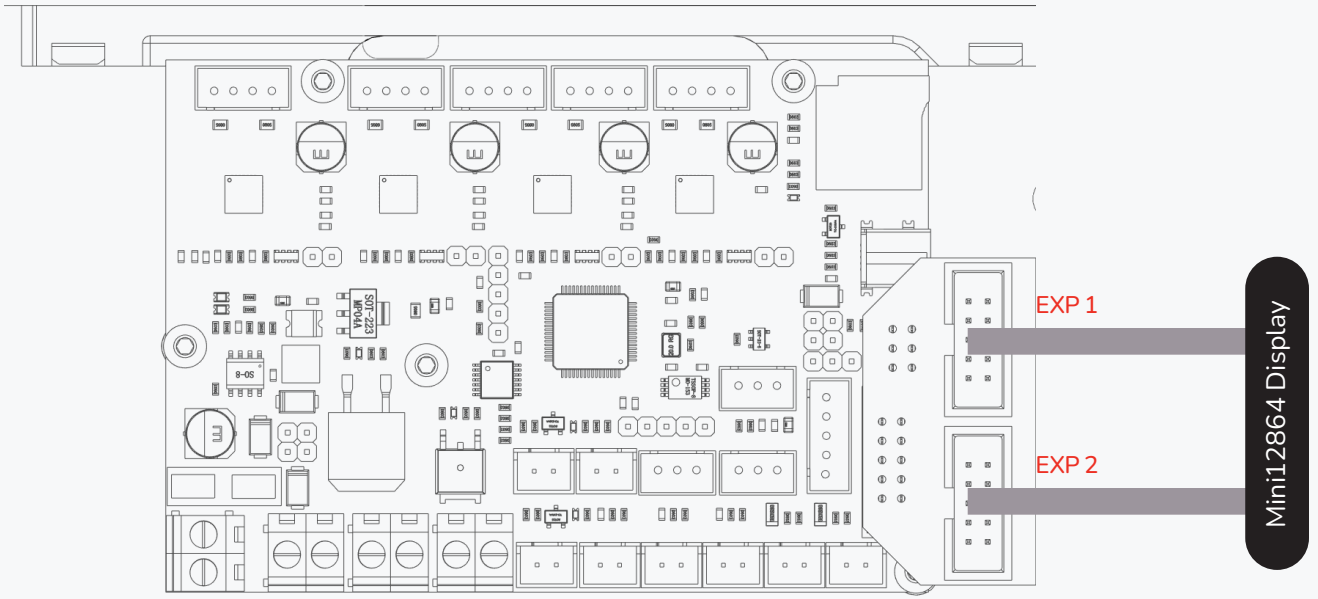


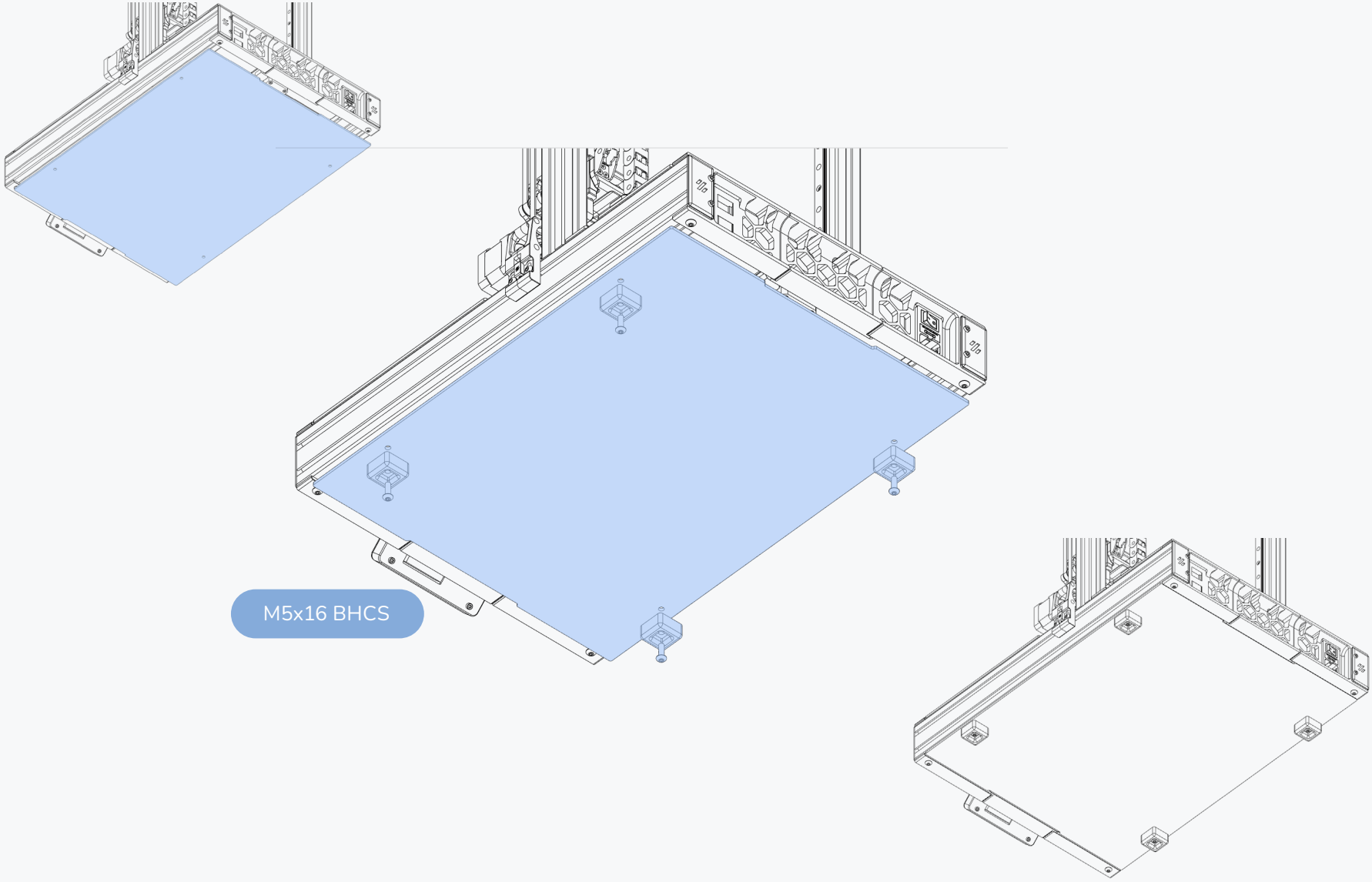


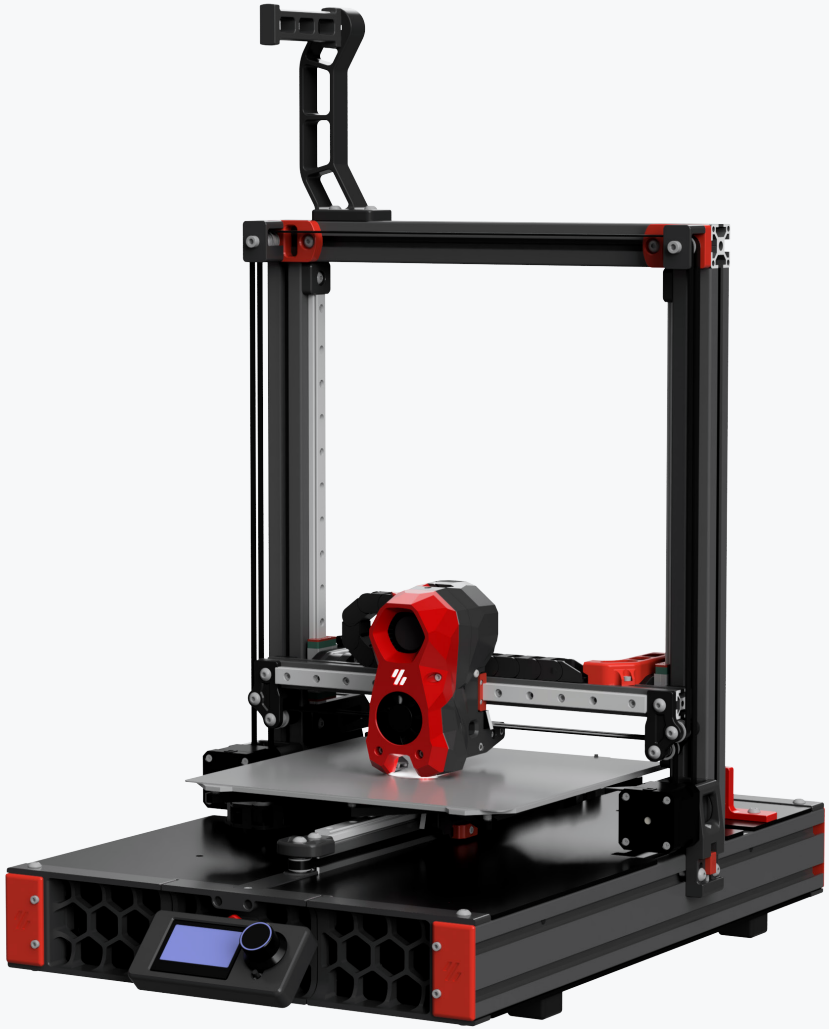
MOTOR CONNECTIONS

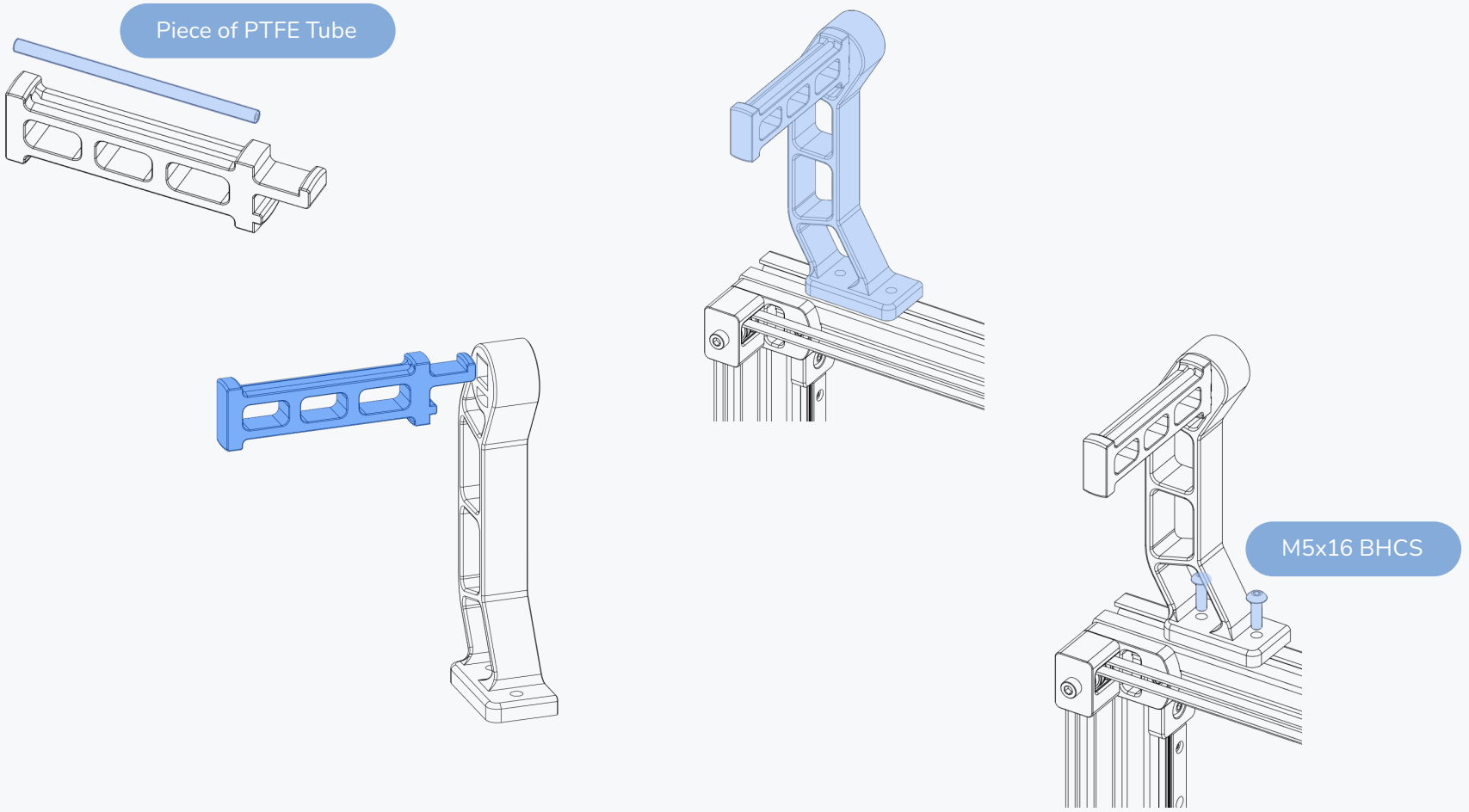
4x 0.25mm²(AWG24) or larger

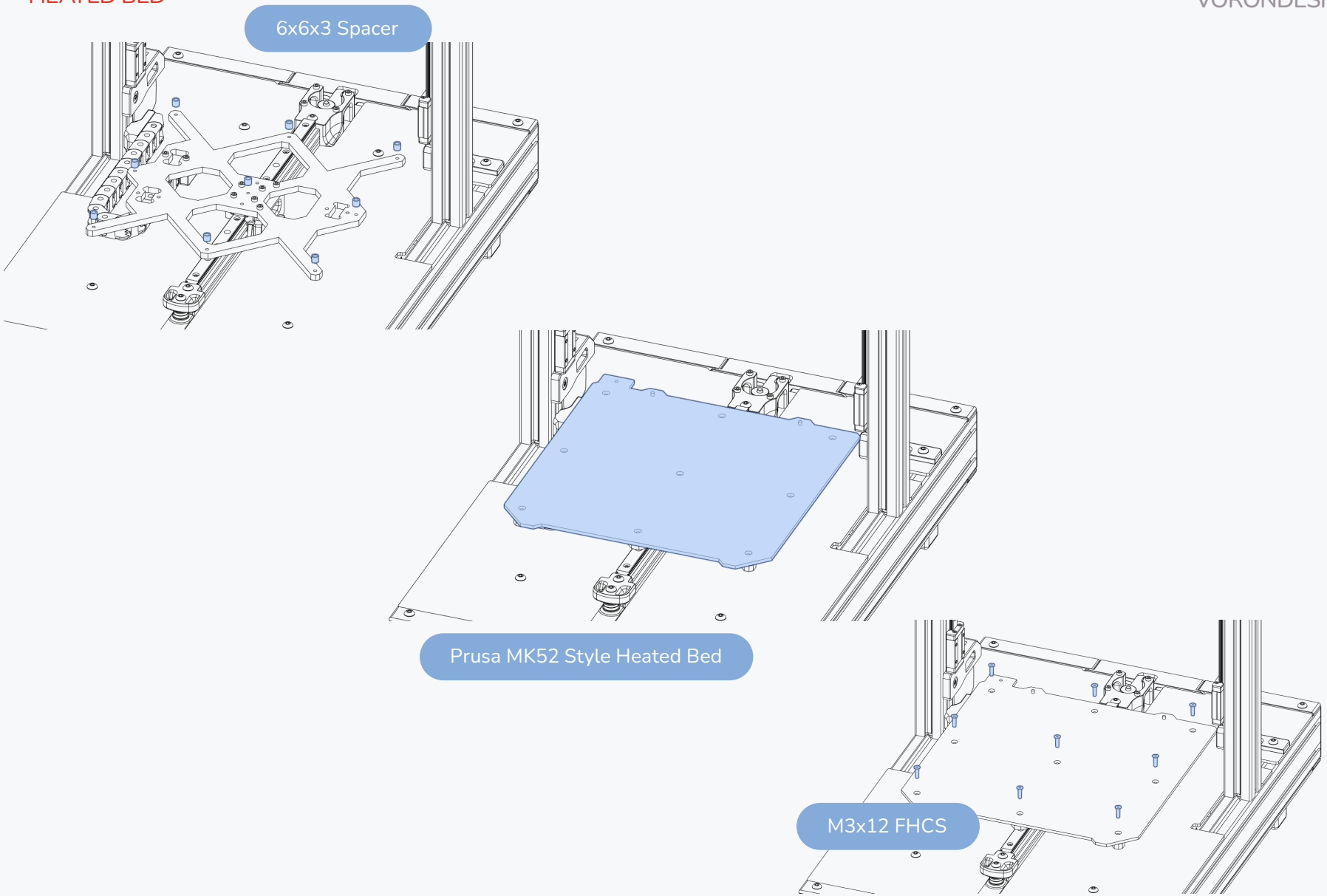

SKR Mini Screen Adapter

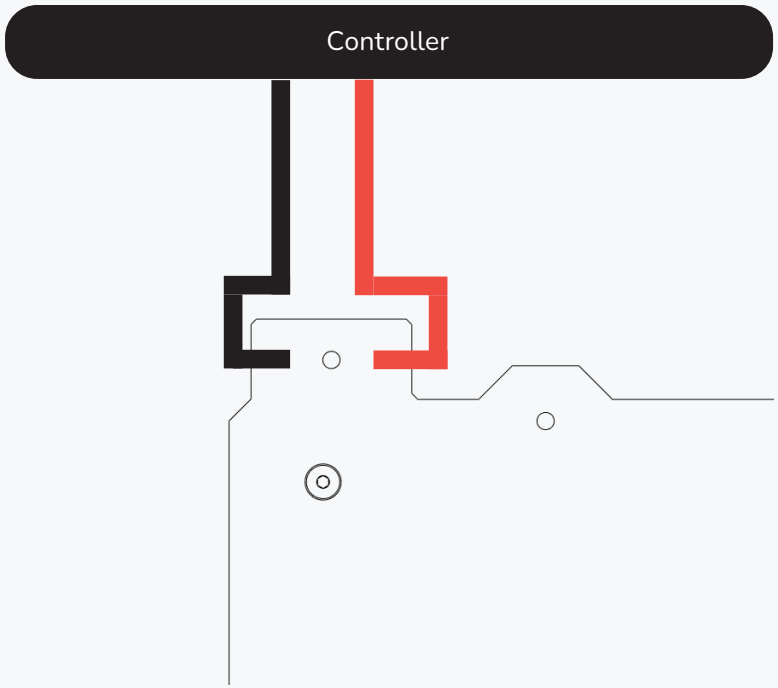

EXP 1
EXP 2
Mini12864 Display

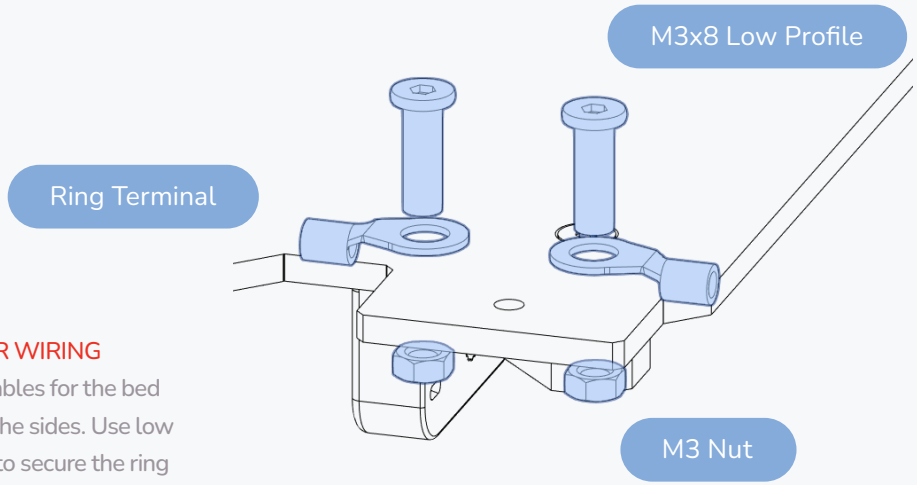

M5x16 BHCS




Piece of PTFE Tube
M5x16 BHCS

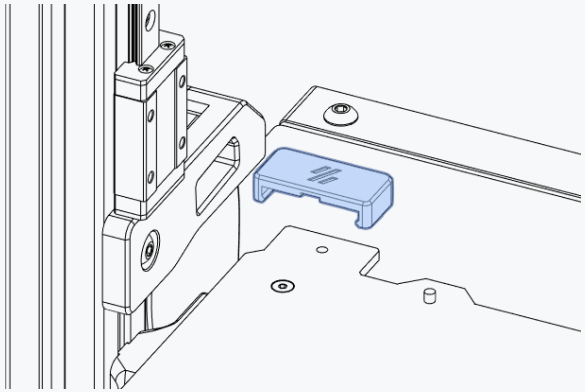

6x6x3 Spacer
Prusa MK52 Style Heated Bed
M3x12 FHCS



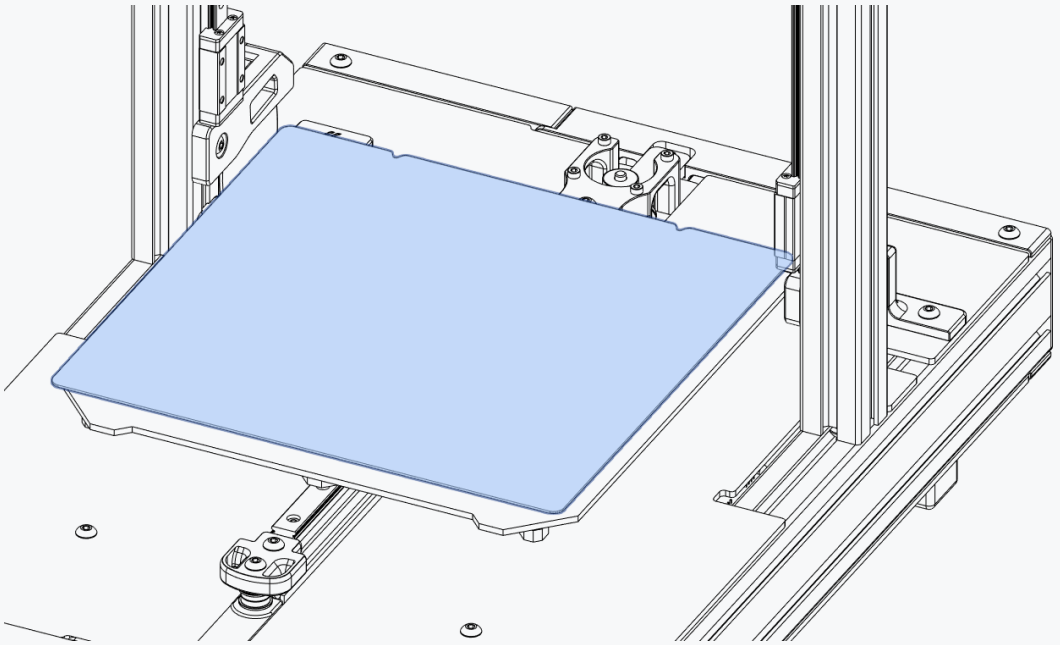


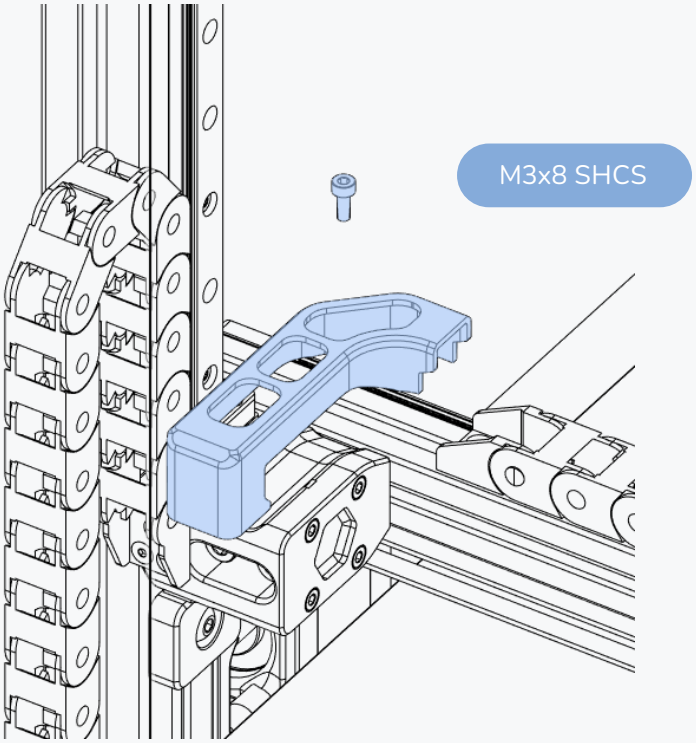
BED HEATER WIRING

Attach the cables for the bed heater from the sides. Use low profile bolts to secure the ring terminals to the heater.



Flex Sheet




M3x8 SHCS

ASSEMBLY COMPLETED! ... NEXT STEP: SETUP & CALIBRATION

This manual is designed to be a simple reference manual for the build process.
For details on the setup of the electronics and other initial steps of your new printer please visit our documentation available on github and docs.vorondesign.com.

https://docs.vorondesign.com

https://github.com/VoronDesign/Voron-Switchwire

HOW TO GET HELP

If you need assistance with your build, we're here to help. Head on over to our Discord group and post your questions. This is our primary medium to help VORON Users and we have a great community that can help you out if you get stuck.

https://discord.gg/voron

Website
https://vorondesign.com

Github
https://github.com/vorondesign

Discord
https://discord.gg/voron